MATH 1025: Applied Linear Algebra

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Preface

These are the first edition of these lecture notes for MATH 1025 (Applied Linear Algebra). Consequently, there may be several typographical errors, missing exposition on necessary background, and more advanced topics for which there will not be time in class to cover. Future iterations of these notes will hopefully be fairly self-contained provided one has the necessary background. If you come across any typos, errors, omissions, or unclear expositions, please feel free to contact me so that I may continually improve these notes.

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Chapter 1

Systems of Linear Equations

Linear algebra begins with one of the most fundamental problems in mathematics: solving systems of linear equations. Many real-world problems involve several quantities that are related to one another through linear relationships. Determining unknown quantities from these relationships appears naturally in areas such as engineering, economics, computer science, physics, biology, statistics, and data science.

For example, a company may wish to determine how many products were manufactured based on production constraints, a transportation network may need to optimize traffic flow through intersections, or a scientist may wish to model interactions between variables in an experiment. In each of these situations, the underlying mathematics can often be expressed as a system of linear equations.

A linear equation is an equation in which each variable appears only to the first power and is not multiplied together with other variables. A collection of such equations forms a linear system. The main goal of this chapter is to develop systematic methods for analyzing and solving these systems.

1.1 Linear Systems

In many applications of mathematics, we are interested in determining unknown quantities that satisfy several conditions simultaneously. These conditions are often expressed as equations involving multiple variables. When the equations are linear, the resulting collection is called a system of linear equations.

Systems of linear equations appear naturally in numerous areas of science and engineering. For example:

- economists use systems to model supply and demand,
- engineers analyze electrical circuits using linear systems,
- computer graphics relies heavily on matrix computations,
- and network analysts study traffic flow using systems of equations.

The main goal of this section is to introduce linear systems and understand what it means to solve them.

1.1.1 Linear Equations

Definition 1.1.1. A *linear equation* in the variables $x_1, x_2, ..., x_n$ is an equation of the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

where

- $a_1, a_2, ..., a_n \in \mathbb{R}$ are constants called the *coefficients*,
- b is a constant called the *constant term*,
- the variables appear only to the first power.

Remark 1.1.2. A linear equation

- does not contain products of variables.
- does not contain square roots of variables.
- does not contain variables in denominators,
- and does not contain powers higher than one.

Example 1.1.3.

a) The equation $2x - 3y + 5z = 7$ is linear.

b) The equation $x^2 + y = 4$ is not linear because the variable x is squared.

c) The equation $xy + z = 1$ is not linear because the variables are multiplied together.

d) The equation $\frac{1}{x} + y = 3$ is not linear because the variable appears in the denominator.

1.1.2 Systems of Linear Equations

Definition 1.1.4. A *system of linear equations* is a collection of linear equations involving the same variables. A general system of m linear equations in n variables has the form

$$\begin{aligned} a_{1,1}x_1 + a_{1,2}x_2 + \cdots + a_{1,n}x_n &= b_1 \\ a_{2,1}x_1 + a_{2,2}x_2 + \cdots + a_{2,n}x_n &= b_2 \\ &\vdots \\ a_{m,1}x_1 + a_{m,2}x_2 + \cdots + a_{m,n}x_n &= b_m \end{aligned}$$

Here, m is the number of equations, n is the number of equations, $a_{i,j}$ denotes the coefficient of x_j in the ith equation, and b_i is the constant term in the ith equation.

Example 1.1.5. The system

$$\begin{aligned} x + 2y &= 5 \\ 3x - y &= 4 \end{aligned}$$

is a system of two linear equations in two variables.

1.1.3 Solutions of Linear Equations

Definition 1.1.6. A *solution* of linear equations is a collection of values for the variables that satisfies every equation in the system simultaneously.

Example 1.1.7. Consider the system

$$\begin{aligned} x + 2y &= 5 \\ 3x - y &= 4 \end{aligned}$$

We claim that $x = \frac{13}{7}$ and $y = \frac{11}{7}$ is a solution.

Indeed, substitute into the first equation:

$$\frac{13}{7} + 2\left(\frac{11}{7}\right) = \frac{13}{7} + \frac{22}{7} = \frac{35}{7} = 5$$

Substitute into the second equation:

$$3\left(\frac{13}{7}\right) - \frac{11}{7} = \frac{39}{7} - \frac{11}{7} = \frac{28}{7} = 4$$

Thus, the ordered pair $\left(\frac{13}{7}, \frac{11}{7}\right)$ satisfies both equations, and is therefore, a solution.

1.1.4 Geometric Interpretation

Systems of linear equations can often be interpreted geometrically.

In two variables, a linear equation in two variables represents a line in the plane. Thus, solving a system of two equations corresponds to finding the intersection points of lines. Possible situations:

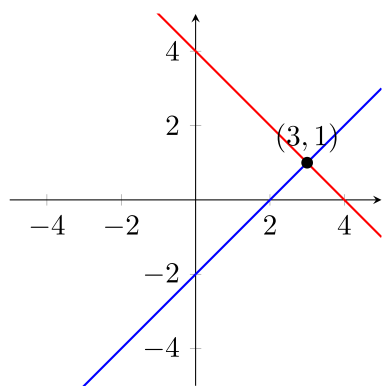
1. the lines intersect once.
2. the lines are parallel.
3. the lines coincide.

Example 1.1.8.

1. Consider

$$x + y = 4$$
$$x - y = 2$$

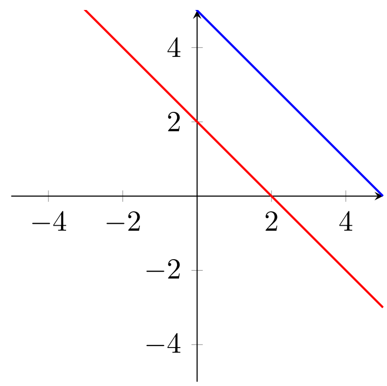
These two lines intersect at exactly one point. Therefore, the system has a unique solution.



2. Consider

$$x + y = 2$$
$$x + y = 5$$

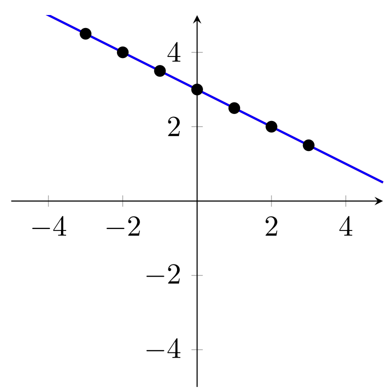
These equations represent parallel lines. Since parallel lines never intersect, the system has no solution.



3. Consider

$$x + 2y = 6$$
$$2x + 4y = 12$$

   The second equation is simply twice the first equation. Thus, both equations represent the same line, so there are infinitely many solutions.



1.1.5 Consistent and Inconsistent Systems

Definition 1.1.9. A system of linear equations is called *consistent* if it has at least one solution. A system is called *inconsistent* if it has no solution.

Theorem 1.1.10. *A system of linear equations has exactly one of the following:*

1. *a unique solution.*
2. *infinitely many solutions.*
3. *no solution.*

Remark 1.1.11. A linear system can never have exactly two solutions, exactly three solutions, or any other finite number greater than one.

1.1.6 Matrix Form of a Linear System

Linear systems can be written compactly using matrices.

Example 1.1.12. Consider the system

$$\begin{aligned} x + 2y &= 5, \\ 3x - y &= 4. \end{aligned}$$

This system can be written in matrix form as

$$\begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$$

Definition 1.1.13. The matrix containing only the coefficients of variables is called the *coefficient matrix*. For the previous example, the coefficient matrix is

$$\begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$$

Definition 1.1.14. The matrix formed by appending the constants column to the coefficient matrix is called the *augmented matrix*. For the previous example,

$$\left[\begin{array}{cc|c} 1 & 2 & 5 \\ 3 & -1 & 4 \end{array}\right]$$

1.2 Gaussian Elimination and Row Echelon Form

In the previous section, we introduced systems of linear equations and discussed what it means for a system to have one solution, infinitely many solutions, or no solution at all. However, as systems become larger and more complicated, solving them by simple substitution or elimination becomes inefficient.

In this section, we develop a systematic procedure called Gaussian elimination, which allows us to solve linear systems efficiently using matrices and elementary row operations.

The central idea is to transform a given system into a simpler equivalent system whose solutions are easier to determine. This process leads to important matrix forms known as:

- row echelon form (REF)
- row reduced echelon form (RREF)

These forms allow us to

- determine whether a system is consistent,
- identify free variables,
- describe all possible solutions clearly.

Gaussian elimination is one of the most important algorithms in mathematics and plays a fundamental role throughout linear algebra, numerical analysis, engineering, computer science, and data science.

1.2.1 Elementary Row Operations

To simplify systems systematically, we use operations that do not change the solution set of the system.

Definition 1.2.1. The following operations are called *elementary row operations*:

1. Row interchange: swap two rows.
$$R_i \leftrightarrow R_j$$
2. Row scaling: multiply a row by a nonzero constant.
$$R_i \to cR_i, \quad c \neq 0$$
3. Row replacement: replace a row by itself plus a multiple of another row.
$$R_i \to R_i + cR_j$$

Remark 1.2.2. Elementary row operations preserve the solutions of the system. Therefore, any system obtained using these operations is equivalent to the original system.

1.2.2 Gaussian Elimination and Row Echelon Form (REF)

Definition 1.2.3. *Gaussian elimination* is a process of applying elementary row operations to an augmented matrix in order to transform it into a simpler form. The goal is typically to obtain

- row echelon form
- reduced row echelon form

Definition 1.2.4. A matrix is in *row echelon form* if

1. All nonzero rows are above any rows consisting entirely of zeros.
2. The leading entry of each nonzero row is to the right of the leading entry in the above row.
3. All entries below a leading entry are zero.

Definition 1.2.5. The first nonzero entry in a row is called the *leading entry* or *pivot*.

Example 1.2.6. The matrix

$$\begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & 1 & 4 & 5 \\ 0 & 0 & 2 & 1 \end{bmatrix}$$

is in row echelon form. The pivots occur in columns 1, 2, and 3.

Example 1.2.7. The matrix

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & 1 & 4 \end{bmatrix}$$

is not in row echelon form because the pivot in the third row is to the left of the pivot above it.

1.2.3 Reduced Row Echelon Form (RREF)

Definition 1.2.8. A matrix is in *reduced row echelon form* if

1. It is already in row echelon form.
2. Every pivot equals 1.
3. Each pivot is the only nonzero entry in its column.

Example 1.2.9. The matrix

$$\begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

is in reduced row echelon form.

Example 1.2.10. The matrix

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

is not in reduced row echelon form because there is a nonzero entry above the pivot in column 2.

1.2.4 Solving Systems Using Gaussian Elimination

We now solve systems systematically using row operations.

Example 1.2.11. Solve the system

$$\begin{aligned} x + 2y - z &= 3, \\ 2x + y + z &= 7, \\ -x + y + 2z &= 2. \end{aligned}$$

Solution

First write the augmented matrix

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 2 & 1 & 1 & 7 \\ -1 & 1 & 2 & 2 \end{array}\right]$$

We first eliminate entries below the first pivot. Thus, we use the row operations

$$\begin{aligned} &R_2 \to R_2 - 2R_1, \\ &R_3 \to R_3 + R_1 \end{aligned}$$

to get

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & -3 & 3 & 1 \\ 0 & 3 & 1 & 5 \end{array}\right]$$

Next, we eliminate below the second pivot. We use the operation

$$R_3 \to R_3 + R_2$$

to get

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & -3 & 3 & 1 \\ 0 & 0 & 4 & 6 \end{array}\right]$$

The third row gives

$$4z = 6$$

so

$$z = \frac{3}{2}$$

The second row gives

$$-3y + 3z = 1$$

Substitute $z = \frac{3}{2}$ to get

$$\begin{aligned} -3y + \frac{9}{2} &= 1 \\ -3y &= -\frac{7}{2} \\ y &= \frac{7}{6} \end{aligned}$$

Finally, the first row gives

$$x + 2y - z = 3$$

Substitute $y = \frac{7}{6}$ and $z = \frac{3}{2}$ to get

$$\begin{aligned} x + 2\left(\frac{7}{6}\right) - \frac{3}{2} &= 3 \\ x + \frac{7}{3} - \frac{3}{2} &= 3 \\ x + \frac{5}{6} &= 3 \\ x &= \frac{13}{6} \end{aligned}$$

The solution is

$$\left(\frac{13}{6}, \frac{7}{6}, \frac{3}{2}\right).$$

Once a system is in row echelon form, the variables can be solved form the bottom equation and working upward. This process is called *back substitution*.

1.2.5 Inconsistent Systems

Gaussian elimination also helps determine whether a system has no solution.

Example 1.2.12. Consider the system

$$\begin{aligned} x + y &= 2, \\ 2x + 2y &= 5. \end{aligned}$$

The augmented matrix is

$$\left[\begin{array}{cc|c} 1 & 1 & 2 \\ 2 & 2 & 5 \end{array}\right]$$

Apply $R_2 \to R_2 - 2R_1$ to get

$$\left[\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 0 & 1 \end{array}\right]$$

The second row corresponds to $0 = 1$, which is impossible.

Therefore, the system is inconsistent, and has no solution.

1.2.6 Free Variables and Infinitely Many Solutions

Example 1.2.13. Solve the system

$$\begin{aligned} x + 2y + z &= 4, \\ 2x + 4y + 2z &= 8 \end{aligned}$$

Solution

The augmented matrix is

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 2 & 4 & 2 & 8 \end{array}\right]$$

Apply $R_2 \to R_2 - 2R_1$ to get

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Since there is only one equation involving three variables, some variables may be chosen freely.

Let $y = s$ and $z = t$. Then

$$x = 4 - 2s - t$$

Thus, the solutions are

$$(x, y, z) = (4 - 2s - t, s, t)$$

where $s, t \in \mathbb{R}$. Therefore, the system has infinitely many solutions.

Definition 1.2.14. A variable corresponding to a pivot column is called a *pivot column*. A variable corresponding to a non-pivot column is called a *free variable*.

Remark 1.2.15. Free variables can take arbitrary values, leading to infinitely many solutions.

1.3 Homogeneous Systems

In the previous section, we developed systematic methods for solving systems of linear equations using Gaussian elimination and row echelon form. We now turn our attention to a special and extremely important class of systems called homogeneous systems.

A homogeneous system is distinguished by the fact that all constant terms are zero. Although this may appear to be a small modification, homogeneous systems possess important structural properties that make them fundamentally different from general systems.

One of the most important facts is that homogeneous systems are always consistent because the zero solution always exists. However, depending on the number of variables and pivots, a homogeneous system may also possess infinitely many nontrivial solutions.

1.3.1 Homogeneous Systems

Definition 1.3.1. A system of linear equations is called *homogeneous* if all constant terms are zero. A general homogeneous system has the form

$$\begin{aligned} a_{1,1}x_1 + a_{1,2}x_2 + \cdots + a_{1,n}x_n &= 0 \\ a_{2,1}x_1 + a_{2,2}x_2 + \cdots + a_{2,n}x_n &= 0 \\ &\vdots \\ a_{m,1}x_1 + a_{m,2}x_2 + \cdots + a_{m,n}x_n &= 0 \end{aligned}$$

Example 1.3.2. The system

$$x + 2y - z = 0,$$

$$3x - y + 4z = 0$$

is homogeneous.

Example 1.3.3. The system

$$\begin{aligned} x + y &= 2, \\ 2x - y &= 5 \end{aligned}$$

is not homogeneous because the constants on the right side are not zero.

1.3.2 The Trivial Solution

Definition 1.3.4. The solution $x_1 = x_2 = \cdots = x_n = 0$ is called the *trivial solution.*

Theorem 1.3.5. *Every homogeneous system has at least one solution.*

Proof. Since all constant terms are zero, substituting $x_1 = x_2 = \cdots = x_n = 0$ into every equation produces $0 = 0$. Thus, the trivial solution always satisfies the system. Therefore, every homogeneous system is consistent. □

Remark 1.3.6. A homogeneous system can never be inconsistent because the trivial solution always exists.

1.3.3 Nontrivial Solutions

Definition 1.3.7. Any solution of a homogeneous system other than the trivial solution is called a *nontrivial solution.*

Example 1.3.8. Consider the system

$$\begin{aligned} x + y &= 0, \\ 2x + 2y &= 0 \end{aligned}$$

The second equation is simply twice the first equation, so the system reduces to

$$\begin{aligned} x + y &= 0 \\ x &= -y \end{aligned}$$

Let $y = t$, so that $x = -t$. Thus, the solutions are $(x, y) = (-t, t)$, where $t \in \mathbb{R}$.

If $t \neq 0$, the solution is nontrivial. For example, $(-1, 1)$ is a nontrivial solution.

1.3.4 Solving Homogeneous Systems

Homogeneous systems are solved using the same row reduction techniques developed in the previous section.

Example 1.3.9. Solve the homogeneous system

$$\begin{aligned} x + 2y - z &= 0, \\ 2x + 5y + z &= 0. \end{aligned}$$

Solution

Write the augmented matrix

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 2 & 5 & 1 & 0 \end{array}\right]$$

Eliminate below the pivot by applying $R_2 \to R_2 - 2R_1$ to get

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 0 & 1 & 3 & 0 \end{array}\right]$$

The matrix corresponds to

$$\begin{aligned} x + 2y - z &= 0, \\ y + 3z &= 0. \end{aligned}$$

From the second equation, we get

$$y = -3z$$

Let $z = t$ so that

$$y = -3t$$

Substituting into the first equation, we have

$$\begin{aligned} x + 2(-3t) - t &= 0 \\ x - 7t &= 0 \\ x &= 7t \end{aligned}$$

Therefore, the solutions are $(x, y, z) = (7t, -3t, t)$, where $t \in \mathbb{R}$. Since infinitely many values of t are possible, the system has infinitely many solutions.

The existence of free variables determines whether nontrivial solutions exist.

Theorem 1.3.10. *If a homogeneous system has at least one free variable, then it has infinitely many solutions.*

Proof. A free variable may be assigned any real number. Since infinitely many choices are possible, infinitely many solutions are obtained. ☐

1.3.5 Existence of Nontrivial Solutions

One of the most important results concerning homogeneous systems is the following theorem.

Theorem 1.3.11. *A homogeneous system with more variables than equations must have at least one variable. Consequently, it must have infinitely many solutions.*

Proof. Suppose a homogeneous system has m equations and n variables, where $n > m$. A pivot can occur in at most one column of each row, so there can be at most m pivot columns. Since there are n variables, and $n > m$, at least one variable cannot correspond to a pivot column. Therefore, at least one free variable exists. By Theorem 1.3.10, the system has infinitely many solutions. ☐

Remark 1.3.12. This theorem is extremely important in linear algebra and will reappear frequently throughout the course.

1.3.6 Parametric and Vector Form of Solutions

Solutions involving free variables are often written using parameters and can also be written using vectors.

Example 1.3.13. Solve

$$\begin{aligned} x + y + z &= 0, \\ 2x + 2y + 2z &= 0 \end{aligned}$$

Solution

The second equation is redundant, so we have

$$\begin{aligned} x + y + z &= 0 \\ x &= -y - z \end{aligned}$$

Let $y = s$ and $z = t$. Then

$$x = -s - t$$

Therefore,

$$(x, y, z) = (-s - t, s, t)$$

where $s, t \in \mathbb{R}$. This is called the *parametric form* of the solution set.

Equivalently, we can write in *vector form*

$$(x, y, z) = s(-1, 1, 0) + t(-1, 0, 1)$$

This representation will become extremely important later when we study vector spaces and span.

1.3.7 Homogeneous Systems in Matrix Form

A homogeneous system may be written compactly as

$$A\vec{x} = \vec{0}$$

where A is the coefficient matrix, $\vec{x}$ is the vector of variables, and $\vec{0}$ is the zero vector.

Example 1.3.14. The system

$$\begin{aligned} x + 2y - z &= 0, \\ 3x - y + 4z &= 0 \end{aligned}$$

can be written as

$$\begin{bmatrix} 1 & 2 & -1 \\ 3 & -1 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

1.4 Network Flow Models

One of the strengths of linear algebra is its ability to model and solve real-world problems. In many situations, quantities move through a network:

- cars travel through roads,
- passengers move through transit systems,
- water flows through pipes,
- electricity moves through circuits,

- information travels through communication networks.

Thse situations can be analyzed using systems of linear equations.

In this section, we study *network flow models*, which describe how quantities move through interconnected systems. The central idea is based on a conservation principle: The total flow entering a node equals the total flow leaving the node.

This principle naturally produces systems of linear equations. By solving these systems, we can determine unknown traffic patterns, transportation flows, or resource distributions.

1.4.1 Networks and Nodes

Definition 1.4.1. A *network* is a collection of points connected by paths. The points are called *nodes* and the connecting paths are called *edges*.

Examples of networks include road systems, subway systems, airline routes, water pipelines, internet communication systems, and electrical circuits.

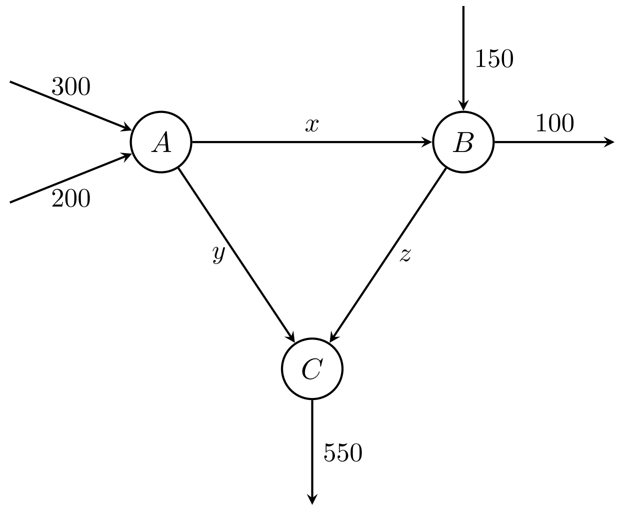
1.4.2 Flow Conservation

The key principle governing network flow models is conservation of flow.

Definition 1.4.2. At each node, the total amount entering a node must equal the total amount leaving the node.

This principle is sometimes called the conservation of mass, conservation of traffic, or flow balance.

Example 1.4.3. Consider the following road network. Cars move through an intersection system according to the diagram below.



Suppose x, y, and z represent unknown traffic flows measured in cars per hour. The network satisfies the following conditions:

- At Node A: $300 + 200 = x + y$
- At Node B: $x + 150 = z + 100$
- At Node C: $y + z = 550$

We obtain the system

$$\begin{aligned} x + y &= 500, \\ x - z &= -50, \\ y + z &= 550 \end{aligned}$$

Writing the augmented matrix of the system gives

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 500 \\ 1 & 0 & -1 & -50 \\ 0 & 1 & 1 & 550 \end{array}\right]$$

We eliminate the entries using row operations. Apply the operations

$$\begin{aligned} R_2 &\to R_2 - R_1, \\ R_3 &\to R_3 + R_2 \end{aligned}$$

so that we get

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 500 \\ 0 & -1 & -1 & -550 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

The system reduces to

$$\begin{aligned} x + y &= 500, \\ y + z &= 550. \end{aligned}$$

There is one free variable. Let $z = t$ so that $y = 550 - t$. Substitute into the first equation to get

$$\begin{aligned} x + (550 - t) &= 500 \\ x &= t - 50 \end{aligned}$$

The traffic flows are

$$(x, y, z) = (t - 50, 550 - t, t)$$

where t is a free parameter.

Different values of t correspond to different possible traffic patterns that satisfy the flow conditions. However, in practice, traffic flow cannot be negative, so we require $x \geq 0$, $y \geq 0$, and $z \geq 0$. This imposes restrictions on t.

Example 1.4.4. We now consider a transportation example inspired by the TTC. Suppose passengers move between major TTC stations:

- Eglinton,
- St. George,
- Bloor–Yonge,

Let x, y, and z represent passenger flow rates (in passengers per hour). Suppose the following conditions hold:

- Bloor–Yonge: 700 passengers/hour enter while x and y leave. Then

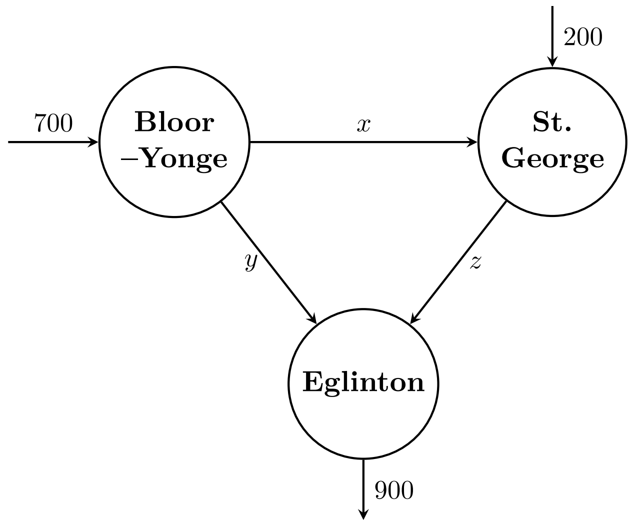
$$x + y = 700$$

- St. George: flow x enters, 200 passengers/hour additionally enter, and z exists. Then

$$x + 200 = z$$

- Eglinton: flows y and z enter, and 900 passengers/hour leave. Then

$$y + z = 900$$



We solve the system

$$\begin{aligned} x + y &= 700, \\ x - z &= -200, \\ y + z &= 900 \end{aligned}$$

Write the augmented matrix

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 700 \\ 1 & 0 & -1 & -200 \\ 0 & 1 & 1 & 900 \end{array}\right]$$

We eliminate the entries using row operations. Apply the operations

$$\begin{aligned} R_2 &\to R_2 - R_1, \\ R_3 &\to R_3 + R_2 \end{aligned}$$

so that we get

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 700 \\ 0 & -1 & -1 & -900 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Let $z = t$ so that $y = 900 - t$. Substitute into the first equation,

$$\begin{aligned} x + (900 - t) &= 700 \\ x &= t - 200 \end{aligned}$$

Therefore, the passenger flows are

$$(x, y, z) = (t - 200, 900 - t, t)$$

Chapter 2

Matrices

In Chapter 1, we studied systems of linear equations and developed systematic methods for solving them using Gaussian elimination and row operations. As the size and complexity of systems increase, however, it becomes increasingly useful to organize information efficiently. This motivates the introduction of one of the most important objects in linear algebra: the matrix.

Matrices provide a compact and powerful way to represent systems of equations, organize data, describe transformations, and perform computations efficiently. They appear naturally in many areas of mathematics, science, engineering, economics, computer graphics, machine learning, statistics, and data science.

For example:

- a spreadsheet of numerical data can be represented as a matrix,
- images on a computer are stored as matrices of pixel values,
- transportation networks may be modeled using matrices,
- and systems of equations can be solved using matrix operations.

In this chapter, we develop the basic algebra of matrices and study how matrices can be manipulated and combined.

2.1 Matrix Addition and Scalar Multiplication

In Chapter 1, we used matrices as a convenient way to organize systems of linear equations. In this chapter, we study matrices as mathematical objects in their own right and develop the algebraic operations that can be performed on them.

The first operations we consider are

- matrix addition,
- matrix subtraction, and
- scalar multiplication.

These operations allow us to combine matrices and scale their entries in a natural way. They form the foundation for matrix algebra and will be used repeatedly throughout the course.

2.1.1 Matrices

Definition 2.1.1. A *matrix* is a rectangular array of numbers arranged in rows and columns. A general $m \times n$ matrix has the form

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & & \vdots \\ a_{m,1} & a_{m,2} & \cdots & a_{m,n} \end{bmatrix}$$

where m is the number of rows, n is the number of columns, and $a_{i,j}$ denotes the entry in row i, column j.

Example 2.1.2. The matrix

$$A = \begin{bmatrix} 1 & -2 & 4 \\ 0 & 5 & 7 \end{bmatrix}$$

is a 2×3 matrix. Its entries include

$$a_{1,1} = 1, \quad a_{1,2} = -2, \quad a_{2,3} = 7$$

Example 2.1.3. The matrix

$$B = \begin{bmatrix} 3 \\ -1 \\ 5 \end{bmatrix}$$

is a 3×1 matrix. Such matrices are often called *column matrices* or *column vectors*.

2.1.2 Equality of Matrices

Definition 2.1.4. Two matrices are *equal* if

1. they have the same dimensions, and

2. all corresponding entries are equal.

In other words, $A = B$ if and only if $a_{i,j} = b_{i,j}$ for all i and j.

Example 2.1.5. Let

$$A = \begin{bmatrix} x & 2 \\ 3 & y \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

If $A = B$, then the corresponding entries must match. Thus, $x = 1$ and $y = 4$.

2.1.3 Matrix Addition

Definition 2.1.6. Two matrices may be added only if they have the same dimension. That is, if $A = [a_{i,j}]$ and $B = [b_{i,j}]$ are both $m \times n$ matrices, then

$$A + B = [a_{i,j} + b_{i,j}]$$

In other words, matrices are added entry-by-entry.

Example 2.1.7. Let

$$A = \begin{bmatrix} 1 & 3 \\ -2 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & -1 \\ 7 & 2 \end{bmatrix}$$

Then

$$A + B = \begin{bmatrix} 1+5 & 3+(-1) \\ -2+7 & 4+2 \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 5 & 6 \end{bmatrix}$$

Example 2.1.8. the matrices

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

cannot be added because they do not have the same dimensions.

Similar to matrix addition, we also have matrix subtraction.

Definition 2.1.9. If A and B have the same dimensions, then

$$A - B = \left[a_{i,j} - b_{i,j}\right]$$

Example 2.1.10. Let

$$A = \begin{bmatrix} 4 & 1 \\ 7 & -2 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & 5 \\ 1 & 6 \end{bmatrix}$$

Then

$$A - B = \begin{bmatrix} 4-3 & 1-5 \\ 7-1 & -2-6 \end{bmatrix} = \begin{bmatrix} 1 & -4 \\ 6 & -8 \end{bmatrix}$$

2.1.4 Scalar Multiplication

Definition 2.1.11. Let $A = [a_{i,j}]$ be a matrix and let c be a real number. The matrix cA is obtained by multiplying every entry of A by c. That is,

$$cA = [ca_{i,j}]$$

Example 2.1.12. Let

$$A = \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix}$$

Compute $3A$.

Solution

Multiply every entry of A by 3 to get

$$3A = \begin{bmatrix} 3 \cdot 2 & 3 \cdot (-1) \\ 3 \cdot 3 & 3 \cdot 4 \end{bmatrix} = \begin{bmatrix} 6 & -3 \\ 9 & 12 \end{bmatrix}$$

Example 2.1.13. Let

$$A = \begin{bmatrix} 1 & -2 & 3 \\ 0 & 4 & -1 \end{bmatrix}$$

Compute $-2A$.

Solution

Multiply every entry of A by -2 to obtain

$$-2A = \begin{bmatrix} -2 \cdot 1 & -2 \cdot (-2) & -2 \cdot 3 \\ -2 \cdot 0 & -2 \cdot 4 & -2 \cdot (-1) \end{bmatrix} = \begin{bmatrix} -2 & 4 & -6 \\ 0 & -8 & 2 \end{bmatrix}$$

2.1.5 Algebraic Properties of Matrix Addition and Scalar Multiplication

Matrix addition and scalar multiplication satisfy many properties similar to ordinary arithmetic.

Theorem 2.1.14. *Suppose A, B, and C are matrices of the same dimension. Then*

1. *Commutative Property:* $A + B = B + A$
2. *Associative Property:* $(A + B) + C = A + (B + C)$
3. *Additive Property: There exists a zero matrix O such that $A + O = A$.*

4. *Additive Inverse: For every matrix A, there exists a matrix $-A$ such that $A + (-A) = O$.*

Definition 2.1.15. A matrix whose entries are all zero is called the *zero matrix*. Some examples include

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Example 2.1.16. Let

$$A = \begin{bmatrix} 2 & 5 \\ -1 & 4 \end{bmatrix}$$

Then

$$A + O = \begin{bmatrix} 2 & 5 \\ -1 & 4 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 2+0 & 5+0 \\ -1+0 & 4+0 \end{bmatrix} = \begin{bmatrix} 2 & 5 \\ -1 & 4 \end{bmatrix} = A$$

Theorem 2.1.17. *Let A and B be matrices of the same dimension, and let $c, d \in \mathbb{R}$. Then*

1. $c(A + B) = cA + cB$
2. $(c + d)A = cA + dA$
3. $(cd)A = c(dA)$
4. $1A = A$.

Example 2.1.18. Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & -1 \\ 2 & 0 \end{bmatrix}$$

Compute $2(A + B)$.

Solution

First compute

$$A + B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 5 & -1 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 6 & 1 \\ 5 & 4 \end{bmatrix}$$

Then

$$2(A + B) = 2\begin{bmatrix} 6 & 1 \\ 5 & 4 \end{bmatrix} = \begin{bmatrix} 12 & 2 \\ 10 & 8 \end{bmatrix}$$

Also,

$$2A + 2B = 2\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + 2\begin{bmatrix} 5 & -1 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 12 & 2 \\ 10 & 8 \end{bmatrix}$$

Therefore, $2(A + B) = 2A + 2B$.

2.1.6 Applications of Matrix Addition and Scalar Multiplication

Matrices are widely used to organize and manipulate data. Examples include

- financial spreadsheets,
- population models,
- image processing,
- transportation data, and
- computer graphics.

Example 2.1.19. Suppose a company tracks quarterly revenues (in thousands of dollars) for two stores.

$$A = \begin{bmatrix} 120 & 135 \\ 140 & 150 \end{bmatrix}, \quad B = \begin{bmatrix} 90 & 100 \\ 110 & 225 \end{bmatrix}$$

Then $A + B$ represents the combined revenues

$$A + B = \begin{bmatrix} 210 & 235 \\ 250 & 275 \end{bmatrix}$$

If prices for store A, for example, increase by 10%, then

$$1.1A = \begin{bmatrix} 132 & 148.5 \\ 154 & 165 \end{bmatrix}$$

represents the adjusted revenue matrix.

2.2 Matrix Multiplication

In the previous section, we introduced matrix addition and scalar multiplication. These operations allowed us to combine matrices entry-by-entry and scale matrices by real numbers. We now study one of the most important operations in all of linear algebra: matrix multiplication.

Matrix multiplication is fundamentally different from ordinary multiplication of numbers. Unlike addition, matrix multiplication:

- is not always defined,
- depends on the dimensions of the matrices,
- generally not commutative.

Despite these differences, matrix multiplication is one of the most powerful tools in mathematics because it allows matrices to:

- represent systems of equations,
- describe geometric transformations,
- encode networks,
- model dynamical systems,
- perform computations efficiently.

2.2.1 When Matrix Multiplication is Defined

Suppose A is an $m \times n$ matrix, and B is an $n \times p$ matrix. Then the product AB is defined.

The general rule is: the number of columns of the first matrix must be equal the number of rows of the second matrix.

$$(m \times n)(n \times p) \to (m \times p)$$

The resulting matrix will have the same number of rows as the first matrix, and the same number of columns as the second matrix.

Example 2.2.1. If A is 2×3 and B is 3×4, then AB is defined and will be a 2×4 matrix.

Example 2.2.2. If A is 2×3 and B is 2×2, then AB is not defined because $3 \neq 2$.

2.2.2 Definition of Matrix Multiplication

Definition 2.2.3. Suppose $A = [a_{i,j}]$ is an $m \times n$ matrix and $B = [b_{i,j}]$ is an $n \times p$ matrix. Then the product AB is the $m \times p$ matrix whose entries are

$$a_{i,1}b_{1,j} + a_{i,2}b_{2,j} + \cdots + a_{i,n}b_{n,j}$$

To compute an entry of AB,

1. Take a row from A,
2. Take a row from B,
3. Multiply corresponding entries
4. Add the products.

Suppose, for example, A is a 2×3 matrix and B is a 3×2 matrix given by

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \end{bmatrix}, \quad B = \begin{bmatrix} b_{1,1} & b_{1,2} \\ b_{2,1} & b_{2,2} \\ b_{3,1} & b_{3,2} \end{bmatrix}$$

Computing the $(1,1)$ entry of AB,

$$AB = \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \end{bmatrix} \begin{bmatrix} b_{1,1} & b_{1,2} \\ b_{2,1} & b_{2,2} \\ b_{3,1} & b_{3,2} \end{bmatrix} = \begin{bmatrix} a_{1,1}b_{1,1} + a_{1,2}b_{2,1} + a_{1,3}b_{3,1} & \\ & \end{bmatrix}$$

Then compute the $(1,2)$ entry of AB

$$AB = \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \end{bmatrix} \begin{bmatrix} b_{1,1} & b_{1,2} \\ b_{2,1} & b_{2,2} \\ b_{3,1} & b_{3,2} \end{bmatrix} = \begin{bmatrix} a_{1,1}b_{1,1} + a_{1,2}b_{2,1} + a_{1,3}b_{3,1} & a_{1,1}b_{1,2} + a_{1,2}b_{2,2} + a_{1,3}b_{3,2} \\ & \end{bmatrix}$$

Then compute the $(2,1)$ entry of AB

$$AB = \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \end{bmatrix} \begin{bmatrix} b_{1,1} & b_{1,2} \\ b_{2,1} & b_{2,2} \\ b_{3,1} & b_{3,2} \end{bmatrix} = \begin{bmatrix} a_{1,1}b_{1,1} + a_{1,2}b_{2,1} + a_{1,3}b_{3,1} & a_{1,1}b_{1,2} + a_{1,2}b_{2,2} + a_{1,3}b_{3,2} \\ a_{2,1}b_{1,1} + a_{2,2}b_{2,1} + a_{2,3}b_{3,1} & \end{bmatrix}$$

Then compute the $(2,2)$ entry of AB

$$AB = \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \end{bmatrix} \begin{bmatrix} b_{1,1} & b_{1,2} \\ b_{2,1} & b_{2,2} \\ b_{3,1} & b_{3,2} \end{bmatrix} = \begin{bmatrix} a_{1,1}b_{1,1} + a_{1,2}b_{2,1} + a_{1,3}b_{3,1} & a_{1,1}b_{1,2} + a_{1,2}b_{2,2} + a_{1,3}b_{3,2} \\ a_{2,1}b_{1,1} + a_{2,2}b_{2,1} + a_{2,3}b_{3,1} & a_{2,1}b_{1,2} + a_{2,2}b_{2,2} + a_{2,3}b_{3,2} \end{bmatrix}$$

Therefore,

$$AB = \begin{bmatrix} a_{1,1}b_{1,1} + a_{1,2}b_{2,1} + a_{1,3}b_{3,1} & a_{1,1}b_{1,2} + a_{1,2}b_{2,2} + a_{1,3}b_{3,2} \\ a_{2,1}b_{1,1} + a_{2,2}b_{2,1} + a_{2,3}b_{3,1} & a_{2,1}b_{1,2} + a_{2,2}b_{2,2} + a_{2,3}b_{3,2} \end{bmatrix}$$

Example 2.2.4. Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

Compute AB.

Solution

Computing the $(1,1)$ entry

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} -1 & \\ & \end{bmatrix}$$

Computing the $(1,2)$ entry

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ & \end{bmatrix}$$

Computing the $(2,1)$ entry

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ -1 & \end{bmatrix}$$

Finally, computing the $(2,2)$ entry

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ -1 & 1 \end{bmatrix}$$

Example 2.2.5. Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 \\ 2 & -1 \\ 3 & 4 \end{bmatrix}$$

Determine the dimensions $(2 \times 3)(3 \times 2)$. Thus, AB is a 2×2 matrix.

Computing the $(1,1)$ entry

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & -1 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 14 & \\ & \end{bmatrix}$$

Compute the $(1,2)$ entry

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & -1 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 14 & 10 \\ & \end{bmatrix}$$

Computing the $(2,1)$ entry

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & -1 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 14 & 10 \\ 32 & \end{bmatrix}$$

Finally, computing the $(2,2)$ entry,

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & -1 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 14 & 10 \\ 32 & 19 \end{bmatrix}$$

Therefore, the matrix AB is given by

$$AB = \begin{bmatrix} 14 & 10 \\ 32 & 19 \end{bmatrix}$$

2.2.3 Matrix Multiplication is Not Commutative

One of the most important facts about matrix multiplication is the following.

Theorem 2.2.6. *If A and B are both square matrices of the same dimensions, then in general,*

$$AB \neq BA$$

Example 2.2.7. Let

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & 0 \\ 1 & 4 \end{bmatrix}$$

Then

$$AB = \begin{bmatrix} 5 & 8 \\ 1 & 4 \end{bmatrix}$$

and

$$BA = \begin{bmatrix} 3 & 6 \\ 1 & 6 \end{bmatrix}$$

Therefore, $AB \neq BA$.

This is one of the major differences between matrix multiplication and ordinary multiplication of numbers.

2.2.4 Algebraic Properties of Matrix Multiplication

Theorem 2.2.8. *Suppose all matrix operations below are defined. Then*

1. *Associative Property:* $A(BC) = (AB)C$
2. *Left Distributive Property:* $A(B + C) = AB + AC$
3. *Right Distributive Property:* $(A + B)C = AC + BC$
4. *Scalar Compatibility:* $c(AB) = (cA)B = A(cB)$

Although multiplication is associative and distributive, it is generally not commutative.

Theorem 2.2.9. *If multiplication is defined, then*

$$AO = O, \quad 0A = 0$$

Example 2.2.10. Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad O = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Then

$$AO = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

2.2.5 Matrix Multiplication and Systems of Equations

One of the main motivations for matrix multiplication comes from systems of equations.

Example 2.2.11. Consider the system

$$\begin{aligned} x + 2y &= 5, \\ 3x - y &= 4 \end{aligned}$$

This can be written as

$$\begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$$

The matrix multiplication on the left side produces

$$\begin{bmatrix} 1x + 2y \\ 3x - y \end{bmatrix} = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$$

Thus, equality of matrices gives the equations

$$\begin{aligned} 1x + 2y &= 5, \\ 3x - 1y &= 4 \end{aligned}$$

2.2.6 Row Vectors and Column Vectors

Definition 2.2.12. A matrix with one row is called a *row vector*, and a matrix with one column is called a *column vector*.

For example,

$$\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$$

is a row vector, while

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

is a column vector.

2.2.7 Dot Product Interpretation

Matrix multiplication is closely related to the dot product.

Example 2.2.13. Compute

$$\begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

We obtain

$$(1)(4) + (2)(5) + (3)(6) = 4 + 10 + 18 = 32$$

Therefore Compute

$$\begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = 32$$

This is called the *dot product* of the vectors.

2.2.8 Applications of Matrix Multiplication

Matrix multiplication often appears in computer graphics, economics, machine learning, cryptography, network analysis, engineering, and scientific computing.

Example 2.2.14. Suppose

$$A = \begin{bmatrix} 10 & 5 \\ 7 & 8 \end{bmatrix}$$

represents quantities sold of two products at two stores, and suppose

$$B = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$$

represents prices. Then the matrix AB computes the total revenue at each store. Computing

$$AB = \begin{bmatrix} 10 & 5 \\ 7 & 8 \end{bmatrix} \begin{bmatrix} 4 \\ 6 \end{bmatrix} = \begin{bmatrix} 70 \\ 76 \end{bmatrix}$$

Therefore, the first store earns \$70, and the second store earns \$76.

2.3 The Transpose

In the previous section, we introduced matrix multiplication and studied its algebraic properties. We now consider another important matrix operation called the transpose.

The transpose operation interchanges the rows and columns of a matrix. Although the operation itself is simple, it plays a fundamental role throughout linear algebra and its applications.

2.3.1 Definition of the Transpose

Definition 2.3.1. Let $A = [a_{i,j}]$ be an $m \times n$ matrix. The *transpose* of A, denoted by A^T is the $n \times m$ matrix obtained by interchanging the rows and columns of A. Equivalently,

$$A^T = [a_{i,j}]^T = [a_{j,i}]$$

The transpose allows the rows of a matrix A to become the columns of A^T, and the columns of a matrix A to become the rows of A^T.

Example 2.3.2. Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

Since A is a 2×3 matrix, its transpose will be a 3×2 matrix. Interchanging rows and columns gives

$$A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

Example 2.3.3. Let

$$B = \begin{bmatrix} 7 & -1 \\ 0 & 5 \end{bmatrix}$$

Then

$$B^T = \begin{bmatrix} 7 & 0 \\ -1 & 5 \end{bmatrix}$$

The transpose converts row vectors into column vectors, and vice versa.

Example 2.3.4. Let

$$\vec{u} = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$$

Then

$$\vec{u}^T = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Example 2.3.5. Let

$$\vec{v} = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

Then

$$\vec{v}^T = \begin{bmatrix} 4 & 5 & 6 \end{bmatrix}$$

2.3.2 Double Transpose

Transposing once swaps rows and columns. Transposing again restores the original arrangement.

Theorem 2.3.6. *For any matrix A,*

$$(A^T)^T = A$$

Example 2.3.7. Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

First compute

$$A^T = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

Now transpose again

$$(A^T)^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}^T = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = A$$

2.3.3 Algebraic Properties of the Transpose

The transpose operation satisfies several important algebraic properties.

Theorem 2.3.8. *Suppose A and B are matrices of the same dimension and $c \in \mathbb{R}$. Then*

1. *Transpose of a sum:* $(A + B)^T = A^T + B^T$
2. *Transpose of a scalar multiple:* $(cA)^T = cA^T$

One of the most important transpose properties involves matrix multiplication.

Theorem 2.3.9. *If A and B are matrices for which AB is defined, then*

$$(AB)^T = B^T A^T$$

Notice that the order reverses. This is extremely important.

Example 2.3.10. Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 \\ 5 & 2 \end{bmatrix}$$

First compute

$$AB = \begin{bmatrix} 10 & 5 \\ 20 & 11 \end{bmatrix}$$

Then

$$(AB)^T = \begin{bmatrix} 10 & 20 \\ 5 & 11 \end{bmatrix}$$

Now compute

$$B^T = \begin{bmatrix} 0 & 5 \\ 1 & 2 \end{bmatrix}, \quad A^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

Multiply to get

$$B^T A^T = \begin{bmatrix} 10 & 5 \\ 20 & 11 \end{bmatrix}$$

Therefore, $(AB)^T = B^T A^T$.

2.3.4 Symmetric Matrices

Transpose operations lead naturally to an important class of matrices.

Definition 2.3.11. A square matrix A is called *symmetric* if

$$A^T = A$$

Example 2.3.12. The matrix

$$A = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}$$

is symmetric because

$$A^T = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}^T = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix} = A$$

Example 2.3.13. The matrix

$$B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

is not symmetric because

$$B^T = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \neq B$$

2.3.5 Skew-Symmetric Matrices

Another type of class involving transpose operations is called the skew-symmetric matrix.

Definition 2.3.14. A square matrix A is called *skew-symmetric* if

$$A^T = -A$$

Example 2.3.15. The matrix

$$A = \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$$

is skew-symmetric because

$$A^T = \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix} = -\begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix} = -A$$

2.3.6 Transpose and Dot Products

Transpose notation provides a compact way to write dot products.

Example 2.3.16. Let

$$\vec{u} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

Then

$$\vec{u}^T\vec{v} = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = (1)(4) + (2)(5) + (3)(6) = 32$$

2.3.7 Applications of the Transpose

The transpose operation appears in many applications.

Example 2.3.17. Suppose rows represent students and columns represent test scores

$$A = \begin{bmatrix} 85 & 90 & 88 \\ 78 & 81 & 84 \end{bmatrix}$$

Then

$$A^T = \begin{bmatrix} 85 & 78 \\ 90 & 81 \\ 88 & 84 \end{bmatrix}$$

switches the perspective: rows now represent tests, and columns represent students.

2.4 The Identity Matrix and Inverse Matrices

In ordinary arithmetic, the number "1" has a special property:

$$1 \cdot a = a \cdot 1 = a$$

for every real number a.

Similarly, every nonzero real number has a reciprocal $a^{-1} = \frac{1}{a}$ which satisfies

$$aa^{-1} = a^{-1}a = 1$$

Matrices possess analogous concepts known as the identity matrix and inverse matrices. These ideas are extremely important because inverse matrices allow us to solve systems of linear equations, reverse matrix transformations, and simplify matrix equations.

2.4.1 The Identity Matrix

Definition 2.4.1. A square matrix whose diagonal entries are all 1 and whose off-diagonal entries are all 0 is called an *identity matrix*. The identity matrix of size $n \times n$ is denoted by

$$I_n = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

Example 2.4.2. The following are identity matrices of sizes 2×2 and 3×3.

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Identity matrices are only defined for square matrices.

2.4.2 Identity Property of Matrix Multiplication

Theorem 2.4.3. *Let A be an $m \times n$ matrix. Then*

$$I_m A = A \quad \textit{and} \quad AI_n = A$$

That is, multiplying the identity matrix leaves a matrix unchanged. Thus, the identity matrix plays the same role as the number "1" in ordinary arithmetic.

Example 2.4.4. Let

$$A = \begin{bmatrix} 2 & 5 \\ 1 & -3 \end{bmatrix}$$

Then

$$I_2 A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 5 \\ 1 & -3 \end{bmatrix} = \begin{bmatrix} 2 & 5 \\ 1 & -3 \end{bmatrix} = A$$

Similarly,

$$A I_2 = \begin{bmatrix} 2 & 5 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 5 \\ 1 & -3 \end{bmatrix} = A$$

2.4.3 Inverse Matrices

Definition 2.4.5. Let A be an $n \times n$ matrix. A matrix A^{-1} is called the *inverse* of A if

$$AA^{-1} = A^{-1}A = I_n$$

Remark 2.4.6.

1. Only square matrices can have inverses.
2. Not every square matrix is invertible.
3. If an inverse exists, it is unique.

2.4.4 Invertible and Singular Matrices

Definition 2.4.7. A square matrix that possesses an inverse is called *invertible* or *nonsingular*. A square matrix that does not possess an inverse is called *singular*.

Example 2.4.8. Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix}, \quad B = \begin{bmatrix} 7 & -2 \\ -3 & 1 \end{bmatrix}$$

Determine whether B is the inverse of A.

Solution

We compute AB and BA to compare. First see that

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix} \begin{bmatrix} 7 & -2 \\ -3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2$$

and

$$BA = \begin{bmatrix} 7 & -2 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix} = I_2$$

Since $AB = BA = I_2$ we conclude that $B = A^{-1}$.

2.4.5 Solving Systems Using Inverses

Consider the matrix equation

$$A\vec{x} = \vec{b}$$

If A is invertible, multiply both sides on the left by A^{-1} to get

$$\begin{aligned} A^{-1}A\vec{x} &= A^{-1}\vec{b} \\ I_n\vec{x} &= A^{-1}\vec{b} \\ \vec{x} &= A^{-1}\vec{b} \end{aligned}$$

Example 2.4.9. Solve

$$\begin{aligned} x + 2y &= 5, \\ 3x + 4y &= 11 \end{aligned}$$

Solution

We first write the system in matrix form given as

$$A\vec{x} = \vec{b}$$

where

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad \vec{b} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

Suppose

$$A^{-1} = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix}$$

Then

$$\vec{x} = A^{-1}\vec{b} = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} 5 \\ 11 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

2.4.6 Properties of Inverses

Theorem 2.4.10. *Suppose A and B are invertible matrices of the same size. Then*

1. $(A^{-1})^{-1} = A$
2. $(AB)^{-1} = B^{-1}A^{-1}$
3. $I_n^{-1} = I_n$
4. $(A^T)^{-1} = (A^{-1})^T$

Remark 2.4.11. Notice again that the order reverses

$$(AB)^{-1} = B^{-1}A^{-1}$$

This is similar to the transpose property

$$(AB)^T = B^T A^T$$

Some matrices are singular.

Example 2.4.12. Consider

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

Notice that the second row is twice the first row. Thus, the rows are linearly dependent, and the matrix does not have an inverse.

A singular matrix collapses dimensions. That is, lines may collapse to points, and planes may collapse to lines. Such transformations cannot be reversed.

2.4.7 The Cancellation Property

Theorem 2.4.13. *If A is invertible and $AB = AC$, then $B = C$.*

Proof. Multiply both sides on the left by A^{-1} to get

$$\begin{aligned} A^{-1}AB &= A^{-1}AC \\ IB &= IC \\ B &= C \end{aligned}$$

□

2.5 Finding the Inverse Matrix

In the previous section, we introduced inverse matrices and saw that inverse matrices allow us to solve systems of linear equations using

$$\vec{x} = A^{-1}\vec{B}$$

However, this raises an important question: How do we actually compute the matrix?

In this section, we develop systematic methods for finding inverse matrices. The main technique uses Gaussian elimination together with augmented matrices.

2.5.1 The Augmented Matrix Method

The primary method for finding inverses uses row reduction.

Theorem 2.5.1. *If a square matrix A is invertible, then*

$$[A \mid I]$$

can be row reduced to

$$\left[I \mid A^{-1}\right]$$

We perform row operations simultaneously on the matrix A and the identity matrix. When the left side becomes the identity matrix, the right side becomes the inverse.

Example 2.5.2. Find the inverse of

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Solution

Form the augmented matrix

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 4 & 0 & 1 \end{array}\right]$$

Eliminate below the first pivot by applying the row operation $R_2 \to R_2 - 3R_1$ to get

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & -2 & -3 & 1 \end{array}\right]$$

Now create a pivot of 1 in row 2 by applying the operation $R_2 \to -\frac{1}{2}R_2$

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \end{array}\right]$$

Next, eliminate above the pivot by applying the row operation $R_1 \to R_1 - 2R_2$ to get

$$\left[\begin{array}{cc|cc} 1 & 0 & -2 & 1 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \end{array}\right]$$

The left side is now the identity matrix, so the inverse of A is given by

$$A^{-1} = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix}$$

Of course, we should always verify our answer.

Example 2.5.3. Verify that the matrix A^{-1} found in Example 2.5.2 is the inverse of A.

Solution

Compute

$$AA^{-1} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2$$

Therefore, A^{-1} is indeed, the inverse of A.

Not every matrix is invertible.

Example 2.5.4. Determine whether

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

is invertible.

Solution

Form the augmented matrix given as

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 2 & 4 & 0 & 1 \end{array}\right]$$

Eliminate below the pivot by applying $R_2 \to R_2 - 2R_1$

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 0 & -2 & 1 \end{array}\right]$$

The left side cannot be transformed into the identity matrix. Therefore, A does not have an inverse. Thus, A is singular.

2.5.2 Inverse Formula for 2×2 Matrices

For 2×2 matrices, there is a direct formula.

Theorem 2.5.5. *Let*

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Then A is invertible whenever $ad - bc \neq 0$, and

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

The quantity $ad - bc$ is called the *determinant* of A, which we will study in detail later.

Example 2.5.6. Find the inverse of

$$A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$$

Solution

Computing the determinant, we have

$$ad - bc = (2)(3) - (1)(5) = 6 - 5 = 1$$

Since the determinant is nonzero, the matrix is invertible. Then applying the formula, we have

$$A^{-1} = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$$

2.5.3 Solving Systems Using Inverses

Example 2.5.7. Solve

$$\begin{aligned} 2x + y &= 7, \\ 5x + 3y &= 19. \end{aligned}$$

Solution

Forming the matrix

$$\begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 7 \\ 19 \end{bmatrix}$$

Then we compute

$$\vec{x} = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}^{-1} \begin{bmatrix} 7 \\ 19 \end{bmatrix}$$

Computing the inverse, we have

$$\begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}^{-1} = \frac{1}{(2)(3) - (1)(5)} \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix} = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$$

Then matrix multiplying the right side

$$\vec{x} = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix} \begin{bmatrix} 7 \\ 19 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

2.5.4 Geometric Interpretation of Inverses

Invertible matrices correspond to reversible transformations. This includes rotations, reflections, and scalings by nonzero factors. Singular matrices collapse dimensions and cannot be reversed.

Example 2.5.8. The matrix

$$\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

scales all vectors by a factor fo 2. Its inverse

$$\begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$$

reverses the scaling.

2.6 Elementary Matrices

In Chapter 1, we introduced elementary row operations and used them extensively in Gaussian elimination to solve systems of linear equations. In Section 2.5, we also used row operations to compute inverse matrices.

In this section, we study these operations from a matrix perspective through the concept of elementary matrices.

2.6.1 Elementary Matrices

Definition 2.6.1. An *elementary matrix* is a matrix obtained by performing a single elementary row operation on an identity matrix.

The important idea is to start with the identity matrix I_n, and apply one row operation. The resulting matrix is elementary.

Example 2.6.2. Construct the elementary matrix that interchanges rows 1 and 2 of a 2×2 matrix.

Solution

Starting with

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Swap rows 1 and 2 using the elementary matrix

$$E = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

This is an elementary matrix.

Example 2.6.3. Construct the elementary matrix that scales the second row of a 2×2 matrix by 3.

Solution

Starting with the identity matrix I_2, multiply row 2 by 3 using the elementary matrix

$$E = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$$

Example 2.6.4. Construct the elementary matrix corresponding to $R_2 \to R_2 + 4R_1$.

Solution

Start with the 2×2 identity matrix I_2, and add 4 times row 1 to row 2.

$$E = \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix}$$

2.6.2 Elementary Matrices Perform Row Operations

One of the most important facts about elementary matrices is the following theorem.

Theorem 2.6.5. *Let E be an elementary matrix obtained from the identity matrix by a row operation. Then multiplying by a matrix A on the left by EA performs the same row operation on A.*

Left multiplication by an elementary matrix corresponds to row operations.

Example 2.6.6. Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Use an elementary matrix to interchange the rows.

Solution

Swap rows of I_2 to get the elementary matrix

$$E = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Then multiplying A by E on the left, we have

$$EA = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$$

The rows of A have been interchanged.

Example 2.6.7. Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Use an elementary matrix to multiply row 2 by 2.

Solution

Multiply row 2 of I_2 by 2 to get the elementary matrix

$$E = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \tag{2.1}$$

Then multiplying A by E on the left, we have

$$EA = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 6 & 8 \end{bmatrix}$$

The second row has been multiplied by 2.

Example 2.6.8. Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Use an elementary matrix to perform $R_2 \to R_2 - 3R_1$.

Solution

Starting with the 2×2 identity matrix I_2, add -3 times row 1 to row 2 to get the elementary matrix

$$E = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}$$

Then multiplying A by E on the left, we have

$$EA = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & -2 \end{bmatrix}$$

Thus, the row operation has been performed.

2.6.3 Inverses of Elementary Matrices

Every elementary matrix is invertible. Moreover, the inverse of an elementary matrix is also elementary. Each row operation can also be reversed. That is,

- Swapping rows can be undone by swapping again.
- Multiplying by 3 can be undone by multiplying by $\frac{1}{3}$.
- Adding $4R_1$ can be undone by subtracting $4R_1$.

Example 2.6.9. Let

$$E = \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix}$$

This corresponds to $R_2 \to R_2 + 4R_1$. To reverse the operation, perform $R_2 \to R_2 - 4R_1$ to get

$$E^{-1} = \begin{bmatrix} 1 & 0 \\ -4 & 1 \end{bmatrix}$$

2.6.4 Products of Elementary Matrices

Repeating row operations correspond to multiply by several elementary matrices.

Theorem 2.6.10. *Suppose a sequence of elementary row operations transforms A into B. Then*

$$B = E_k E_{k-1} \cdots E_1 A$$

where each E_i is an elementary matrix.

Gaussian elimination may therefore by interpreted entirely using matrix multiplication. This is one of the foundational ideas of linear algebra.

Recall that

$$A\vec{x} = \vec{b}$$

Applying row operations corresponds to multiplying both sides by elementary matrices

$$EA\vec{x} = E\vec{x}$$

Thus, row reduction, Gaussian elimination, and elementary matrices are all deeply connected.

2.7 LU Factorization

In previous sections, we studied Gaussian elimination and elementary row operations as systematic methods for solving systems of linear equations. Although Gaussian elimination is extremely effective, repeatedly performing row operations on large systems can become computationally expensive.

One of the most important ideas in applied linear algebra is that a matrix can often be decomposed into simpler matrices whose structure is easier to work with. In this section, we study one of the most useful decompositions in linear algebra: the LU factorization.

2.7.1 Triangular Matrices

Before introducing LU factorization, we first define triangular matrices.

Definition 2.7.1. A square matrix is called *upper triangular* if all entries below the main diagonal are zero. That is, $u_{i,j} = 0$ whenever $i > j$.

Example 2.7.2. The matrix

$$U = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 3 & 5 \\ 0 & 0 & 4 \end{bmatrix}$$

is upper triangular.

Definition 2.7.3. A square matrix is called *lower triangular* if all entries above the main diagonal are zero. That is, $\ell_{i,j} = 0$ whenever $i < j$.

Example 2.7.4. The matrix

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ -1 & 4 & 5 \end{bmatrix}$$

is lower triangular.

Recall the Gaussian elimination process. Suppose

$$A = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$$

To eliminate the entry below the pivot, $4 \to 4 - 2(2)$. This corresponds to the row operation $R_2 \to R_2 - 2R_1$. Gaussian elimination transforms A into an upper triangular matrix. LU factorization records the elimination steps, and the resulting triangular matrix.

Definition 2.7.5. A matrix A has an *LU factorization* if

$$A = LU$$

where L is lower triangular, and U is upper triangular.

In many cases, the diagonal entries of L are chosen to be 1.

Example 2.7.6. Find the LU factorization of

$$A = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$$

Solution

Eliminate the entry below the pivot $R_2 \to 2R_1$. We obtain

$$U = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$$

The multiplier used was 2. Then we construct the lower triangular matrix

$$L = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$$

Therefore, the final factorization is

$$A = LU = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$$

Example 2.7.7. Find the LU factorization of

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 4 & 9 & 7 \end{bmatrix}$$

Solution

We first eliminate below the first pivot by using the row operations $R_2 \to R_2 - 2R_1$ and $R_3 \to R_3 - 4R_1$ to get

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 3 \end{bmatrix}$$

The multipliers are 2 and 4. Next, we eliminate below the second pivot by using the row operation $R_3 \to R_3 - R_2$. Then

$$U = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix}$$

The multiplier used is 1. Place the multipliers into a lower triangular matrix

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 1 & 1 \end{bmatrix}$$

Therefore,

$$A = LU = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix}$$

2.7.2 Solving Systems Using LU Factorization

LU factorization simplifies solving

$$A\vec{x} = \vec{b}$$

Since $A = LU$, we obtain

$$LU\vec{x} = \vec{b}$$

Introduce $U\vec{x} = \vec{y}$, then solve

$$\begin{aligned} L\vec{y} &= \vec{b}, \\ U\vec{x} &= \vec{y} \end{aligned}$$

Definition 2.7.8. Solving a lower triangular system beginning from the top equation downward is called *forward substitution*. Similarly, solving an upper triangular system beginning from the bottom equation upward is called *back substitution*.

Example 2.7.9. Solve $A\vec{x} = \vec{b}$, where

$$A = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}, \quad \vec{b} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

Solution

We already know $A = LU$ with

$$L = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}, \quad U = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$$

First solve the system $L\vec{y} = \vec{b}$, or

$$\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

This gives the system

$$\begin{aligned} y_1 &= 5, \\ 2y_1 + y_2 &= 11. \end{aligned}$$

Substituting y_1 into the second equation, we have

$$\begin{aligned} 2(5) + y_2 &= 11 \\ y_2 &= 1 \end{aligned}$$

Therefore, $\vec{y} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$.

Next, using $\vec{y}$, we solve $U\vec{x} = \vec{y}$, or

$$\begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$$

This gives the system

$$\begin{aligned} 2x_1 + x_2 &= 5, \\ x_2 &= 1 \end{aligned}$$

Substituting x_2 into the first equation, we have

$$\begin{aligned} 2x_1 + 1 &= 4 \\ x_1 &= 2 \end{aligned}$$

Therefore, the solution to the system

$$\begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

is $\vec{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$.

LU factorization is important because it avoids repeating Gaussian elimination, it improves computational efficiency, and it is widely used in numerical linear algebra.

Chapter 3

Linear Transformations

In the previous chapter, we developed the algebra of matrices and studied matrix operations, inverse matrices, elementary matrices, and matrix factorization. While matrices are powerful and computational tools, their true significance lies in what they represent.

One of the central ideas of linear algebra is that matrices describe functions between vector spaces. These special functions are called *linear transformations.*

Linear transformations arise whenever quantities are transformed while preserving the fundamental operations of addition and scalar multiplication. Many examples include:

- rotations of geometric figures,
- reflections and projections,
- stretching and compressing objects,
- coordinate transformations,
- image processing,
- computer graphics,
- robotics,
- machine learning,
- and many physical systems.

Linear transformations provide a mathematical framework for understanding how vectors and geometric objects change their various operations. They allow us to move beyond solving systems of equations and begin studying the structure of vector spaces themselves.

A key theme of this chapter is the deep connection between matrices and linear transformations. Every matrix determines a linear transformation, and every linear transformation can be represented by a matrix once appropriate coordinates are chosen. This correspondence allows us to use matrix techniques to analyze geometric and algebraic problems.

3.1 Linear Transformations

In the previous chapter, we studied matrices and their algebraic properties. We learned how matrices can be added, multiplied, inverted, and used to solve systems of linear equations. We now shift our focus from matrices themselves to what matrices represent.

One of the central ideas of linear algebra is the concept of a linear transformation. A linear transformation is a function that maps vectors to vectors while preserving the operations of vector addition and scalar multiplication.

The importance of linear transformations lies in the fact that they preserve linear structure. Because of this property, they can be analyzed using the tools of linear algebra.

3.1.1 Functions Between Euclidean Spaces

Recall that a function assigns exactly one output to each input.

Definition 3.1.1. A *transformation* (or function) from $\mathbb{R}^n$ to $\mathbb{R}^m$ is a rule that assigns each vector $\vec{x} \in \mathbb{R}^n$ to a unique vector $T(\vec{x}) \in \mathbb{R}^m$. We write

$$T : \mathbb{R}^n \to \mathbb{R}^m$$

Example 3.1.2. Define $T : \mathbb{R}^2 \to \mathbb{R}^2$ by

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 2x \\ 3y \end{bmatrix}$$

Then

$$T\left(\begin{bmatrix} 1 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ 6 \end{bmatrix}, \quad T\left(\begin{bmatrix} -1 \\ 4 \end{bmatrix}\right) = \begin{bmatrix} -2 \\ 12 \end{bmatrix}$$

Example 3.1.3. Define $T : \mathbb{R}^2 \to \mathbb{R}^3$ by

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x + y \\ x - y \\ 2y \end{bmatrix}$$

Then

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$$

3.1.2 Definition of a Linear Transformation

Not every transformation is linear. The defining feature of a linear transformation is that it preserves vector addition and scalar multiplication.

Definition 3.1.4. A transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ is called a *linear transformation* if, for all vectors $\vec{u}, \vec{v} \in \mathbb{R}$ and all scalars $c \in \mathbb{R}$, the following two properties hold:

1. Additivity: $T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v})$.
2. Scalar Multiplication: $T(c\vec{u}) = cT(\vec{u})$.

To prove that a transformation is linear, both properties must be verified.

Example 3.1.5. Consider $T : \mathbb{R}^2 \to \mathbb{R}^2$ by

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 2x \\ 3y \end{bmatrix}$$

Determine whether T is linear.

Solution

We first verify additivity. Let $\vec{u} = (x_1, y_1)$ and $\vec{v} = (x_2, y_2)$. Then

$$\vec{u} + \vec{v} = (x_1 + x_2, y_1 + y_2)$$

Applying T, we have

$$T(\vec{u} + \vec{v}) = \begin{bmatrix} 2(x_1 + x_2) \\ 3(y_1 + y_2) \end{bmatrix} = \begin{bmatrix} 2x_1 + 2x_2 \\ 3y_1 + 3y_2 \end{bmatrix} = \begin{bmatrix} 2x_1 \\ 3y_1 \end{bmatrix} + \begin{bmatrix} 2x_2 \\ 3y_2 \end{bmatrix} = T(\vec{u}) + T(\vec{v})$$

Next, let $\vec{u} = (x, y)$ and $c \in \mathbb{R}$. Then

$$c\vec{u} = (cx, cy)$$

Applying T, we have

$$T(c\vec{u}) = \begin{bmatrix} 2(cx) \\ 3(cy) \end{bmatrix} = \begin{bmatrix} c(2x) \\ c(3y) \end{bmatrix} = c\begin{bmatrix} 2x \\ 3y \end{bmatrix} = cT(\vec{u})$$

Therefore, since both properties hold, T is a linear transformation.

Example 3.1.6. Consider $T : \mathbb{R}^2 \to \mathbb{R}^2$ defined by

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x+1 \\ y \end{bmatrix}$$

Determine whether T is linear.

Solution

A useful shortcut is to examine

$$T\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Since $(1,0) \neq (0,0)$, T cannot be linear.

Every linear transformation must map the zero vector to the zero vector.

Theorem 3.1.7. *If $T : \mathbb{R}^n \to \mathbb{R}^m$ is linear, then*

$$T(\vec{0}_{\mathbb{R}^n}) = \vec{0}_{\mathbb{R}^m}$$

where $\vec{0}_{\mathbb{R}^n}$ and $\vec{0}_{\mathbb{R}^m}$ are the zero vectors of $\mathbb{R}^n$ and $\mathbb{R}^m$, respectively.

Proof. Using scalar multiplication, for any $\vec{u} \in \mathbb{R}^n$, since T is linear,

$$T(\vec{0}_{\mathbb{R}^n}) = T(0\vec{u}) = 0T(\vec{u}) = \vec{0}_{\mathbb{R}^m}$$

□

3.1.3 Linear Transformations Preserve Linear Combinations

One of the most important properties of linear transformations is that they preserve linear combinations.

Theorem 3.1.8. *Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformations. Then for any vectors $\vec{v}_1, \vec{v}_2, ..., \vec{v}_k \in \mathbb{R}^n$ and scalars $c_1, c_2, ..., c_k \in \mathbb{R}$, we have*

$$T(c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_k\vec{v}_k) = c_1T(\vec{v}_1) + c_2T(\vec{v}_2) + \cdots + c_kT(\vec{v}_k)$$

That is, linear transformations distribute across addition, scalar multiplication, and therefore, all linear combinations.

3.1.4 Matrix Transformations

Many important linear transformations arise from matrices.

Definition 3.1.9. Let A be an $m \times n$ matrix. Define $T_A : \mathbb{R}^n \to \mathbb{R}^m$ by

$$T_A(\vec{x}) = A\vec{x}$$

Such a transformation is called a *matrix transformation*.

Example 3.1.10. Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$$

Let $T_A : \mathbb{R}^2 \to \mathbb{R}^2$ be the matrix transformation on A. Then if $\vec{x} = \begin{bmatrix} x \\ y \end{bmatrix}$, then

$$T_A\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x + 2y \\ 3x - y \end{bmatrix}$$

Theorem 3.1.11. *If $T_A(\vec{x}) = A\vec{x}$, then T_A is linear.*

Proof. Let $\vec{x}, \vec{y} \in \mathbb{R}^n$. Then

$$T_A(\vec{x} + \vec{y}) = A(\vec{x} + \vec{y})$$

Using matrix properties,

$$A(\vec{x} + \vec{y}) = A\vec{x} + A\vec{y} = T_A(\vec{x}) + T_A(\vec{y})$$

Therefore,

$$T_A(\vec{x} + \vec{y}) = T_A(\vec{x}) + T_A(\vec{y})$$

Similarly, let $c \in \mathbb{R}$. Then

$$T_A(c\vec{x}) = A(c\vec{x}) = cA\vec{x} = cT_A(\vec{x})$$

Therefore, T is linear. $\square$

3.2 The Matrix of a Linear Transformation

In the previous section, we introduced linear transformations and studied their defining properties. We saw that every matrix defines a linear transformation through the rule

$$T_A(\vec{x}) = A\vec{x}$$

This naturally leads to an important question: Does every linear transformation come from a matrix?

The answer is yes. One of the most important rules in linear algebra is that every linear transformation between Euclidean spaces can be represented by a matrix. This establishes a deep connection between functions, vectors, and matrices.

As a consequence, studying matrices is equivalent to studying linear transformations, and matrix algebra becomes a powerful tool for analyzing transformations.

3.2.1 Standard Basis Vectors

To understand how a linear transformation acts, we first introduce the standard basis vectors.

Definition 3.2.1. The *standard basis vectors* of $\mathbb{R}^n$ are

$$\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad \vec{e}_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \quad \cdots, \quad \vec{e}_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

Example 3.2.2. The standard basis for $\mathbb{R}^2$ is

$$\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \vec{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

and the standard basis for $\mathbb{R}^3$ is

$$\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \vec{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \vec{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Every vector in $\mathbb{R}^n$ can be written as a linear combination of the standard basis vectors.

Example 3.2.3. Let

$$\vec{v} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

Then

$$\vec{v} = 3\vec{e}_1 + 5\vec{e}_2 = 3 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 5 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Example 3.2.4. Let

$$\vec{v} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Then

$$\vec{v} = 1\vec{e}_1 + 2\vec{e}_2 + 3\vec{e}_3$$

A remarkable fact follows from linearity.

Theorem 3.2.5. *A linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ is completely determined by its values on the standard basis vectors $\vec{e}_1, \vec{e}_2, ..., \vec{e}_n$.*

Proof. Let $x_1\vec{e}_1 + x_2\vec{e}_2 + \cdots + x_n\vec{e}_n$. Since T is linear,

$$T(x_1\vec{e}_1 + x_2\vec{e}_2 + \cdots + x_n\vec{e}_n) = x_1T(\vec{e}_1) + x_2T(\vec{e}_2) + \cdots + x_nT(\vec{e}_n)$$

Thus, knowing $T(\vec{e}_1), T(\vec{e}_2), ..., T(\vec{e}_n)$ allows us to compute $T(\vec{x})$ for any vector. □

3.2.2 The Matrix of a Linear Transformation

We now arrive at the central result of this section.

Theorem 3.2.6. *Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation. Then there exists a unique $m \times n$ matrix A such that $T(\vec{x}) = A\vec{x}$ for all $\vec{x} \in \mathbb{R}^n$.*

Definition 3.2.7. The matrix A satisfying $T(\vec{x}) = A\vec{s}$ is called the *matrix of the linear transformation T*.

The columns of the matrix are obtained from the images of the standard basis vectors.

Theorem 3.2.8. *Let $T : \mathbb{R}^n \to \mathbb{R}^m$. Then the matrix of T is*

$$A = \begin{bmatrix} \uparrow & \uparrow & & \uparrow \\ T(\vec{e}_1) & T(\vec{e}_2) & \cdots & T(\vec{e}_n) \\ \downarrow & \downarrow & & \downarrow \end{bmatrix}$$

This is one of the most important formulas in this chapter.

Example 3.2.9. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x + 2y \\ 3x - y \end{bmatrix}$$

Find the matrix of T.

Solution

First we compute $T(\vec{e}_1)$ and $T(\vec{e}_2)$

$$T\left(\begin{bmatrix}1\\0\end{bmatrix}\right) = \begin{bmatrix}1\\3\end{bmatrix}, \quad T\left(\begin{bmatrix}0\\1\end{bmatrix}\right) = \begin{bmatrix}2\\-1\end{bmatrix}$$

Then using the columns of $T(\vec{e}_1)$ and $T(\vec{e}_2)$,

$$A = \begin{bmatrix}1 & 2\\3 & -1\end{bmatrix}$$

Example 3.2.10. Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be given by

$$T\left(\begin{bmatrix}x\\y\\z\end{bmatrix}\right) = \begin{bmatrix}x+y\\y-z\end{bmatrix}$$

Find the matrix of T.

Solution

We first compute $T(\vec{e}_1)$, $T(\vec{e}_2)$, and $T(\vec{e}_3)$.

$$T(\vec{e}_1) = \begin{bmatrix}1\\0\end{bmatrix}, \quad T(\vec{e}_2) = \begin{bmatrix}1\\1\end{bmatrix}, \quad T(\vec{e}_3) = \begin{bmatrix}0\\-1\end{bmatrix}$$

Then using the columns of $T(\vec{e}_1)$, $T(\vec{e}_2)$, and $T(\vec{e}_3)$,

$$A = \begin{bmatrix}1 & 1 & 0\\0 & 1 & -1\end{bmatrix}$$

3.2.3 Finding the Transformation from a Matrix

The process also works in reverse.

Example 3.2.11. Let

$$A = \begin{bmatrix}2 & 1\\-1 & 3\end{bmatrix}$$

Find the associated transformation.

Solution

Let $\vec{v} = \begin{bmatrix}x\\y\end{bmatrix}$. Then compute

$$A\vec{v} = \begin{bmatrix}2 & 1\\-1 & 3\end{bmatrix}\begin{bmatrix}x\\y\end{bmatrix} = \begin{bmatrix}2x+y\\-x+3y\end{bmatrix}$$

3.2.4 Matrix Multiplication and Composition

Suppose $A\vec{x}$ and $B\vec{x}$ represent two linear transformations S and T, respectively. Then applying S first, and then T, using

$$(T \circ S)(\vec{x}) = T(B\vec{x}) = A(B\vec{x}) = (AB)\vec{x}$$

Matrix multiplication corresponds to corresponds to performing one transformation after another. This explains why matrix multiplication is defined the way it is.

3.3 Properties of Linear Transformations

In the previous sections, we introduced linear transformations and learned how every linear transformation can be represented by a matrix. We also saw that linear transformations preserve addition and scalar multiplication, making them one of the central objects of study in linear algebra.

In this section, we investigate several important properties of linear transformations and learn how these properties can be used to simplify computations and solve problems.

One of the most useful features of linearity is that a transformation is completely determined by its action on a basis. As a result, we can often compute the image of complicated vectors without knowing an explicit formula for the transformation.

3.3.1 Computing Images Using Linearity

Often, we know the values of a transformation on a few vectors and wish to determine the image of another vector.

Example 3.3.1. Suppose $T : \mathbb{R}^2 \to \mathbb{R}^2$ is linear and satisfies

$$T\left(\begin{bmatrix}1\\0\end{bmatrix}\right) = \begin{bmatrix}2\\3\end{bmatrix}, \quad T\left(\begin{bmatrix}0\\1\end{bmatrix}\right) = \begin{bmatrix}-1\\4\end{bmatrix}$$

Find $T\left(\begin{bmatrix}3\\2\end{bmatrix}\right)$.

Solution

Write the vector $\begin{bmatrix}3\\2\end{bmatrix}$ as a linear combination of $\vec{e}_1$ and $\vec{e}_2$

$$\begin{bmatrix}3\\2\end{bmatrix} = 3\begin{bmatrix}1\\0\end{bmatrix} + 2\begin{bmatrix}0\\1\end{bmatrix}$$

Then apply T and use linearity to get

$$\begin{aligned} T\left(\begin{bmatrix}3\\2\end{bmatrix}\right) &= T\left(3\begin{bmatrix}1\\0\end{bmatrix}+2\begin{bmatrix}0\\1\end{bmatrix}\right) \\ &= T\left(3\begin{bmatrix}1\\0\end{bmatrix}\right)+T\left(2\begin{bmatrix}0\\1\end{bmatrix}\right) \\ &= 3T\left(\begin{bmatrix}1\\0\end{bmatrix}\right)+2T\left(\begin{bmatrix}0\\1\end{bmatrix}\right) \\ &= 3\begin{bmatrix}2\\3\end{bmatrix}+2\begin{bmatrix}-1\\4\end{bmatrix} \\ &= \begin{bmatrix}4\\17\end{bmatrix} \end{aligned}$$

However, there could be times that the information we know does not come from the standard basis vectors like in the previous example. The next example shows this.

Example 3.3.2. Suppose $T : \mathbb{R}^2 \to \mathbb{R}^3$ is a linear transformation and satisfies

$$T\left(\begin{bmatrix}1\\2\end{bmatrix}\right)=\begin{bmatrix}1\\0\\2\end{bmatrix}, \quad T\left(\begin{bmatrix}2\\1\end{bmatrix}\right)=\begin{bmatrix}3\\1\\5\end{bmatrix}$$

Find $T\left(\begin{bmatrix}4\\5\end{bmatrix}\right)$.

Solution

We first express the vector $\begin{bmatrix}4\\5\end{bmatrix}$ as a linear combination of the given vectors. Let a, b be such that

$$\begin{bmatrix}4\\5\end{bmatrix}=a\begin{bmatrix}1\\2\end{bmatrix}+b\begin{bmatrix}2\\1\end{bmatrix}$$

This gives a system

$$\begin{aligned} a+2b&=4, \\ 2a+b&=5 \end{aligned}$$

Solving the system gives $a = 2$ and $b = 1$. Therefore

$$\begin{bmatrix}4\\5\end{bmatrix}=2\begin{bmatrix}1\\2\end{bmatrix}+\begin{bmatrix}2\\1\end{bmatrix}$$

Now apply T and using the fact that T is linear,

$$\begin{aligned} T\left(\begin{bmatrix}4\\5\end{bmatrix}\right) &= 2T\left(\begin{bmatrix}1\\2\end{bmatrix}\right) + T\left(\begin{bmatrix}2\\1\end{bmatrix}\right) \\ &= 2\begin{bmatrix}1\\0\\5\end{bmatrix} + \begin{bmatrix}3\\1\\5\end{bmatrix} \\ &= \begin{bmatrix}5\\1\\9\end{bmatrix} \end{aligned}$$

3.3.2 The Identity Transformation

Definition 3.3.3. The *identity transformation* on $\mathbb{R}^n$ is defined by

$$I(\vec{x}) = \vec{x}$$

for every $\vec{x} \in \mathbb{R}^n$.

Example 3.3.4. On $\mathbb{R}^2$,

$$I(x, y) = (x, y)$$

The matrix of the identity transformation is simply the identity matrix I_n. For example,

$$I_2 = \begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}$$

3.3.3 Composition of Linear Transformations

We briefly saw in the previous section how compositions of linear transformations is related to matrix multiplication. Often one transformation is applied after another.

Definition 3.3.5. Let $S : \mathbb{R}^n \to \mathbb{R}^m$ and $T : \mathbb{R}^m \to \mathbb{R}^p$ be linear transformations. The *composition* of T and S is the linear transformation $T \circ S : \mathbb{R}^m \to \mathbb{R}^p$

$$(T \circ S)(\vec{x}) = T(S(\vec{x}))$$

The idea is to first apply S, and then apply T.

Example 3.3.6. Define $S : \mathbb{R}^2 \to \mathbb{R}^2$ and $T : \mathbb{R}^2 \to \mathbb{R}^2$ by

$$S\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x+y \\ y \end{bmatrix}, \quad T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 2x \\ y \end{bmatrix}$$

Find $(T \circ S)\left(\begin{bmatrix} x \\ y \end{bmatrix}\right)$.

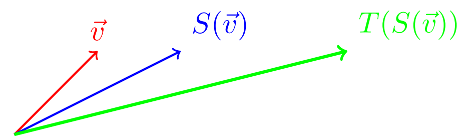
Solution

Denoting $\vec{v} = \begin{bmatrix} x \\ y \end{bmatrix}$. First apply S to get

$$S(\vec{v}) = \begin{bmatrix} x+y \\ y \end{bmatrix}$$

Then apply T to get

$$T(S(\vec{v})) = T\left(\begin{bmatrix} x+y \\ y \end{bmatrix}\right) = \begin{bmatrix} 2(x+y) \\ y \end{bmatrix} = \begin{bmatrix} 2x+2y \\ y \end{bmatrix}$$



3.3.4 Inverse Transformations

Sometimes transformations can be reversed.

Definition 3.3.7. A linear transformation $T : \mathbb{R}^n \to \mathbb{R}^n$ is *invertible* if there exists a linear transformation T^{-1} such that

$$T^{-1}(T(\vec{x})) = T(T^{-1}(\vec{x})) = \vec{x}$$

for every vector $\vec{x}$.

Example 3.3.8. Consider

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 2x \\ 2y \end{bmatrix}$$

This doubles every coordinate. Define

$$T^{-1}\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} \frac{x}{2} \\ \frac{y}{2} \end{bmatrix}$$

Then

$$T^{-1}\left(T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right)\right) = T^{-1}\left(\begin{bmatrix} 2x \\ 2y \end{bmatrix}\right) = \begin{bmatrix} \frac{2x}{2} \\ \frac{2y}{2} \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$$

Therefore, T^{-1} is the inverse transformation.

3.3.5 Inverse Matrices and Inverse Transformations

Suppose $T(\vec{x}) = A\vec{x}$. If A is invertible, then $T^{-1}(\vec{x}) = A^{-1}\vec{x}$. Consequently, T is also invertible.

Example 3.3.9. Let

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

Then

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 2x \\ 3y \end{bmatrix}$$

Since

$$A^{-1} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{3} \end{bmatrix}$$

the inverse transformation is

$$T^{-1}\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} \frac{x}{2} \\ \frac{y}{3} \end{bmatrix}$$

3.4 Special Linear Transformations in $\mathbb{R}^2$

In the previous sections, we introduced linear transformations, studied their matrix representations, and examined their algebraic properties. We now focus on some of the most important linear transformations in the plane $\mathbb{R}^2$.

One of the advantages of working in $\mathbb{R}^2$ is that transformations can be visualized geometrically. Instead of viewing a matrix simply as a collection of numbers, we can think of it as a rule that moves, rotates, reflects, stretches, or compresses points in the plane.

3.4.1 Scaling Transformations

Matrices describe geometric actions.

Definition 3.4.1. A *scaling transformation* multiplies coordinates by fixed constants. The scaling can be *uniform* where both coordinates are multiplied by the same factor, or *nonuniform* where the coordinates are multiplied by different factors.

Consider the linear transformation

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} kx \\ ky \end{bmatrix}$$

for some scaling factor k. Its matrix is

$$A = \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix}$$

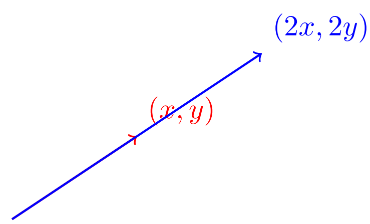
Example 3.4.2. Consider

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

Then

$$T(\vec{v}) = \begin{bmatrix} 2x \\ 2y \end{bmatrix}$$

This doubles all distances from the origin.



Different scaling factors may also be used.

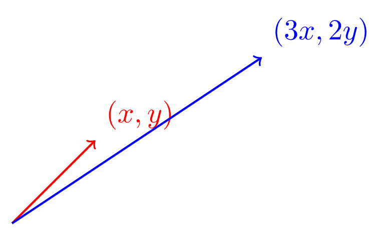
Example 3.4.3. Consider

$$A = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$$

Then

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} ax \\ by \end{bmatrix}$$

For example, consider when $a = 3$ and $b = 2$.



This stretches horizontally by a factor of 3, and vertically by a factor of 2.

3.4.2 Reflection Transformations

Definition 3.4.4. Reflection across the x-axis is described by the transformation

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x \\ -y \end{bmatrix}$$

Its matrix representation is

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

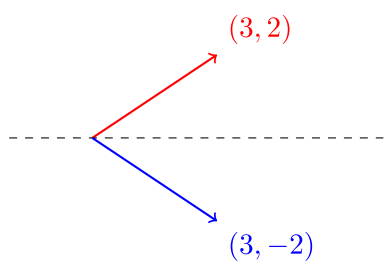
Example 3.4.5. Consider

$$A = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$$

Then

$$T(\vec{v}) = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$$

describes the reflection across the x-axis.



Indeed, observe that

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$$

Definition 3.4.6. Reflection across the y-axis is described by the transformation

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} -x \\ y \end{bmatrix}$$

Its matrix representation is

$$A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

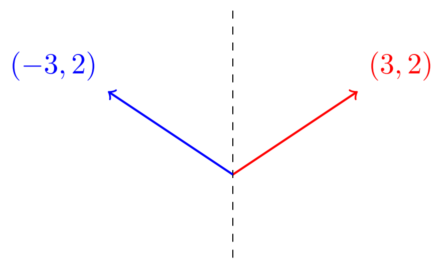
Example 3.4.7. Consider

$$A = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$$

Then

$$T(\vec{v}) = \begin{bmatrix} -3 \\ 2 \end{bmatrix}$$

describes the reflection across the y-axis.



Indeed, observe that

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} -3 \\ 2 \end{bmatrix}$$

Reflections can also be along the line $y = x$ as well.

Definition 3.4.8. Reflection along the line $y = x$ is described by the transformation

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} y \\ x \end{bmatrix}$$

Its matrix representation is

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

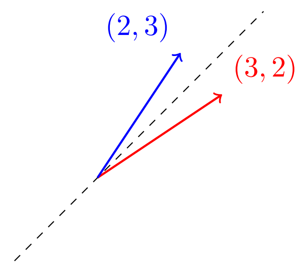
Example 3.4.9. Consider

$$A = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$$

Then

$$T(\vec{v}) = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

describes the reflection along the line $y = x$.



Indeed, observe that

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

3.4.3 Projection Transformations

A projection removes part of a vector.

Definition 3.4.10. Projection onto the x-axis is described by the linear transformation

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x \\ 0 \end{bmatrix}$$

Its corresponding matrix representation is

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

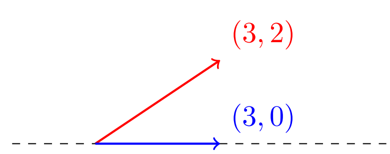
Example 3.4.11. Consider

$$A = \begin{bmatrix} 3 & 0 \\ 0 & 0 \end{bmatrix}$$

Then

$$T\left(\begin{bmatrix} 3 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$$

describes the projection onto the x-axis.



Definition 3.4.12. Projection onto the y-axis is described by the linear transformation

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 0 \\ y \end{bmatrix}$$

Its corresponding matrix representation is

$$A = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

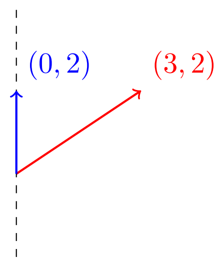
Example 3.4.13. Consider

$$A = \begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix}$$

Then

$$T\left(\begin{bmatrix} 3 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

describes the projection onto the y-axis.



3.4.4 Rotation Transformations

Rotations are among the most important geometric transformations.

Definition 3.4.14. A counterclockwise rotation through an angle θ is represented by

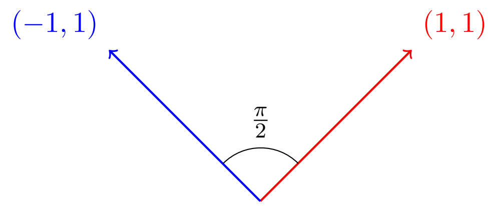
$$R_\theta = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

Example 3.4.15. Since $\cos\left(\frac{\pi}{2}\right) = 0$ and $\sin\left(\frac{\pi}{2}\right) = 1$, the matrix becomes

$$R_{\frac{\pi}{2}} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

Suppose we apply $R_{\frac{\pi}{2}}$ to the vector $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Then

$$R_{\frac{\pi}{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

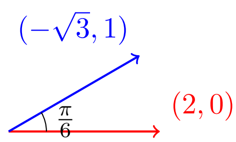


Example 3.4.16. Since $\cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$ and $\sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$, the matrix becomes

$$R_{\frac{\pi}{6}} = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$$

Suppose we apply $R_{\frac{\pi}{6}}$ to the vector $\begin{bmatrix} 2 \\ 0 \end{bmatrix}$. Then

$$R_{\frac{\pi}{6}} \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \begin{bmatrix} \sqrt{3} & -1 \\ 1 & \sqrt{3} \end{bmatrix}$$



3.4.5 Sheer Transformations

A shear transformation shifts points in one direction while leaving another unchanged.

Definition 3.4.17. The *horizontal sheer* is the linear transformation described by

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x + ky \\ y \end{bmatrix}$$

The corresponding matrix representation is given by

$$A = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$$

The *vertical sheer* is the linear transformation described by

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x \\ y + kx \end{bmatrix}$$

The corresponding matrix representation is given by

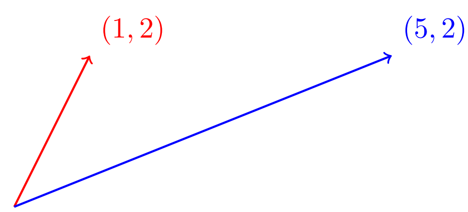
$$A = \begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}$$

Example 3.4.18. Let $k = 2$. Then

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

Applying to the vector $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ gives

$$T\left(\begin{bmatrix} 1 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$

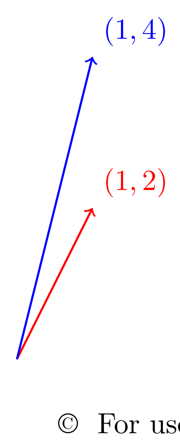


Similarly, if

$$A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$$

Applying to the vector $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ gives

$$T\left(\begin{bmatrix} 1 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$$



3.4.6 Composition of Transformations

Transformations can also be combined.

Example 3.4.19. Suppose a vector $\begin{bmatrix} x \\ y \end{bmatrix}$ is reflected across the x-axis, and then stretched by a factor of 2. The reflection matrix over the x-axis is

$$R = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

and the scaling matrix is

$$S = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

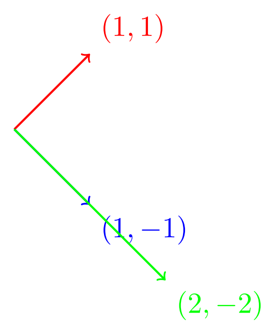
The composite transformation is SR, where

$$RS = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}$$

Thus, the linear transformation is then described by

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 2x \\ -2y \end{bmatrix}$$

For example, suppose we apply to the vector $(1, 1)$. Reflection over the x-axis gives $(1, -1)$, and then scale by a factor of 2 gives $(2, -2)$.



Chapter 4

Determinants

In previous chapters, we studied systems of linear equations, matrices, and linear transformations. We learned that matrices provide a powerful way to represent and analyze linear systems, and we saw that invertible matrices play a fundamental role in solving equations and describing reversible transformations.

A natural question now arises: How can we determine whether a matrix is invertible? One answer is provided by one of the most important numerical quantities associated with a square matrix: the determinant.

The determinant is a single real number assigned to every square matrix. Although it may initially appear to be just another computational tool, the determinant contains a remarkable amount of information about a matrix and the transformation it represents.

The determinant can be used to:

- determine whether a matrix is invertible,
- detect whether a system of equations has a unique solution,
- compute inverse matrices,
- measure changes in area and volume,
- analyze geometric transformations.

From a geometric perspective, determinants describe how a linear transformation changes size and orientation. For example, if a transformation doubles all areas in the plane, its determinant has magnitude 2. If a transformation reverses orientation, such as a reflection, its determinant is negative.

4.1 Computing Determinants

In Chapter 3, we learned that matrices can be used to represent linear transformations. Some transformations can be reversed, while some cannot. Similarly, some matrices are invertible and others are not.

To determine whether a matrix is invertible, we introduce one of the most important concepts in linear algebra: the determinant.

A determinant is a single number associated with a square matrix. Although it may initially appear to be merely a computational tool, the determinant contains important information about a matrix and the transformation it represents.

4.1.1 Determinants of 2×2 Matrices

We begin with 2×2 and gradually extend the definition to larger matrices using minors and cofactors.

Definition 4.1.1. Let

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

The *determinant* of A, denoted by $\det(A)$, or $|A|$, is defined by

$$\det(A) = ad - bc$$

Example 4.1.2. Find the determinant

$$\det \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$$

Solution

Using the definition,

$$\det \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix} = (2)(4) - (3)(1) = 8 - 3 = 5$$

Example 4.1.3. Find the determinant

$$\det \begin{bmatrix} 5 & 2 \\ 10 & 4 \end{bmatrix}$$

Solution

Using the definition,

$$\det \begin{bmatrix} 5 & 2 \\ 10 & 4 \end{bmatrix} = (5)(4) - (2)(10) = 20 - 20 = 0$$

A determinant equal to zero often indicates that the matrix is not invertible. This idea will be explored later in the chapter.

4.1.2 Minors

Computing determinants of larger matrices requires a new method. Consider, for example,

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{bmatrix}$$

The determinant can be computed by expanding along a row or column. Before introducing the general method, we define minors and cofactors.

Definition 4.1.4. Let A be an $n \times n$ matrix. The *minor* of entry $a_{i,j}$, denoted by $M_{i,j}$, is the determinant of the matrix obtained by deleting row i and column j.

Example 4.1.5. Consider

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Find $M_{1,1}$.

Solution

Delete row 1 and column 1

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

to get the matrix

$$\begin{bmatrix} 5 & 6 \\ 8 & 9 \end{bmatrix}$$

Then

$$M_{1,1} = \det \begin{bmatrix} 5 & 6 \\ 8 & 9 \end{bmatrix} = (5)(9) - (6)(8) = 45 - 48 = -3$$

Example 4.1.6. Consider the same matrix above. Find $M_{2,2}$.

Solution

Delete row 2 and column 2

$$A = \begin{bmatrix} 1 & \color{red}{2} & 3 \\ \color{red}{4} & \color{red}{5} & \color{red}{6} \\ 7 & \color{red}{8} & 9 \end{bmatrix}$$

to get the matrix

$$\begin{bmatrix} 1 & 3 \\ 7 & 9 \end{bmatrix}$$

Then

$$M_{2,2} = \det \begin{bmatrix} 1 & 3 \\ 7 & 9 \end{bmatrix} = (1)(9) - (3)(7) = 9 - 21 = -12$$

4.1.3 Cofactors and Cofactor Expansion

The signs of determinant expansions alternate according to a pattern.

Definition 4.1.7. The *cofactor* of $a_{i,j}$ is

$$c_{i,j} = (-1)^{i+j} M_{i,j}$$

The signs follow the checkerboard pattern:

$$\begin{bmatrix} + & - & + & - & + & - & \cdots \\ - & + & - & + & - & + & \cdots \\ + & - & + & - & + & - & \cdots \\ - & + & - & + & - & + & \cdots \\ + & - & + & - & + & - & \cdots \\ - & + & - & + & - & + & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \end{bmatrix}$$

Example 4.1.8. From Example 4.1.5, we have $M_{1,1} = -3$. Since $(-1)^{1+1} = 1$, we obtain

$$c_{1,1} = (1)(-3) = -3$$

Example 4.1.9. From Example 4.1.6, we have $M_{2,2} = -12$. Since $(-1)^{2+2} = 1$, we obtain

$$c_{2,2} = (1)(-12) = -12$$

We now define determinants of larger matrices.

Theorem 4.1.10. *Let A be an $n \times n$ matrix. The determinant of A may be computed using*

$$\det(A) = a_{i,1}c_{i,1} + a_{i,2}c_{i,2} + \cdots + a_{i,n}c_{i,n}$$

for any row i. Similarly, the determinant of A may also be computed using

$$\det(A) = a_{1,j}c_{1,j} + a_{2,j}c_{2,j} + \cdots + a_{n,j}c_{n,j}$$

for any column j.

A determinant may be expanded along any row or any column. In practice, choose a row or column containing many zeros.

Example 4.1.11. Find the determinant

$$\det \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 1 & 0 & 6 \end{bmatrix}$$

Solution

Expand along the first row. For $a_{1,1} = 1$, we have

$$M_{1,1} = \det \begin{bmatrix} 4 & 5 \\ 0 & 6 \end{bmatrix} = 24$$

Since the sign is positive, $c_{1,1} = 24$. Thus, the contribution is $a_{1,1}c_{1,1} = 24$.

For $a_{1,2} = 2$, we have

$$M_{1,2} = \det \begin{bmatrix} 0 & 5 \\ 1 & 6 \end{bmatrix} = -5$$

Since the sign is negative, $c_{1,2} = 5$. Thus, the contribution is $a_{1,2}c_{1,2} = (2)(5) = 10$.

Finally, for $a_{1,3} = 3$, we have

$$M_{1,3} = \det \begin{bmatrix} 0 & 4 \\ 1 & 0 \end{bmatrix} = -4$$

Since the sign is positive, we have $c_{1,3} = -4$. Thus, the contribution is $a_{1,3}c_{1,3} = (3)(-4) = -12$.

Therefore, summing all the contributions, we have

$$\det(A) = 24 + 10 - 12 = 22$$

Example 4.1.12. Compute

$$\det \begin{bmatrix} 2 & 1 & 0 \\ 3 & 0 & 4 \\ 1 & 0 & 5 \end{bmatrix}$$

Solution

Notice the second column contains two zeros. Expand along the second column. Only one term survivces

$$\det(A) = 1 \cdot c_{1,2}$$

Compute

$$M_{1,2} = \det \begin{bmatrix} 3 & 4 \\ 1 & 5 \end{bmatrix} = 15 - 4 = 11$$

Since the sign is negative, we have $c_{1,2} = -11$. Therefore,

$$\det(A) = -11$$

The same process works for 4×4, 5×5, and larger matrices.

Example 4.1.13. Find

$$\det \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 5 & 0 & 0 & 1 \end{bmatrix}$$

Solution

A cofactor expansion along the second row immediately reduces the problem to a 3×3 determinant. Only one term survives

$$\det(A) = 3 \cdot c_{2,2}$$

First observe that by deleting row 2 and column 2, we have the matrix

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 4 & 0 \\ 5 & 0 & 1 \end{bmatrix}$$

Notice that the second row contains two zeros. Expand along row 2. Only one term survives

$$M_{2,2} = \det \begin{bmatrix} 1 & 0 & 2 \\ 0 & 4 & 0 \\ 5 & 0 & 1 \end{bmatrix} = 4 \cdot c_{3,3}$$

Compute

$$M_{3,3} = \det \begin{bmatrix} 1 & 2 \\ 5 & 1 \end{bmatrix} = (1)(1) - (2)(5) = 1 - 10 = -9$$

Since $(-1)^{3+3} = 1$, then $c_{3,3} = (1)(-9) = -9$. Therefore,

$$M_{2,2} = \det \begin{bmatrix} 1 & 0 & 2 \\ 0 & 4 & 0 \\ 5 & 0 & 1 \end{bmatrix} = (4)(-9) = -36$$

Since $(-1)^{2+2} = 1$, then $c_{2,2} = (1)(-36) = -36$. Therefore,

$$\det(A) = 3 \cdot c_{2,2} = (3)(-36) = -108$$

So far, we have mostly chosen rows or columns that make the determinant easier to compute, especially rows or columns containing zeros. However, cofactor expansion works along any row or column, even when there are no zeros. In the following example, we will expand along the first row and carefully keep track of both the cofactor signs and the entries in the row.

Example 4.1.14. Find

$$A = \begin{bmatrix} 2 & -1 & 3 \\ 4 & 7 & 5 \\ -2 & 6 & 1 \end{bmatrix}$$

Solution

Expanding along the first row gives

$$\det(A) = (-1)^{1+1}(2) \det \begin{bmatrix} 7 & 5 \\ 6 & 1 \end{bmatrix} + (-1)^{1+2}(-1) \det \begin{bmatrix} 4 & 5 \\ -2 & 1 \end{bmatrix} + (-1)^{1+3}(3) \begin{bmatrix} 4 & 7 \\ -2 & 6 \end{bmatrix}$$

$$= (1)(2)(-23) + (-1)(-1)(14) + (1)(3)(38)$$
$$= -46 + 14 + 114 = 82$$

4.2 Properties and Applications of Determinants

In Section 4.1, we developed methods for computing determinants using minors, cofactors, and cofactor expansion. While these techniques allow us to calculate determinants, they do not yet explain why determinants are important.

In this section, we study the fundamental properties of determinants and explore their applications. These properties allow determinants to be computed more efficiently and reveal their deep connections with invertibility, inverse matrices, and geometry.

4.2.1 Determinants and Elementary Row Operations

One of the most important properties of determinants is how they behave under row operations. Each row operation affects the determinant in a specific way.

Theorem 4.2.1. *If two rows of a matrix are interchanged, then the determinant changes sign. That is,*

$$\det(B) = -\det(A)$$

where B is obtained from A by swapping two rows.

Example 4.2.2. Consider

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Then

$$\det(A) = (1)(4) - (2)(3) = -2$$

If we swap the rows,

$$B = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$$

Then

$$\det(B) = (3)(2) - (4)(1) = 2$$

Notice that

$$\det(B) = -\det(A)$$

Theorem 4.2.3. *If a row is multiplied by a scalar c, then the determinant is multiplied by c.*

Example 4.2.4. Consider

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

which has determinant -2. Multiplying row 1 by 5 we have

$$B = \begin{bmatrix} 5 & 10 \\ 3 & 4 \end{bmatrix}$$

Then

$$\det(B) = (5)(4) - (10)(3) = 20 - 30 = -10$$

Notice that $-10 = 5(-2)$.

Theorem 4.2.5. *If a multiple of one row is added to another row, the determinant remains unchanged.*

Example 4.2.6. Consider

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

which has determinant -2. Apply $R_2 \to R_2 - 3R_1$ to get

$$B = \begin{bmatrix} 1 & 2 \\ 0 & -2 \end{bmatrix}$$

Compute

$$\det(B) = (1)(-2) - (2)(0) = -2$$

Thus, $\det(B) = \det(A)$.

4.2.2 Determinants of Triangular Matrices

Many determinants can be computed directly.

Theorem 4.2.7. *If A is an upper triangular or lower triangular matrix, then*

$$\det(A) = a_{1,1}a_{2,2}\cdots a_{n,n}$$

That is, we simply multiply the entries on the main diagonal.

Example 4.2.8. Let

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 0 & 5 & 4 \\ 0 & 0 & 7 \end{bmatrix}$$

Compute $\det(A)$.

Solution

Since A is upper triangular,

$$\det(A) = (2)(5)(7) = 70$$

Example 4.2.9. Let

$$A = \begin{bmatrix} 3 & 0 & 0 \\ 2 & 4 & 0 \\ 5 & 1 & 2 \end{bmatrix}$$

Compute $\det(A)$.

Solution

Since A is lower triangular,

$$\det(A) = (3)(4)(2) = 24$$

4.2.3 Special Determinant Properties

The following results are extremely useful.

Theorem 4.2.10. *If a matrix A contains a row of zeros or a row of columns, then*

$$\det(A) = 0$$

This result follows from cofactor expanding along the row or column of zeros.

Example 4.2.11. Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 4 & 5 & 6 \end{bmatrix}$$

Then $\det(A) = 0$.

Theorem 4.2.12. *If two rows of a matrix are identical, then*

$$\det(A) = 0$$

Example 4.2.13. Let

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix}$$

Compute $\det(A)$.

Solution

Observe that

$$\det(A) = (1)(2) - (2)(1) = 0$$

This verifies the conclusion of Theorem 4.2.12.

Theorem 4.2.14. *If one row is a multiple of another row, then*

$$\det(A) = 0$$

Example 4.2.15. Let

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

Compute $\det(A)$.

Solution

Observe that

$$\det(A) = (1)(4) - (2)(2) = 0$$

This verifies the conclusion of Theorem 4.2.14.

4.2.4 Determinants and Matrix Multiplication

One of the most important determinant properties is the following.

Theorem 4.2.16. *For square matrices A and B,*

$$\det(AB) = \det(A)\det(B)$$

Example 4.2.17. Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}$$

Compute $\det(A)$, $\det(B)$, $\det(A)\det(B)$ and $\det(AB)$.

Solution

Here, we have $\det(A) = -2$ and $\det(B) = 10$. Therefore,

$$\det(A)\det(B) = (-2)(10) = -20$$

Also, observe that

$$AB = \begin{bmatrix} 2 & 10 \\ 6 & 20 \end{bmatrix}$$

so then

$$\det(AB) = (2)(20) - (6)(10) = 40 - 60 = -20$$

Therefore, we have that

$$\det(A)\det(B) = \det(AB)$$

4.2.5 Determinants and Invertibility

We now arrive at one of the most important results in linear algebra.

Theorem 4.2.18. *A square matrix A is invertible if and only if* $\det(A) \neq 0$.

The determinant completely determines whether a matrix has an inverse.

Example 4.2.19. Determine whether

$$A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$$

is invertible.

Solution

Computing the determinant,

$$\det(A) = (2)(3) - (1)(5) = 1$$

Therefore, by Theorem 4.2.18, A is invertible.

Example 4.2.20. Determine whether

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

is invertible.

Solution

Computing the determinant

$$\det(A) = (1)(4) - (2)(2) = 0$$

Therefore, by Theorem 4.2.18, A is not invertible.

4.2.6 The Adjoint Matrix

Determinants can also be used to compute inverse matrices.

Definition 4.2.21. The *cofactor matrix* of A is the matrix whose entries are the cofactors:

$$C = [c_{i,j}]$$

Definition 4.2.22. The *adjoint matrix* (or *adjugate matrix*) is

$$\operatorname{adj}(A) = C^T$$

The adjugate is useful to determine the inverse of a matrix.

Theorem 4.2.23. *If* $\det(A) \neq 0$, *then*

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A)$$

For small matrices, this formula is practical. For larger matrices, row reduction is usually more efficient.

Example 4.2.24. Let

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Show using Theorem 4.2.23 that

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A)$$

Solution

The determinant of A is given by

$$\det(A) = ad - bc$$

Computing each minor, we have

$$M_{1,1} = d, \quad M_{1,2} = c, \quad M_{2,1} = b, \quad M_{2,2} = a$$

Therefore, the cofactors are

$$\begin{aligned} c_{1,1} &= (-1)^{1+1} M_{1,1} = d, \\ c_{1,2} &= (-1)^{1+2} M_{1,2} = -c, \\ c_{2,1} &= (-1)^{2+1} M_{2,1} = -b, \\ c_{2,2} &= (-1)^{2+2} M_{2,2} = a \end{aligned}$$

Therefore, the cofactor matrix is

$$C = \begin{bmatrix} d & -c \\ -b & a \end{bmatrix}$$

and so

$$\text{adj}(A) = C^T = \begin{bmatrix} d & -c \\ -b & a \end{bmatrix}^T = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Therefore, combining with the determinant and using Theorem 4.2.23, we have

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

4.2.7 Geometric Interpretations of the Determinant

Determinants measure changes in area and volume.

Theorem 4.2.25. *For the matrix*

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

the quantity $|\det(A)|$ *represents the area of the parallelogram determined by the column vectors.*

Example 4.2.26. Consider

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$$

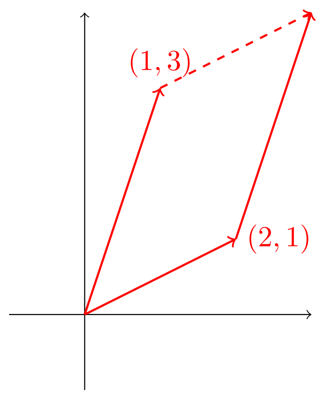
Find the area of the parallelogram.

Solution

We have

$$\det(A) = (2)(3) - (1)(1) = 6 - 1 = 5$$

The corresponding parallelogram has area 5.



If $\det(A) > 0$, the orientation is preserved. Otherwise, if $\det(A) < 0$, the orientation is reversed. Reflections are common examples of transformations with negative determinants.

Theorem 4.2.27. *For a 3×3 matrix A, $|\det(A)|$ is the volume of the parallelepiped formed by the column vectors.*

Example 4.2.28. Let

$$A = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{bmatrix}$$

Compute $|\det(A)|$.

Solution

Here, we have $\det(A) = -4$, so the associated volume is $|\det(A)| = 4$.

4.2.8 Cramer's Rule

Determinants can be used to solve systems of equations.

Theorem 4.2.29 (**Cramer's Rule**). *Consider $A\vec{x} = \vec{b}$. If $\det(A) \neq 0$, then*

$$x_i = \frac{\det(A_i)}{\det(A)}$$

where A_i is obtained by replacing column i of A with $\vec{b}$.

Example 4.2.30. Solve the system

$$\begin{aligned} 2x + y &= 5, \\ x - y &= 1 \end{aligned}$$

Solution

Let

$$A = \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix}$$

Then calculating the determinant of A

$$\det(A) = (2)(-1) - (1)(1) = -2 - 1 = -3$$

Replace column 1 of A by $\vec{b}$ to get

$$A_1 = \begin{bmatrix} 5 & 1 \\ 1 & -1 \end{bmatrix}$$

Then

$$\det(A_1) = (5)(-1) - (1)(1) = -5 - 1 = -6$$

Therefore,

$$x = \frac{\det(A_1)}{\det(A)} = \frac{-6}{-3} = 2$$

Next, replace column 2 of A by $\vec{b}$ to get

$$A_2 = \begin{bmatrix} 2 & 5 \\ 1 & 1 \end{bmatrix}$$

Then

$$\det(A_2) = (2)(1) - (5)(1) = -3$$

Therefore,

$$y = \frac{\det(A_2)}{\det(A)} = \frac{-3}{-3} = 1$$

Chapter 5

Eigenvalues and Eigenvectors

In the previous chapters, we studied systems of linear equations, matrices, linear transformations, and determinants. We learned how matrices can represent transformations of vectors and how determinants can be used to determine whether those transformations are invertible.

We now turn to one of the most important topics in linear algebra: eigenvalues and eigenvectors.

Suppose a matrix transformation acts on a vector in the plane or in space. In general, the transformation changes both the direction and the magnitude of the vector. However, certain special vectors behave differently.

For some vectors, a transformation changes only their length while leaving their direction unchanged. These special vectors are called eigenvectors, and the factors by which they are stretched or compressed are called eigenvalues.

The relationship between an eigenvalue λ and an eigenvector $\vec{x}$ is described by the equation

$$A\vec{x} = \lambda\vec{x}$$

This equation lies at the heart of the chapter.

One reason eigenvalues are so important is that they reveal hidden structure within a matrix. They allow complicated matrix transformations to be understood in terms of simple stretching and compressing actions.

5.1 Eigenvalues and Eigenvectors

In previous chapters, we studied matrices and linear transformations. We learned that a matrix transformation generally changes both the magnitude and direction of a vector.

However, some vectors behave in a special way under a transformation.

For certain nonzero vectors, a matrix transformation changes only the length of the vector while leaving its direction unchanged. These special vectors are called eigenvectors, and the corresponding scale factors are called eigenvalues.

Eigenvalues and eigenvectors reveal important information about the behaviour of a matrix and the transformation it represents

Consider a matrix

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

Apply A ti the vector $\vec{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ to get

$$A\vec{x} = \begin{bmatrix} 2 \\ 0 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Notice that the direction did not change, only the magnitude changed. Similarly,

$$A \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix} = 3 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

These vectors are examples of eigenvectors.

5.1.1 Eigenvalues and Eigenvectors

Definition 5.1.1. Let A be an $n \times n$ matrix. A nonzero vector $\vec{x}$ is called an *eigenvector* of A if

$$A\vec{x} = \lambda\vec{x}$$

for some scalar λ. The scalar λ is called an *eigenvalue* of A.

The fundamental equation is

$$A\vec{x} = \lambda\vec{x}$$

Here, eigenvalues must be nonzero, and the eigenvalue may be positive, negative, or zero, may also contain several eigenvalues, and different eigenvalues generally correspond to different directions.

Example 5.1.2. Let

$$A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}$$

Determine whether $\vec{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ is an eigenvector.

Solution

Compute

$$A\vec{x} = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \end{bmatrix}$$

Since

$$\begin{bmatrix} 4 \\ 0 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

we have $A\vec{x} = 4\vec{x}$. Thus, $\lambda = 4$ is an eigenvalue of A and $\vec{x}$ is an eigenvector corresponding to $\lambda = 4$.

Example 5.1.3. Let

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$$

Determine whether $\vec{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ is an eigenvector.

Solution

Compute

$$A\vec{x} = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix} = 2\vec{x}$$

Therefore, $\lambda = 2$ is an eigenvalue of A, and $\vec{x}$ is an eigenvector corresponding to $\lambda = 2$.

5.1.2 Finding Eigenvalues

The main problem is: Given a matrix A, determine all eigenvalues. We begin with

$$A\vec{x} = \lambda\vec{x}$$

Move everything to one side to get

$$A\vec{x} - \lambda\vec{x} = \vec{0}$$

Since $\lambda\vec{x} = \lambda I_n \vec{x}$, we obtain

$$(A - \lambda I_n)\vec{x} = \vec{0}$$

Observe that an eigenvector must satisfy

$$(A - \lambda I_n)\vec{x} = \vec{0}$$

with $\vec{x} \neq \vec{0}$. This means the homogeneous system has a nontrivial solution.

Recall that a homogeneous system has a nontrivial solution if and only if the coefficient matrix is unique. Thus,

Definition 5.1.4. The equation

$$\det(A - \lambda I_n) = 0$$

is called the *characteristic equation* of A, and the polynomial $\det(A - \lambda I_n)$ is called the *characteristic polynomial*.

Example 5.1.5. Suppose

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Then the characteristic polynomial is

$$\det(A - \lambda I_2) = \det \begin{bmatrix} a - \lambda & b \\ c & d - \lambda \end{bmatrix} = (a - \lambda)(d - \lambda) - bc$$

Setting this equal to zero gives the characteristic equation.

Example 5.1.6. Find the eigenvalues of

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

Solution

First compute

$$\det(A - \lambda I_2) = \det \begin{bmatrix} 2 - \lambda & 0 \\ 0 & 3 - \lambda \end{bmatrix} = (2 - \lambda)(3 - \lambda)$$

Set $\det(A - \lambda I_2) = 0$ so that

$$(2 - \lambda)(3 - \lambda) = 0$$

Therefore, the eigenvalues are $\lambda = 2$ or $\lambda = 3$.

Once an eigenvalue is known, we substitute it into

$$(A - \lambda I_n)\vec{x} = \vec{0}$$

and find the corresponding eigenvector $\vec{x}$.

Example 5.1.7. Find an eigenvector corresponding to each eigenvalue for

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

Solution

For $\lambda = 2$, we compute

$$A - 2I_2 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

So we solve

$$(A - 2I_n)\vec{x} = \vec{0}$$

which gives

$$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This gives $y = 0$, and thus, x is free. Choose $x = 1$. Then $\vec{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ is an eigenvector of A corresponding to $\lambda = 2$. Any nonzero multiple is also an eigenvector.

For $\lambda = 3$, we compute

$$A - 3I_2 = \begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix}$$

So we solve

$$(A - 3I_2)\vec{x} = \vec{0}$$

which gives

$$\begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This gives $x = 0$, and thus, y is free. Choose $y = 1$. Then $\vec{x} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ is an eigenvector of A corresponding to $\lambda = 3$.

Example 5.1.8. Find all eigenvalues and corresponding eigenvectors of

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

Solution

First compute

$$\begin{aligned} \det(A - \lambda I_2) = \det \begin{bmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{bmatrix} &= (4-\lambda)(3-\lambda) - 2 \\ &= 12 - 7\lambda + \lambda^2 - 2 \\ &= \lambda^2 - 7\lambda + 10 \\ &= (\lambda - 5)(\lambda - 2) \end{aligned}$$

Set $\det(A - \lambda I_2) = 0$, then the characteristic equation is

$$(\lambda - 5)(\lambda - 2) = 0$$

Therefore, the eigenvalues are $\lambda = 5$ and $\lambda = 2$.

For $\lambda = 5$, we solve

$$\begin{gathered} (A - 5I_2)\vec{x} = \vec{0} \\ \begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \end{gathered}$$

which gives $x = y$. Choose $x = 1$ so the eigenvector corresponding to $\lambda = 5$ is

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

For $\lambda = 2$, we solve

$$\begin{gathered} (A - 2I_2)\vec{x} = \vec{0} \\ \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \end{gathered}$$

Thus, $2x + y = 0$. Choose $x = 1$ so that $y = -2$. Then the eigenvector corresponding to $\lambda = 2$ is

$$\vec{v}_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

5.1.3 Special Cases

Sometimes, the characteristic polynomial has repeated roots. For example, if we get the characteristic equation

$$(\lambda - 3)^2 = 0$$

Then $\lambda = 3$ is a repeated eigenvalue.

Definition 5.1.9. If a characteristic equation has an eigenvalue λ that is repeated n times, then we say that λ is an *eigenvalue of algebraic multiplicity n.*

Example 5.1.10. Find all eigenvalues and eigenvectors of

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 1 \\ 0 & 1 & 3 \end{bmatrix}$$

Solution

First compute

$$\begin{aligned} \det(A) = \det \begin{bmatrix} 2-\lambda & 0 & 0 \\ 0 & 3-\lambda & 1 \\ 0 & 1 & 3-\lambda \end{bmatrix} &= (2-\lambda)[(3-\lambda)^2 - 1] \\ &= (2-\lambda)((3-\lambda)-1)((3-\lambda)+1) \\ &= (2-\lambda)(2-\lambda)(4-\lambda) \\ &= (2-\lambda)^2(4-\lambda) \end{aligned}$$

So the eigenvalues of A are $\lambda = 2$ with algebraic multiplicity 2, and $\lambda = 4$.

For $\lambda = 2$, we solve

$$(A - 2I_3)\vec{x} = \vec{0}$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

This gives $y + z = 0$, so $y = -z$. We also have a free variable x, so let $x = 1$ and $z = 1$. Then $y = -1$. Therefore, an eigenvector corresponding to $\lambda = 2$ is

$$\vec{v}_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

For $\lambda = 4$, we solve

$$(A - 4I_3)\vec{x} = \vec{0}$$
$$\begin{bmatrix} -2 & 0 & 0 \\ 0 & -1 & 1 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

This gives $-2x = 0$, so $x = 0$. Also, $-y + z = 0$, so $y = z$. Let $z = 1$ so that $y = 1$. Then an eigenvector corresponding to $\lambda = 4$ is

$$\vec{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

If $\lambda = 0$ is a eigenvalue, then $\det(A) = 0$, so the matrix A is singular.

Example 5.1.11. Find the eigenvalues and eigenvectors of

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

Solution

First compute

$$\det(A - \lambda I_2) = \deg \begin{bmatrix} 1-\lambda & 0 \\ 0 & 0-\lambda \end{bmatrix} = -\lambda(1-\lambda)$$

Set the characteristic equation to zero, and we get eigenvalues $\lambda = 0$ and $\lambda = 1$.

When $\lambda = 0$, we solve

$$(A - 0I_2)\vec{x} = \vec{0}$$
$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

which gives $x = 0$. Since y is free, choose $y = 1$, so an eigenvector corresponding to $\lambda = 0$ is

$$\vec{v}_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

When $\lambda = 1$, we solve

$$(A - I_2)\vec{x} = \vec{0}$$

$$\begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

which gives $y = 0$. Since x is free, choose $x = 1$, so an eigenvector corresponding to $\lambda = 1$ is

$$\vec{v}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

5.2 Diagonalization and Powers of Matrices

In Section 5.1, we introduced eigenvalues and eigenvectors and learned how to find them using the characteristic equation. We saw that eigenvectors identify special directions that remain unchanged under a matrix transformation, while eigenvalues describe how vectors in those directions are stretched or compressed.

In this section, we study one of the most powerful applications of eigenvalues and eigenvectors: diagonalization.

The main idea behind diagonalization is to replace a complicated matrix with a much simpler matrix that has the same essential behaviour.

5.2.1 Diagonal Matrices

Definition 5.2.1. A square matrix is called *diagonal* if all entries off the main diagonal are zero. That is,

$$D = \begin{bmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{bmatrix}$$

Example 5.2.2. The following matrices are diagonal.

$$\begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 7 \end{bmatrix}$$

Diagonal matrices are easy to work with because multiplication only affects the diagonal entries.

5.2.2 Powers of Diagonal Matrices

One major advantage of diagonal matrices is that their powers are easy to compute.

Theorem 5.2.3. *If D is a diagonal matrix given by*

$$D = \begin{bmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{bmatrix}$$

Then D^k is also a diagonal matrix with

$$D^k = \begin{bmatrix} d_1^k & 0 & \cdots & 0 \\ 0 & d_2^k & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n^k \end{bmatrix}$$

Example 5.2.4. Let

$$D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

Then

$$D^5 = \begin{bmatrix} 2^5 & 0 \\ 0 & 3^5 \end{bmatrix} = \begin{bmatrix} 32 & 0 \\ 0 & 243 \end{bmatrix}$$

5.2.3 Diagonalization

Consider

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

Computing A^{20} directly would require matrix multiplication 20 times! Wouldn't it be easier if we could somehow replace A with a diagonal matrix? This motivates the concept of diagonalization.

Definition 5.2.5. Two matrices A and B are said to be *similar* if there exists an invertible matrix P such that

$$A = PBP^{-1}$$

Similar matrices represent the same linear transformation expressed in different coordinate systems.

Definition 5.2.6. A matrix A is *diagonalizable* if there exists an invertible matrix P and a diagonal matrix D such that

$$A = PDP^{-1}$$

Here, the matrix D is a simpler version of A, and the matrix P changes coordinates.

The key idea to constructing a diagonalization is surprisingly simple. We simply need to know what the eigenvalues and eigenvectors of A are.

Theorem 5.2.7. *Suppose A has n distinct eigenvectors $\vec{v}_1, \vec{v}_2, ..., \vec{v}_n$ with corresponding eigenvalues $\lambda_1, \lambda_2, ..., \lambda_n$. Then A is diagonalizable. Furthermore,*

$$P = \begin{bmatrix} \uparrow & \uparrow & & \uparrow \\ \vec{v}_1 & \vec{v}_2 & \cdots & \vec{v}_n \\ \downarrow & \downarrow & & \downarrow \end{bmatrix}$$

and

$$D = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}$$

Example 5.2.8. Diagonalize

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

Solution

From Example 5.1.8, the eigenvalues are $\lambda_1 = 5$ and $\lambda_2 = 2$, and the corresponding eigenvectors are

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \vec{v}_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

We first construct the matrix P. Place eigenvectors as columns

$$P = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}$$

Then we construct the matrix D be placing the eigenvalues on the diagonal.

$$D = \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix}$$

Therefore,

$$A = PDP^{-1}$$

where

$$P = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}, \quad D = \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix}$$

5.2.4 When Is a Matrix Diagonalizable?

Not every matrix can be diagonalized. An important consequence is that if a matrix has n distinct eigenvalues, then it is automatically diagonalizable.

Example 5.2.9. The matrix

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}$$

is diagonalizable because it has two distinct eigenvalues.

5.2.5 Power Matrices

The main reason we diagonalize matrices is to simplify powers. Suppose

$$A = PDP^{-1}$$

then

$$A^2 = AA = PDP^{-1}PDP^{-1}$$

Since $P^{-1}P = I_n$, we obtain

$$A^2 = PDI_nDP^{-1} = PDDP^{-1} = PD^2P^{-1}$$

Similarly,

$$A^3 = PD^3P^{-1}$$

and in general,

$$A^k = PD^kP^{-1}$$

This is a very useful tool because instead of computing A^k directly, we only need to compute D^k. Since D is diagonal, this is easy.

Example 5.2.10. Let $A = PDP^{-1}$, where

$$D = \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix}$$

Compute D^4.

Solution

Raise each diagonal entry to the fourth power:

$$D^4 = \begin{bmatrix} 5^4 & 0 \\ 0 & 2^4 \end{bmatrix} = \begin{bmatrix} 625 & 0 \\ 0 & 16 \end{bmatrix}$$

Example 5.2.11. Suppose $A = PDP^{-1}$ with

$$D = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$$

Compute A^{10}.

Solution

Using

$$A^{10} = PD^{10}P^{-1}$$

compute

$$D^{10} = \begin{bmatrix} 3^{10} & 0 \\ 0 & 1^{10} \end{bmatrix} = \begin{bmatrix} 59049 & 0 \\ 0 & 1 \end{bmatrix}$$

Therefore,

$$A^{10} = P \begin{bmatrix} 59049 & 0 \\ 0 & 1 \end{bmatrix} P^{-1}$$

Chapter 6

Vector Geometry

In previous chapters, we studied systems of linear equations, matrices, linear transformations, determinants, and eigenvalues. Much of our focus was algebraic: solving equations, manipulating matrices, and analyzing transformations through symbolic calculations.

While linear algebra is often developed algebraically, it is also deeply connected to geometry. Many of the concepts we have encountered can be interpreted geometrically using vectors, lengths, angles, lines, and planes. Understanding these geometric ideas provides valuable intuition and helps us visualize many of the abstract concepts introduced earlier in the course.

This chapter focuses on the geometry of vectors in two- and three-dimensional space. We will develop tools that allow us to measure distances, compute angles, describe geometric objects, and analyze relationships between vectors.

One of the central themes of this chapter is that algebra and geometry are closely connected.

6.1 Length of a Vector

In previous chapters, we studied vectors primarily from an algebraic perspective. We learned how to add vectors, multiply vectors by scalars, form linear combinations, and analyze vector spaces.

In geometry, however, vectors also possess physical meaning. A vector can represent:

- a displacement,
- a velocity,

- a force,
- a direction,
- or a position in space.

One important geometric property of a vector is its length.

The length of a vector tells us how large the vector is, regardless of its direction.

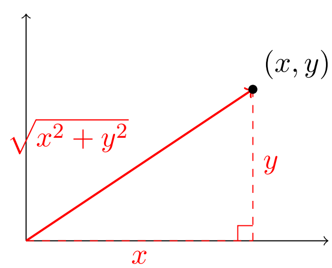
In this section, we introduce the length (or norm) of a vector and develop formulas for computing distances in $\mathbb{R}^2$, $\mathbb{R}^3$, and $\mathbb{R}^n$.

6.1.1 The Length of a Vector in $\mathbb{R}^2$

Consider a vector $\vec{v} = \begin{bmatrix} x \\ y \end{bmatrix}$. Geometrically, this vector corresponds to a point (x, y) in the plane. The vector, together with the coordinate axes, forms a right triangle. Using Pythagorean Theorem,

$$x^2 + y^2 = (\text{length})^2$$

Thus, the length is $\sqrt{x^2 + y^2}$.



Definition 6.1.1. The *length* (or *norm*) of a vector $\vec{v} = \begin{bmatrix} x \\ y \end{bmatrix}$ is

$$\|\vec{v}\| = \sqrt{x^2 + y^2}$$

Example 6.1.2. Find the length of $\vec{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$.

Solution

Using the formula,

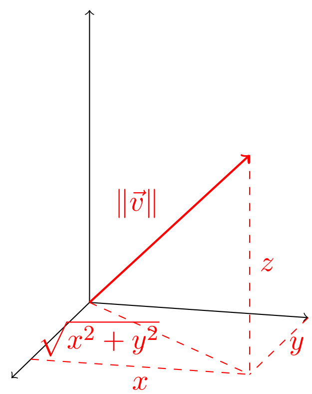
$$\|\vec{v}\| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$

6.1.2 The Length of a Vector in $\mathbb{R}^3$

The same idea extends naturally to three dimensions.

Definition 6.1.3. For $\vec{v} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, the length is

$$\|\vec{v}\| = \sqrt{x^2 + y^2 + z^2}$$



Example 6.1.4. Find the length of $\vec{v} = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}$.

Solution

Using the formula,

$$\|\vec{v}\| = \sqrt{2^2 + (-1)^2 + 2^2} = \sqrt{4+1+4} = \sqrt{9} = 3$$

6.1.3 The Length of a Vector in $\mathbb{R}^n$

The formula extends to vectors with any number of components.

Definition 6.1.5. Let $\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$. The length (or Euclidean norm) of $\vec{v}$ is

$$\|\vec{v}\| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$$

Example 6.1.6. Find the length of $\vec{v} = \begin{bmatrix} 1 \\ 2 \\ 2 \\ 1 \end{bmatrix}$.

Solution

Using the formula,

$$\|\vec{v}\| = \sqrt{1^2 + 2^2 + 2^2 + 1^2} = \sqrt{1 + 4 + 4 + 1} = \sqrt{10}$$

6.1.4 Properties of Length

The length of a vector satisfies several important properties.

Theorem 6.1.7.

1. *$\|\vec{v}\| = 0$ if and only if $\vec{v} = \vec{0}$.*
2. *For every vector $\vec{v}$, $\|\vec{v}\| \geq 0$.*
3. *For any $\vec{v}$ and $c \in \mathbb{R}$, $\|c\vec{v}\| = |c|\|\vec{v}\|$.*

The first point is saying that the zero vector is the only vector with zero length, the second point is saying that lengths cannot be negative.

Example 6.1.8. Let $\vec{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$. Since $\|\vec{v}\| = 5$, we have

$$\|2\vec{v}\| = 2(5) = 10$$

and

$$\| - 3\vec{v}\| = 3(5) = 15$$

6.1.5 Unit Vectors

Often, we are interested only in the direction of a vector. A vector of length 1 is called the unit vector.

Definition 6.1.9. A vector $\vec{u}$ is called a *unit vector* if $\|\vec{u}\| = 1$.

Example 6.1.10. The vectors $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ are unit vectors.

Example 6.1.11. The vector $\begin{bmatrix} \frac{3}{5} \\ \frac{4}{5} \end{bmatrix}$ is also a unit vector since

$$\sqrt{\left(\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2} = 1$$

Given a nonzero vector, we can create a unit vector pointing in the same direction.

Theorem 6.1.12. *Let $\vec{v} \neq \vec{0}$. Then*

$$\vec{u} = \frac{\vec{v}}{\|\vec{v}\|}$$

is a unit vector.

Proof. Compute its length,

$$\left\| \frac{\vec{v}}{\|\vec{v}\|} \right\| = \frac{1}{\|\vec{v}\|} \|\vec{v}\| = 1$$

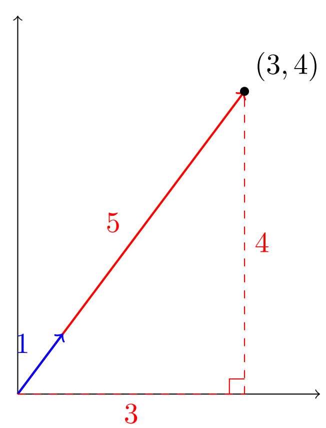
Therefore, it is a unit vector. □

Example 6.1.13. Find a unit vector in the direction of $\vec{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$.

Solution

Computing its length, we get $\|\vec{v}\| = 5$. Then divide the vector by the length

$$\vec{u} = \frac{1}{5} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} \frac{3}{4} \\ \frac{4}{5} \end{bmatrix}$$



6.1.6 The Distance Between Two Points

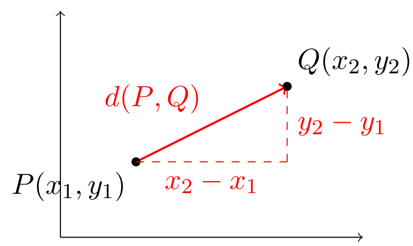
Length allows us to define distance. Suppose $P(x_1, y_1)$ and $Q(x_2, y_2)$. The vector from P to Q is

$$\overrightarrow{PQ} = \begin{bmatrix} x_2 - x_1 \\ y_2 - y_1 \end{bmatrix}$$

The distance between points is the length of this vector.

Definition 6.1.14. The *distance* between $P(x_1, y_1)$ and $Q(x_2, y_2)$ is

$$d(P,Q) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$



Example 6.1.15. Find the distance between $P(1, 2)$ and $Q(4, 6)$.

Solution

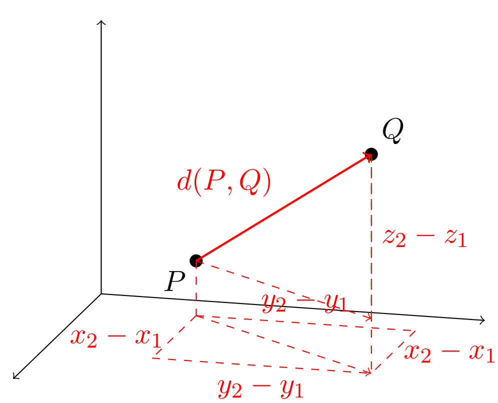
Using the formula,

$$d(P,Q) = \sqrt{(4-1)^2 + (6-2)^2} = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$$

The formula extends naturally to $\mathbb{R}^3$ or $\mathbb{R}^n$.

Definition 6.1.16. For $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$, the distance is

$$d(P,Q) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$



Definition 6.1.17. For $P(x_1, x_2, ..., x_n)$ and $Q(y_1, y_2, ..., y_n)$, the distance is

$$d(P, Q) = \sqrt{(y_1 - x_1)^2 + (y_2 - x_2)^2 + \cdots + (y_n - x_n)^2}$$

Example 6.1.18. Find the distance between $(1, 2, 3)$ and $(4, 6, 3)$.

Solution

Using the formula,

$$d(P, Q) = \sqrt{(4-1)^2 + (6-2)^2 + (3-3)^2} = \sqrt{3^2 + 4^2 + 0^2} = 5$$

6.1.7 The Triangle Inequality

Another property of vector lengths is the triangle inequality.

Theorem 6.1.19 (The Triangle Inequality). *For all vectors $\vec{u}, \vec{v} \in \mathbb{R}^n$,*

$$\|\vec{u} + \vec{v}\| \leq \|\vec{u}\| + \|\vec{v}\|$$

The shortest path between two points is a straight line. If you travel first along $\vec{u}$ and then along $\vec{v}$, the total distance travelled cannot be less than the direct distance represented by $\vec{u} + \vec{v}$. This is exactly the familiar triangle inequality from geometry.

Example 6.1.20. Verify the triangle inequality for $\vec{u} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 0 \\ 4 \end{bmatrix}$.

Solution

First note

$$\|\vec{u} + \vec{v}\| = \left\| \begin{bmatrix} 3 \\ 4 \end{bmatrix} \right\| = 5$$

while

$$\|\vec{u}\| + \|\vec{v}\| = 3 + 4 = 7$$

Thus, $5 \leq 7$.

6.1.8 Distance Between Vectors

Using vector length, we can define distance between vectors.

Definition 6.1.21. The distance between vectors $\vec{u}$ and $\vec{v}$ is

$$d(\vec{u}, \vec{v}) = \|\vec{u} - \vec{v}\|$$

Example 6.1.22. Find the distance between $\vec{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$.

Solution

Computing

$$d(\vec{u}, \vec{v}) = \left\| \begin{bmatrix} -3 \\ -4 \end{bmatrix} \right\| = \sqrt{(-3)^2 + (-4)^2} = 5$$

6.2 The Dot Product

In the previous section, we introduced the length of a vector and the distance between points. These concepts will allow us to measure the size of vectors, but they do not tell us anything about the relationship between two vectors. For example:

- Do these vectors point in the same direction?
- Do they point in opposite directions?
- Are they perpendicular?
- What is the angle between them?

To answer these questions, we introduce one of the most important operations in linear algebra and vector geometry: the dot product.

The dot product combines two vectors to produce a scalar (a real number). Although its definition appears simple, it provides powerful geometric information about vectors.

6.2.1 Definition of the Dot Product

Definition 6.2.1. Let $\vec{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$ be vectors in $\mathbb{R}^n$. The *dot product* of $\vec{u}$ and $\vec{v}$ is

$$\vec{u} \cdot \vec{v} = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n$$

It is important to observe that the dot product produces a scalar, and not a vector.

Example 6.2.2. Compute $\begin{bmatrix}1\\2\end{bmatrix} \cdot \begin{bmatrix}3\\4\end{bmatrix}$.

Solution

Multiply corresponding components and add

$$\begin{bmatrix}1\\2\end{bmatrix} \cdot \begin{bmatrix}3\\4\end{bmatrix} = (1)(3) + (2)(4) = 3 + 8 = 11$$

Example 6.2.3. Compute $\begin{bmatrix}2\\-1\\3\end{bmatrix} \cdot \begin{bmatrix}4\\5\\-2\end{bmatrix}$.

Solution

Multiply corresponding components and add

$$\begin{bmatrix}2\\-1\\3\end{bmatrix} \cdot \begin{bmatrix}4\\5\\-2\end{bmatrix} = (2)(4) + (-1)(5) + (3)(-2) = 8 - 5 - 6 = -3$$

6.2.2 Properties of the Dot Product

The dot product satisfies several useful algebraic properties.

Theorem 6.2.4. *For vectors $\vec{u}$, $\vec{v}$, and $\vec{w}$ in $\mathbb{R}^n$ and $c \in \mathbb{R}$, the following properties hold:*

1. *Commutative Property:* $\vec{u} \cdot \vec{v} = \vec{v} \cdot \vec{u}$
2. *Distributive Property:* $\vec{u} \cdot (\vec{v} + \vec{w}) = \vec{u} \cdot \vec{v} + \vec{u} \cdot \vec{w}$
3. *Scalar Multiplication:* $(c\vec{u}) \cdot \vec{v} = c(\vec{u} \cdot \vec{v})$

6.2.3 Dot Product and Length

The dot product is closely related to vector length.

Theorem 6.2.5. *For every vector $\vec{v}$,*

$$\vec{v} \cdot \vec{v} = \|\vec{v}\|^2$$

Proof. Let $\vec{v} = \begin{bmatrix}v_1\\v_2\\\vdots\\v_n\end{bmatrix}$. Then

$$\vec{v} \cdot \vec{v} = v_1^2 + v_2^2 + \cdots + v_n^2$$

But then

$$\|\vec{v}\| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$$

Squaring both sides gives the result. □

Example 6.2.6. Verify that $\vec{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ satisfies $\vec{v} \cdot \vec{v} = \|\vec{v}\|^2$.

Solution

Compute

$$\vec{v} \cdot \vec{v} = 3^2 + 4^2 = 25$$

Also, $\|\vec{v}\| = 5$. Therefore, $\|\vec{v}\|^2 = 25$. Both values agree.

6.2.4 Geometric Interpretation

The true power of the dot product comes from its geometric interpretation.

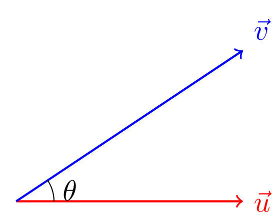
Theorem 6.2.7. *For nonzero vectors $\vec{u}$ and $\vec{v}$,*

$$\vec{u} \cdot \vec{v} = \|\vec{u}\|\|\vec{v}\| \cos(\theta)$$

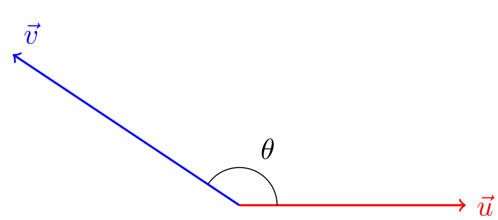
where θ is the angle between the vectors.

The dot product measures how closely two vectors point in the same direction.

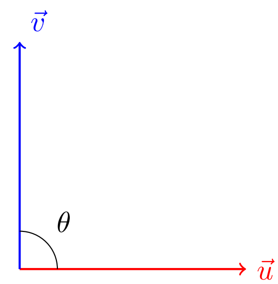
- If $\vec{u} \cdot \vec{v} > 0$, then the angle is acute.



- If $\vec{u} \cdot \vec{v} < 0$, then the angle is obtuse.



- If $\vec{u} \cdot \vec{v} = 0$, then the vectors are perpendicular.



6.2.5 Angle Between Vectors

Rearranging the geometric formula gives

$$\cos(\theta) = \frac{\vec{u} \cdot \vec{v}}{\|\vec{u}\|\|\vec{v}\|}$$

Example 6.2.8. Find the angle between $\vec{u} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

Solution

Computing the dot product and lengths

$$\begin{aligned} \vec{u} \cdot \vec{v} &= 1, \\ \|\vec{u}\| &= 1, \\ \|\vec{v}\| &= \sqrt{2} \end{aligned}$$

Then substitute to get

$$\begin{aligned} \cos(\theta) &= \frac{1}{\sqrt{2}} \\ \theta &= \cos^{-1}\left(\frac{1}{\sqrt{2}}\right) \\ \theta &= \frac{\pi}{4} \end{aligned}$$

One of the most important applications of the dot product is determining perpendicularity.

Definition 6.2.9. Two vectors are *orthogonal* if they are perpendicular.

Theorem 6.2.10. *Two nonzero vectors are orthogonal if and only if* $\vec{u} \cdot \vec{v} = 0$*.*

Example 6.2.11. Determine whether $\vec{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$ are orthogonal.

Solution

Compute

$$\vec{u} \cdot \vec{v} = (1)(2) + (2)(-1) = 2 - 2 = 0$$

Therefore, $\vec{u}$ and $\vec{v}$ are orthogonal.

The same criterion works in higher dimension.

Example 6.2.12. Determine whether $\vec{u} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$ are orthogonal.

Solution

Compute

$$\vec{u} \cdot \vec{v} = (1)(1) + (1)(-1) + (1)(0) = 1 - 1 + 0 = 0$$

Therefore, $\vec{u}$ and $\vec{v}$ are orthogonal.

6.2.6 Scalar and Vector Projections

Another important application of the dot product is measuring how much of one vector lies in the direction of another.

Definition 6.2.13. The *scalar projection* (or *component*) of $\vec{u}$ onto $\vec{v}$ is

$$\text{scal}_{\vec{v}}(\vec{u}) = \frac{\vec{u} \cdot \vec{v}}{\|\vec{v}\|}$$

This quantity is a scalar and represents the signed length of the projection of $\vec{u}$ onto $\vec{v}$.

Definition 6.2.14. The *vector projection* of $\vec{u}$ onto $\vec{v}$ is

$$\text{proj}_{\vec{v}}(\vec{u}) = \frac{\vec{u} \cdot \vec{v}}{\|\vec{v}\|^2}\vec{v}$$

This is a vector pointing in the same direction of $\vec{v}$.

Some observations that we can make about the scalar and vector projections.

1. Since the dot product of two vectors $\vec{u}$ and $\vec{v}$ can be represented as

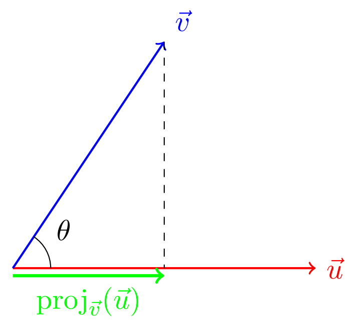
$$\vec{u} \cdot \vec{v} = \|\vec{u}\|\|\vec{v}\| \cos(\theta)$$

   Rearranging the formula gives

$$\text{scal}_{\vec{v}}(\vec{u}) = \frac{\vec{u} \cdot \vec{v}}{\|\vec{v}\|} = \|\vec{u}\| \cos(\theta)$$

2. The vector projection of $\vec{u}$ onto $\vec{v}$ is the scalar projection of $\vec{u}$ onto $\vec{v}$ times the unit vector of $\vec{v}$. That is,

$$\text{proj}_{\vec{v}}(\vec{u}) = \frac{\vec{u} \cdot \vec{v}}{\|\vec{v}\|^2}\vec{v} = \frac{\vec{u} \cdot \vec{v}}{\|\vec{v}\|} \cdot \frac{\vec{v}}{\|\vec{v}\|} = \text{scal}_{\vec{v}}(\vec{u})\frac{\vec{v}}{\|\vec{v}\|}$$



Example 6.2.15. Find the scalar and vector projections of $\vec{u} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ onto $\vec{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

Solution

First compute

$$\begin{aligned} \vec{u} \cdot \vec{v} &= (3)(1) + (4)(0) = 3, \\ \|\vec{v}\| &= \sqrt{1^2 + 0^2} = 1 \end{aligned}$$

Therefore, the scalar projection of $\vec{u}$ onto $\vec{v}$ is

$$\text{scal}_{\vec{v}}(\vec{u}) = \frac{\vec{u} \cdot \vec{v}}{\|\vec{v}\|} = \frac{3}{1} = 3$$

and the vector projection of $\vec{u}$ onto $\vec{v}$ is

$$\text{proj}_{\vec{v}}(\vec{u}) = \text{scal}_{\vec{v}}(\vec{u})\frac{\vec{v}}{\|\vec{v}\|} = 3\begin{bmatrix}1\\0\end{bmatrix} = \begin{bmatrix}3\\0\end{bmatrix}$$

6.2.7 Work Done by a Force

A classical application comes from physics.

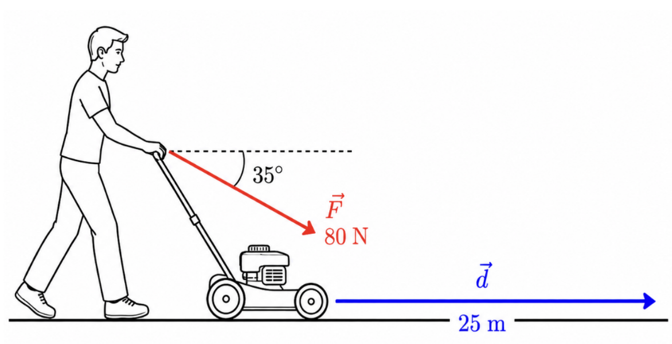
Definition 6.2.16. If a force vector $\vec{F}$ moves an object through displacement $\vec{d}$, then the *work* done is

$$W = \vec{F} \cdot \vec{d}$$

Example 6.2.17. Suppose $\vec{F} = \begin{bmatrix}10\\0\end{bmatrix}$ and $\vec{d} = \begin{bmatrix}3\\4\end{bmatrix}$. Then

$$W = (10)(3) + (0)(4) = 30 \text{ J}$$

Example 6.2.18. A person pushes a lawn mower with a force of 80 N at an angle of 35° below the horizontal. The lawn mower moves 25 m forward. How much work does the person do on the lawn mower?



Solution

We use the formula

$$W = \vec{F} \cdot \vec{d} = \|\vec{F}\|\|\vec{d}\|\cos(\theta)$$

where $\|\vec{F}\| = 80$ N, $\|\vec{d}\| = 25$ m, and $\theta = 35°$. Indeed,

$$W = \|\vec{F}\|\|\vec{d}\|\cos(\theta) = (80)(25)\cos(35°) = 1638.4 \text{ J}$$

6.2.8 The Cauchy–Schwarz Inequality

One of the most important results involving the dot product is the Cauchy–Schwarz inequality.

Theorem 6.2.19 (Cauchy–Schwarz Inequality). *For any vectors $\vec{u}, \vec{v} \in \mathbb{R}^n$,*

$$|\vec{u} \cdot \vec{v}| \leq \|\vec{u}\|\|\vec{v}\|$$

The absolute value of the dot product can never exceed the product of the vector lengths. This theorem guarantees that

$$-1 \leq \frac{\vec{u} \cdot \vec{v}}{\|\vec{u}\|\|\vec{v}\|} \leq 1$$

which explains why the angle formula

$$\cos(\theta) = \frac{\vec{u} \cdot \vec{v}}{\|\vec{u}\|\|\vec{v}\|}$$

always make sense.

Example 6.2.20. Verify the Cauchy–Schwarz inequality if $\vec{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$.

Solution

Computing values,

$$|\vec{u} \cdot \vec{v}| = 11, \quad \|\vec{u}\| = \sqrt{5}, \quad \|\vec{v}\| = 5$$

Thus,

$$11 < 5\sqrt{5}$$

The inequality is satisfied.

6.3 The Cross Product

In Section 6.2, we introduced the dot product and saw how it can be used to determine angles between vectors and identify orthogonal vectors.

The dot product takes two vectors and produces a scalar.

In this section, we introduce another important vector operation called the cross product.

6.3.1 Definition of the Cross Product

The cross product is defined for vectors in $\mathbb{R}^3$ only.

Definition 6.3.1. Let $\vec{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$. The *cross product* of $\vec{u}$ and $\vec{v}$ is

$$\vec{u} \times \vec{v} = \begin{bmatrix} u_2v_3 - u_3v_2 \\ u_3v_1 - u_1v_3 \\ u_1v_2 - u_2v_1 \end{bmatrix}$$

The cross product produces a vector. Thus, $\vec{u} \times \vec{v}$ is a vector in $\mathbb{R}^3$.
The cross product is usually computed using a determinant.

$$\vec{u} \times \vec{v} = \det \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{bmatrix}$$

where

$$\hat{i} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \hat{j} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \hat{k} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

are the unit direction vectors.

It would always be easier to cofactor the expand the first row of the matrix.

Example 6.3.2. Compute

$$\vec{u} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

Solution

Using the determinant formula,

$$\begin{aligned} \vec{u} \times \vec{v} &= \det \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \\ &= \hat{i} \det \begin{bmatrix} 2 & 3 \\ 5 & 6 \end{bmatrix} - \hat{j} \det \begin{bmatrix} 1 & 3 \\ 4 & 6 \end{bmatrix} + \hat{k} \det \begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix} \\ &= \hat{i}(12 - 15) - \hat{j}(6 - 12) + \hat{k}(5 - 8) \end{aligned}$$

$$= -3\hat{i} + 6\hat{j} - 3\hat{k}$$

Therefore,

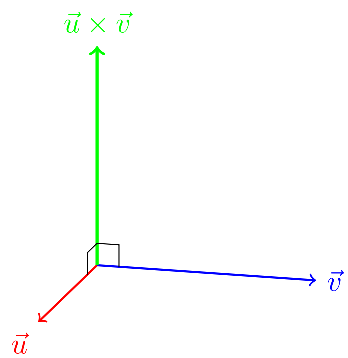
$$\vec{u} \times \vec{v} = \begin{bmatrix} -3 \\ 6 \\ 3 \end{bmatrix}$$

6.3.2 Geometric Interpretation

The cross product has an important geometric interpretation.

Theorem 6.3.3. *The vector $\vec{u} \times \vec{v}$ is perpendicular to both $\vec{u}$ and $\vec{v}$.*

The cross product produces a vector *normal* to the plane containing $\vec{u}$ and $\vec{v}$.



Example 6.3.4. Verify that $\vec{u} \times \vec{v} = \begin{bmatrix} -3 \\ 6 \\ -3 \end{bmatrix}$ is orthogonal to $\vec{u} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$.

Solution

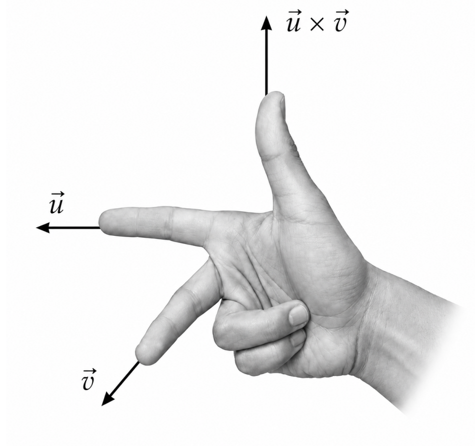
Compute the dot product

$$(\vec{u} \times \vec{v}) \cdot \vec{u} = (-3)(1) + (6)(2) + (-3)(3) = -3 + 12 - 9 = 0$$

6.3.3 The Right-Hand Rule

The cross product has a direction. There are two vectors perpendicular to a given plane: $\hat{n}$ and $-\hat{n}$. The cross product chooses one of these directions using the right-hand rule.

1. Point the fingers of your right hand in the direction of $\vec{u}$.
2. Curl them towards $\vec{v}$.
3. Your thumb points in the direction of $\vec{u} \times \vec{v}$.



6.3.4 Properties of the Cross Product

Theorem 6.3.5. *For vectors $\vec{u}$, $\vec{v}$, and $\vec{w}$ in $\mathbb{R}^3$ and scalar $c \in \mathbb{R}$.*

1. *Anti-Commutative Property:* $\vec{u} \times \vec{v} = -(\vec{v} \times \vec{u})$
2. *Scalar Multiplication:* $(c\vec{u}) \times \vec{v} = c(\vec{u} \times \vec{v})$
3. *Distributive Property:* $\vec{u} \times (\vec{v} + \vec{w}) = \vec{u} \times \vec{v} + \vec{u} \times \vec{w}$

The standard basis vectors satisfies

$$\hat{\imath} \times \hat{\jmath} = \hat{k}, \quad \hat{\jmath} \times \hat{k} = \hat{\imath}, \quad \hat{k} \times \hat{\imath} = \hat{\jmath}$$

Example 6.3.6. $\hat{\jmath} \times \hat{\imath} = -(\hat{\imath} \times \hat{\jmath}) = -\hat{k}$.

6.3.5 Parallel Vectors

The cross product detects whether vectors are parallel.

Theorem 6.3.7. *Two vectors are parallel if and only if*

$$\vec{u} \times \vec{v} = \vec{0}$$

Consequently, if a vector is a scalar multiple of another, then their cross product is the zero vector.

Example 6.3.8. Determine whether $\begin{bmatrix}1\\2\\3\end{bmatrix}$ and $\begin{bmatrix}2\\4\\6\end{bmatrix}$ are parallel.

Solution

Since

$$\begin{bmatrix}2\\4\\6\end{bmatrix} = 2\begin{bmatrix}1\\2\\3\end{bmatrix}$$

the vectors are parallel. Therefore, $\vec{u} \times \vec{v} = \vec{0}$.

6.3.6 Magnitude of the Cross Product

The magnitude of the cross product has an important geometric interpretation.

Theorem 6.3.9. *If θ is the angle between $\vec{u}$ and $\vec{v}$, then*

$$\|\vec{u} \times \vec{v}\| = \|\vec{u}\|\|\vec{v}\|\sin(\theta)$$

The magnitude depends on the length of the vectors and the sine of the angle between them.

6.3.7 Area of a Parallelogram

The magnitude of the cross product gives area.

Theorem 6.3.10. *The area of the parallelogram determined by $\vec{u}$ and $\vec{v}$ is*

$$A = \|\vec{u} \times \vec{v}\|$$

Example 6.3.11. Find the area of the parallelogram determined by $\vec{u} = \begin{bmatrix}1\\0\\0\end{bmatrix}$ and $\vec{v} = \begin{bmatrix}0\\2\\0\end{bmatrix}$.

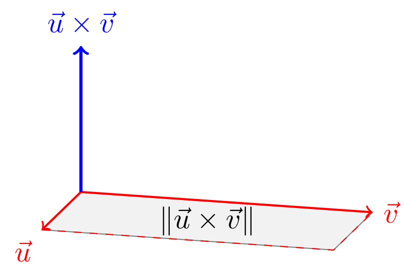
Solution

Compute

$$\vec{u} \times \vec{v} = \det\begin{bmatrix}\hat{i} & \hat{j} & \hat{k}\\1 & 0 & 0\\0 & 0 & 2\end{bmatrix} = \begin{bmatrix}0\\2\\0\end{bmatrix}$$

Thus,

$$A = \|\vec{u} \times \vec{v}\| = 2$$



6.3.8 Area of a Triangle

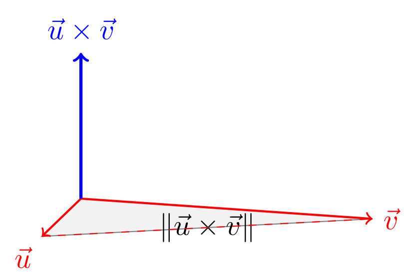
A triangle occupies half of the corresponding parallelogram.

Theorem 6.3.12. *The area of the triangle determined by $\vec{u}$ and $\vec{v}$ is*

$$A = \frac{1}{2}\|\vec{u} \times \vec{v}\|$$

Example 6.3.13. Using the previous example,

$$A = \frac{1}{2}(2) = 1$$



If a triangle has vertices P, Q, and R, then its area is

$$A = \frac{1}{2}\|\overrightarrow{PQ} \times \overrightarrow{PR}\|$$

Example 6.3.14. Find the area of the triangle with vertices $P(0,0,0)$, $Q(2,0,0)$, and $R(0,3,0)$.

Solution

First compute

$$\overrightarrow{PQ} = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}, \quad \overrightarrow{PR} = \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix}$$

Then

$$\overrightarrow{PQ} \times \overrightarrow{PR} = \det \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 0 & 0 \\ 0 & 3 & 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 6 \end{bmatrix}$$

Therefore, the area of the triangle is

$$A = \frac{1}{2}(6) = 3$$

6.3.9 Finding a Normal Vector

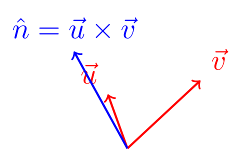
One of the most important applications of the cross product is finding normal vectors to planes.

Example 6.3.15. Find a vector perpendicular to both $\vec{u} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$.

Solution

Compute

$$\hat{n} = \vec{u} \times \vec{v} = \det \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$$



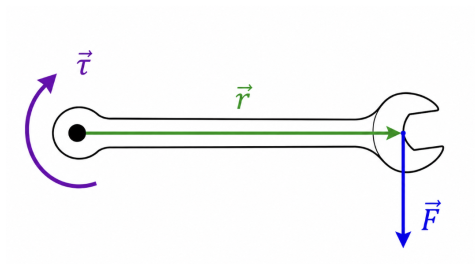
Therefore, the vector is perpendicular to both $\vec{u}$ and $\vec{v}$.

6.3.10 Torque

In physics, *torque* measures the rotational effect of a force. If $\vec{r}$ is the position vector from the axis of rotation to the point where the force is applied, and $\vec{F}$ is the force of the vector, then the torque is

$$\vec{\tau} = \vec{r} \times \vec{F}$$

The direction of $\vec{\tau}$ indicates the axis of rotation, and the magnitude tells us the strength of the rotational effect.



Example 6.3.16. Suppose $\vec{r} = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$ and $\vec{F} = \begin{bmatrix} 0 \\ 5 \\ 0 \end{bmatrix}$. Find the torque.

Solution

Computing

$$\vec{\tau} = \vec{r} \times \vec{F} = \det \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 0 & 0 \\ 0 & 5 & 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 10 \end{bmatrix}$$

The torque points in the positive z direction and has magnitude $\|\vec{\tau}\| = 10$.

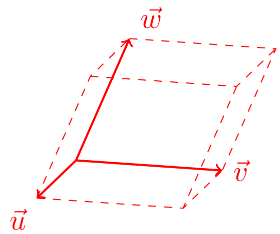
6.3.11 Volume of a Parallelepiped

In two dimensions, the magnitude of the cross product gives the area of a parallelogram. In three dimensions, the cross product can be combined with the dot product to compute volume.

Definition 6.3.17. Let $\vec{u}$, $\vec{v}$, and $\vec{w}$ be vectors in $\mathbb{R}^3$. The *volume of the parallelepiped* determined by these vectors is

$$V = |\vec{u} \cdot (\vec{v} \times \vec{w})|$$

The quantity $u \cdot (\vec{v} \times \vec{w})$ is called the scalar triple product.



The scalar triple product can be computed as

$$\vec{u} \cdot (\vec{v} \times \vec{w}) = \det \begin{bmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{bmatrix}$$

Thus, the volume of a parallelepiped is

$$V = \left| \det \begin{bmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{bmatrix} \right|$$

Example 6.3.18. Find the volume of the parallelepiped determined by

$$\vec{u} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}, \quad \vec{w} = \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix}$$

Solution

Using the determinant formula.

$$V = \left| \det \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} \right| = 6$$

Example 6.3.19. Find the volume of the parallelepiped determined by

$$\vec{u} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad \vec{w} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$$

Solution

Using the determinant formula

$$V = \left| \det \begin{bmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{bmatrix} \right| = \left| \det \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} - 2 \det \begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix} \right| = |(1)(1) - (2)(-2)| = 5$$

6.3.12 Coplanar Vectors

A consequence of the scalar triple product is that three vectors are called *coplanar* if and only if $\vec{u} \cdot (\vec{v} \times \vec{w}) = 0$. Equivalently,

$$\det \begin{bmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{bmatrix} = 0$$

Example 6.3.20. Determine whether

$$\vec{u} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \vec{w} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

are coplanar.

Solution

Compute

$$\vec{u} \cdot (\vec{v} \times \vec{w}) = \det \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} = 0$$

Therefore, the vectors are coplanar.

6.4 Parametric Lines

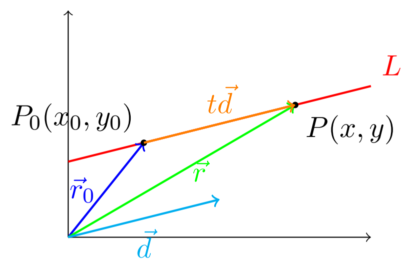
In analytic geometry, one of the most fundamental geometric objects is a line.

In two dimensions, we often describe a line using equations such as $y = mx + b$ or $Ax + By = C$. While these equations are useful in $\mathbb{R}^2$, they become less convenient in higher dimensions.

Linear algebra provides a more powerful approach using vectors. Instead of describing a line by a slope, we describe it using a point on the line and a direction vector. This approach extends naturally to three dimensions and beyond.

6.4.1 The Geometry of a Line

Suppose a line passes through a point $P(x_0, y_0)$ and has direction vector $\vec{d} = \begin{bmatrix} a \\ b \end{bmatrix}$. Every point on the line can be obtained by starting at P and moving some multiple of the direction vector.



Starting from point P, move in the direction of $t\vec{d}$. From here, we get the resulting point $P + t\vec{d}$. The parameter t determines how far we move along the line.

6.4.2 Vector Equation of a Line

Definition 6.4.1. Let $\vec{r}_0$ be the position vector of a point on a line and let $\vec{d}$ be a direction vector. The *vector equation of the line* is

$$\vec{r} = \vec{r}_0 + t\vec{d}$$

where $t \in \mathbb{R}$.

Here, $\vec{r}_0$ fixes the location of the line, $\vec{d}$ determines the direction, and t determines the position along the line.

Example 6.4.2. Find the vector equation of the line passing through $(1, 2)$ with direction vector $\begin{bmatrix} 3 \\ -1 \end{bmatrix}$.

Solution

The position vector of the point is $\vec{r}_0 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and so

$$\vec{r} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} + t \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

6.4.3 Parametric Equations

Expanding the vector equation component wise gives the parametric equations.

Definition 6.4.3. If $\vec{r} = \vec{r}_0 + t\vec{d}$, then the corresponding *parametric equations* are obtained by equating components.

Example 6.4.4. Convert

$$\vec{r} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} + t \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

to parametric form.

Solution

Equating components,

$$\begin{aligned} x &= 1 + 3t, \\ y &= 2 - t \end{aligned}$$

The same idea works in three dimensions.

Example 6.4.5. Find the parametric equations for

$$\vec{r} = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix} + t \begin{bmatrix} 1 \\ 3 \\ -2 \end{bmatrix}$$

Solution

Equating the components, we have

$$\begin{aligned} x &= 2 + t, \\ y &= -1 + 3t, \\ z &= 4 - 2t \end{aligned}$$

6.4.4 Finding a Direction Vector from Two Points

Often, we know two points on a line.

Example 6.4.6. Find the equation of the line through $P(1, 2, 3)$ and $Q(4, 0, 5)$.

Solution

Find the direction vector

$$\overrightarrow{PQ} = Q - P = \begin{bmatrix} 3 \\ -2 \\ 2 \end{bmatrix}$$

Then use the vector equation at point P

$$\vec{r} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + t \begin{bmatrix} 3 \\ -2 \\ 2 \end{bmatrix}$$

Thus, the parametric equations come from equating components

$$\begin{aligned} x &= 1 + 3t, \\ y &= 2 - 2t, \\ z &= 3 + 2t \end{aligned}$$

6.4.5 Determining Whether a Point Lies on a Line

To determine whether a point lies on a line, we first substitute the coordinates into the parametric equations, and check whether the same value of t satisfies every equation.

Example 6.4.7. Determine whether $(7, 1)$ lies on

$$\begin{aligned} x &= 1 + 3t, \\ y &= 2 - t \end{aligned}$$

Solution

Using the x-equation,

$$\begin{aligned} 7 &= 1 + 3t \\ t &= 2 \end{aligned}$$

Substitute into the y-equation

$$2 - 2 = 0 \neq 1$$

But the point has $y = 1$. Therefore, $(7, 1)$ is not on the line.

6.4.6 Symmetric Equations of a Line

Sometimes the parameter can be eliminated.

Definition 6.4.8. If

$$\begin{aligned}x &= x_0 + ta,\\ y &= y_0 + tb,\\ z &= z_0 + tc\end{aligned}$$

then

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$$

is called the *symmetric form of the line.*

Example 6.4.9. Convert

$$\begin{aligned}x &= 1 + 2t,\\ y &= 3 - t,\\ z &= 4 + 5t\end{aligned}$$

to symmetric form.

Solution

Solve each equation for t

$$t = \frac{x-1}{2}, \quad t = 3 - y, \quad \frac{z-4}{5}$$

Then

$$\frac{x-1}{2} = 3 - y = \frac{z-4}{5}$$

6.4.7 Intersection of Two Lines

Two lines intersect if they share a common point. To find intersections, write the parametric equations, set corresponding coordinates equal, and then solve the resulting system.

Example 6.4.10. Determine whether the lines

$$L_1 : \begin{cases} x = 1 + t, \\ y = 2 + t \end{cases}, \quad L_2 : \begin{cases} x = 3 - s, \\ y = s \end{cases}$$

intersect. If so, find the point.

Solution

Set the coordinates equal to each other

$$\begin{aligned} 1+t &= 3-s, \\ 2+t &= s \end{aligned}$$

Substitute the second equation into the first,

$$\begin{aligned} 1+t &= 3-(2+t) \\ 1+t &= 1-t \\ 2t &= 0 \\ t &= 0 \end{aligned}$$

Then $s = 2$. Thus, the lines intersect. The point of intersection is $(1, 2)$.

6.4.8 Parallel Lines

Two lines are parallel whenever they share the same direction vector but different initial point.

Example 6.4.11. Find a vector equation of the line passing through $(2, -1, 3)$ and parallel to

$$\vec{r} = \begin{bmatrix} 1 \\ 4 \\ 0 \end{bmatrix} + t \begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix}$$

Solution

Since the lines are parallel, they have the same direction vector.

$$\vec{r} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} + t \begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix}$$

Example 6.4.12. Determine whether the two lines are parallel

$$L_1 : \vec{r}_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + t \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix},$$

$$L_2 : \vec{r}_2 = \begin{bmatrix} 3 \\ 1 \\ 5 \end{bmatrix} + s \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Solution

Since

$$\begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

the direction vectors are parallel. Therefore, the lines are parallel.

6.5 Planes in $\mathbb{R}^3$, Hyperplanes in $\mathbb{R}^n$

In the previous section, we studied lines and learned how to describe them using vector and parametric equations. A line is a one-dimensional geometric object determined by a point and a direction.

In three-dimensional space, another important geometric object is a plane. A plane can be thought of as a flat two-dimensional surface extending infinitely in all directions within the plane.

Just as a line can be described using a point and a direction vector, a plane can be described using a point on the plane, and a vector perpendicular to the plane. This perpendicular is called a *normal vector*.

Planes are closely connected to systems of linear equations. In fact, many solution sets of linear systems can be interpreted geometrically as planes or intersections of planes.

6.5.1 Normal Vectors

The key to describing a plane is a normal vector.

Definition 6.5.1. A vector $\hat{n}$ is called a *normal vector* to a plane if it is perpendicular to every vector lying in the plane.

Example 6.5.2. The vector

$$\hat{n} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

may serve as a normal vector for some plane. Every direction vector in the plane must satisfy

$$\hat{n} \cdot \vec{d} = 0$$

6.5.2 Equation of a Plane

Suppose a plane passes through the point $P_0(x_0, y_0, z_0)$ and has normal vector $\hat{n} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$.

Let $P(x, y, z)$ be any point on the plane. Then

$$\overrightarrow{P_0P} = P - P_0 = \begin{bmatrix} x - x_0 \\ y - y_0 \\ z - z_0 \end{bmatrix}$$

lies in the plane. Since the normal vector is perpendicular to every vector in the plane,

$$\hat{n} \cdot \overrightarrow{P_0P} = 0$$

Thus,

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

Definition 6.5.3. The *Cartesian equation* of a plane with normal vector $\hat{n} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$ through the point (x_0, y_0, z_0) is

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

Expanding the equation produces the standard form.

Definition 6.5.4. The *standard equation of a plane* is

$$ax + by + cz = d$$

where $\hat{n} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$ is the normal vector.

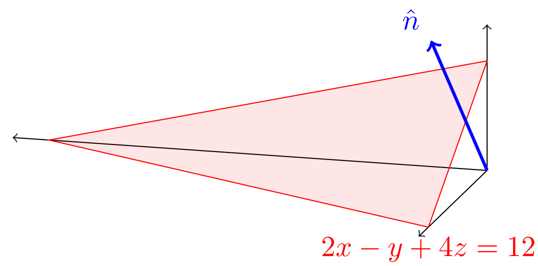
Example 6.5.5. Find the equation of the plane passing through $(1, 2, 3)$ with normal vector $\hat{n} = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}$.

Solution

Using the Cartesian equation of the plane, we have

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

$$\begin{aligned} 2(x-1)-(y-2)+4(z-3)&=0\\ 2x-y+4z-12&=0\\ 2x-y+4z&=12 \end{aligned}$$



6.5.3 Determining Whether a Point Lies on a Plane

Substitute the coordinates into the equation.

Example 6.5.6. Determine whether $(2, 0, 2)$ lies on

$$2x - y + 4z = 12$$

Solution

Substitute

$$2(2) - 0 + 4(2) = 4 + 8 = 12$$

Therefore, $(2, 0, 2)$ lies on the plane.

6.5.4 Finding a Plane Through Three Points

Three noncollinear points determine a unique plane. We first form two direction vectors. Compute their cross product, and then use the resulting normal vector in the plane equation.

Example 6.5.7. Find the equation of the plane through $P(1, 0, 0)$, $Q(0, 1, 0)$, and $R(0, 0, 1)$.

Solution

Construct the direction vectors

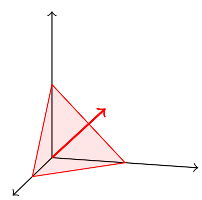
$$\overrightarrow{PQ} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \quad \overrightarrow{PR} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

Then compute their cross product to get the normal vector

$$\hat{n} = \overrightarrow{PQ} \times \overrightarrow{PR} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Using point P

$$\begin{aligned} (x-1) + y + z &= 0 \\ x + y + z &= 1 \end{aligned}$$



6.5.5 Parallel Planes

Two planes are parallel if their normal vectors are parallel.

Theorem 6.5.8. *The planes*

$$\begin{aligned} \pi_1 &: a_1x + b_1y + c_1z = d_1, \\ \pi_2 &: a_2x + b_2y + c_2z = d_2, \end{aligned}$$

are parallel if

$$\begin{bmatrix} a_1 \\ b_1 \\ c_1 \end{bmatrix} = k \begin{bmatrix} a_2 \\ b_2 \\ c_2 \end{bmatrix}$$

for some scalar k.

Example 6.5.9. Determine whether

$$\begin{aligned} \pi_1 &: 2x - y + 4z = 7, \\ \pi_2 &: 4x - 2y + 8z = 3 \end{aligned}$$

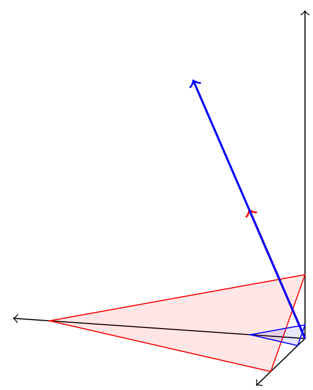
are parallel.

Solution

The normal vectors are

$$\hat{n}_1 = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}, \quad \hat{n}_2 = \begin{bmatrix} 4 \\ -2 \\ 8 \end{bmatrix}$$

Since the second is twice the first, the planes are parallel.



6.5.6 Angle Between Two Planes

The angle between two planes is defined as the angle between their normal vectors.

Theorem 6.5.10. *If $\hat{n}_1$ and $\hat{n}_2$ are normal vectors, then*

$$\cos(\theta) = \frac{\hat{n}_1 \cdot \hat{n}_2}{\|\hat{n}_1\|\|\hat{n}_2\|}$$

Example 6.5.11. Find the angle between

$$\pi_1 : x + y + z = 1,$$
$$\pi_2 : x - y + z = 2$$

Solution

The normal vectors are

$$\hat{n}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad \hat{n}_2 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

Computing their dot product and lengths,

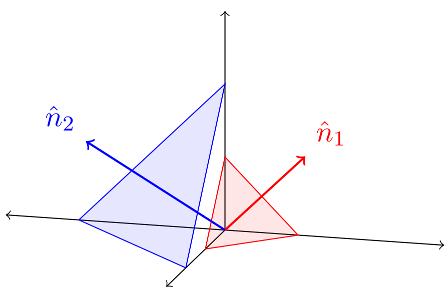
$$\begin{aligned}\hat{n}_1 \cdot \hat{n}_2 &= (1)(1) + (1)(-1) + (1)(1) = 1 - 1 + 1 = 1,\\ \|\hat{n}_1\| &= \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3},\\ \|\hat{n}_2\| &= \sqrt{1^2 + (-1)^2 + 1^2} = \sqrt{3}\end{aligned}$$

Therefore,

$$\cos(\theta) = \frac{1}{3}$$

Thus,

$$\theta = \cos^{-1}\left(\frac{1}{3}\right) \approx 70.5^\circ$$



6.5.7 Distance From a Point to a Plane

This is one of the most important applications of vector geometry.

Theorem 6.5.12. *The distance from $P(x_0, y_0, z_0)$ to the plane $\pi : ax + by + cz = d$ is*

$$D = d(P, \pi) = \frac{|ax_0 + by_0 + cz_0 - d|}{\sqrt{a^2 + b^2 + c^2}}$$

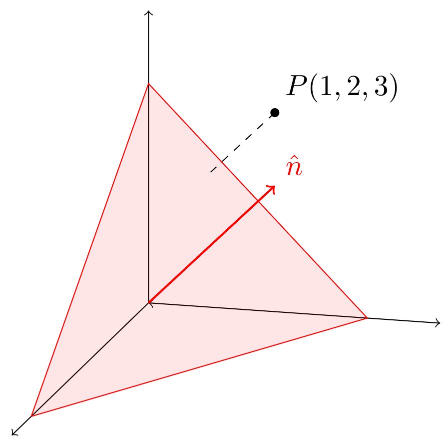
Example 6.5.13. Find the distance from $(1, 2, 3)$ to the plane

$$\pi : x + 2y + 2z = 6$$

Solution

Substitute

$$D = \frac{|1 + 4 + 6 - 6|}{\sqrt{1 + 4 + 4}} = \frac{5}{3}$$



Another example involving distance from a point to the plane is finding the closest point Q to the point P.

Example 6.5.14. Find a point Q on the plane

$$\pi : x + 2y + 2z = 6$$

closest to P.

Solution

The plane has normal vector $\hat{n} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$, so the shortest distance from P to the plane must be parallel to this normal vector. So the line through P perpendicular to the plane is

$$\vec{r} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + t \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

Therefore, the parametric equations are

$$x = 1 + t,$$
$$y = 2 + 2t,$$
$$z = 3 + 2t$$

Substitute into the plane equation

$$(1 + t) + 2(2 + 2t) + 2(3 + 2t) = 6$$

$$\begin{aligned} 1+t+4+4t+6+4t &= 6 \\ 11+9t &= 6 \\ 9t &= -5 \\ t &= -\frac{5}{9} \end{aligned}$$

Now substitute this back into the line

$$\begin{aligned} x &= 1-\frac{5}{9} = \frac{4}{9}, \\ y &= 2+2\left(-\frac{5}{9}\right) = \frac{8}{9}, \\ z &= 3+2\left(-\frac{5}{9}\right) = \frac{17}{9} \end{aligned}$$

Therefore, the closest point on the plane is $Q\left(\frac{4}{9}, \frac{8}{9}, \frac{17}{9}\right)$.

6.5.8 Angle Between a Line and a Plane

Suppose a line has direction vector $\vec{d}$ and a plane has normal vector $\hat{n}$.

Theorem 6.5.15. *Let ϕ be the angle between the line and the plane. Then*

$$\sin(\phi) = \frac{|\vec{d}\cdot\hat{n}|}{\|\hat{d}\|\|\hat{n}\|}$$

Example 6.5.16. Find the angle between the line

$$\vec{r} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + t\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

and the plane

$$x+y+z=0$$

Solution

Both direction and normal vectors are $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. Thus, $\sin(\phi)=1$. Hence, $\phi = 90°$. This line is perpendicular to the plane.

6.5.9 Hyperplanes in $\mathbb{R}^n$

Planes extend naturally to higher dimensions.

Definition 6.5.17. A *hyperplane* in $\mathbb{R}^n$ is the set of solutions to a linear equation

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = d$$

Example 6.5.18. In $\mathbb{R}^2$, we have

$$ax + by = d$$

which is a line.

In $\mathbb{R}^3$, we have

$$ax + by + cz = d$$

which is a plane.

In $\mathbb{R}^4$, we have

$$a_1x_1 + a_2x_2 + a_3x_3 + a_4x_4 = d$$

which is a hyperplane.

Chapter 7

Vector Space $\mathbb{R}^n$

In the previous chapters, we studied systems of linear equations, matrices, linear transformations, and determinants. These topics provided powerful computational tools for solving problems and analyzing linear systems. However, many of the underlying ideas of linear algebra involve understanding the structure of the sets in which vectors live and how vectors relate to one another.

This chapter introduces the vector space $\mathbb{R}^n$, one of the most important objects in linear algebra.

Although vectors are often visualized as arrows in two or three dimensions, the ideas of linear algebra extend naturally to higher-dimensional spaces. The vector space $\mathbb{R}^n$ provides a unified framework for studying vectors in any finite dimension.

7.1 Vectors in $\mathbb{R}^n$

Throughout this course, we have worked extensively with matrices, systems of equations, and linear transformations. In many of these topics, vectors naturally appeared as columns of numbers representing solutions, coordinates, or inputs to matrix transformations.

In this section, we formally introduce vectors in $\mathbb{R}^n$, the fundamental objects of study in linear algebra.

7.1.1 Vectors in $\mathbb{R}^n$

Definition 7.1.1. A *vector* in $\mathbb{R}^n$ is an ordered list of n real numbers. We usually write a vector as a column matrix

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

The numbers $x_1, x_2, ..., x_n$ are called the *components* or *entries* of the vector.

Some familiar examples include those from $\mathbb{R}^2$, $\mathbb{R}^3$, or higher.

$$\begin{bmatrix} 3 \\ -1 \end{bmatrix}, \quad \begin{bmatrix} 2 \\ 5 \\ -4 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 0 \\ 7 \\ -2 \\ 3 \end{bmatrix}$$

Definition 7.1.2. The set of all vectors with n components is called $\mathbb{R}^n$. Thus,

$$\mathbb{R}^2 = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} : x, y \in \mathbb{R} \right\}$$

and

$$\mathbb{R}^3 = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} : x, y, z \in \mathbb{R} \right\}$$

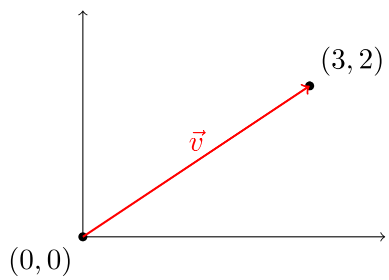
7.1.2 Geometric Interpretation

In low dimensions, vectors may be interpreted geometrically.

Example 7.1.3. A vector $\begin{bmatrix} x \\ y \end{bmatrix}$ may be viewed as a point (x, y), or an arrow from the origin to (x, y). For example, the vector

$$\vec{v} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

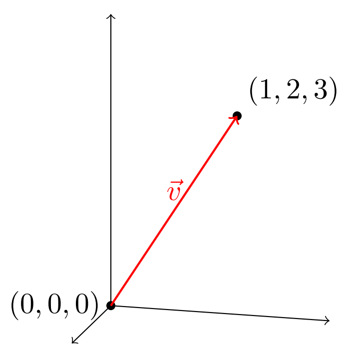
corresponds to the point $(3, 2)$. Geometrically, this vector is represented by an arrow beginning at the origin and ending at $(3, 2)$.



Example 7.1.4. A vector $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ may be viewed as a point or directed line segment in three-dimensional space. For example, the vector

$$\vec{v} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

corresponds to the point $(1, 2, 3)$.



Although we can visualize $\mathbb{R}^2$ and $\mathbb{R}^3$, the algebra developed in this chapter applies equally well to $\mathbb{R}^{10}$, $\mathbb{R}^{100}$, or any higher-dimensional space.

7.1.3 Equality of Vectors

Definition 7.1.5. Two vectors are equal if and only if their corresponding components are equal. That is,

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

if and only if $x_i = y_i$ for every i.

Example 7.1.6. The vectors

$$\begin{bmatrix}1\\2\\3\end{bmatrix} = \begin{bmatrix}1\\2\\3\end{bmatrix}$$

are equal since every component agrees.

Example 7.1.7. The vectors

$$\begin{bmatrix}1\\3\\2\end{bmatrix} \neq \begin{bmatrix}1\\2\\3\end{bmatrix}$$

differ in the second and third components, so they are not equal.

7.1.4 Vector Addition

Vectors of the same dimension may be added componentwise.

Definition 7.1.8. Let $\vec{x} = \begin{bmatrix}x_1\\x_2\\\vdots\\x_n\end{bmatrix}$, $\vec{y} = \begin{bmatrix}y_1\\y_2\\\vdots\\y_n\end{bmatrix}$. Then we define vector addition as

$$\vec{x} + \vec{y} = \begin{bmatrix}x_1 + y_1\\x_2 + y_2\\\vdots\\x_n + y_n\end{bmatrix}$$

Example 7.1.9. Compute

$$\begin{bmatrix}1\\2\end{bmatrix} + \begin{bmatrix}3\\4\end{bmatrix}$$

Solution

Add corresponding components

$$\begin{bmatrix}1\\2\end{bmatrix} + \begin{bmatrix}3\\4\end{bmatrix} = \begin{bmatrix}4\\6\end{bmatrix}$$

Example 7.1.10. Compute

$$\begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix} + \begin{bmatrix} 3 \\ 4 \\ -2 \end{bmatrix}$$

Solution

Add corresponding components

$$\begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix} + \begin{bmatrix} 3 \\ 4 \\ -2 \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \\ 3 \end{bmatrix}$$

Vector addition can be interpreted geometrically. If vectors are represented as arrows, place the tail of the second vector at the head of the first. Then the resulting vector extends from the starting point to the final endpoint. This is called the *head-to-tail rule*.

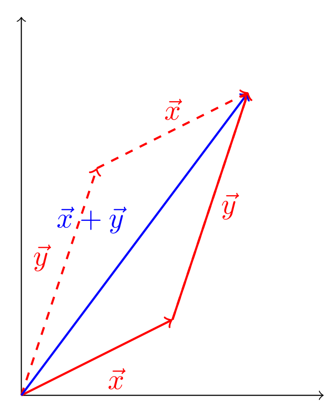
Example 7.1.11. Let

$$\vec{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \vec{y} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

Then

$$\vec{x} + \vec{y} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

Geometrically, the two arrows combine to produce a new vector.



7.1.5 Scalar Multiplication

A vector may be multiplied by a real number.

Definition 7.1.12. Let $c \in \mathbb{R}$ and

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

Then

$$c\vec{x} = \begin{bmatrix} cx_1 \\ cx_2 \\ \vdots \\ cx_n \end{bmatrix}$$

Example 7.1.13. Compute

$$3 \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

Solution

$$3 \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 6 \\ -3 \end{bmatrix}$$

Example 7.1.14. Compute

$$-\frac{1}{2} \begin{bmatrix} 4 \\ 6 \end{bmatrix}$$

Solution

$$-\frac{1}{2} \begin{bmatrix} 4 \\ 6 \end{bmatrix} = \begin{bmatrix} -2 \\ -3 \end{bmatrix}$$

Multiply by a scalar changes the length and possibly the direction of a vector.

- If $c > 0$, the direction remains unchanged.
- If $c < 0$, the direction reverses.

7.1.6 The Zero Vector

Definition 7.1.15. The *zero vector* in $\mathbb{R}^n$ is

$$\vec{0} := \vec{0}_{\mathbb{R}^n} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Familiar examples come from the zero vector in $\mathbb{R}^2$ and $\mathbb{R}^3$

$$\vec{0}_{\mathbb{R}^2} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \vec{0}_{\mathbb{R}^3} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Theorem 7.1.16. *For every vector $\vec{x} \in \mathbb{R}^n$,*

$$\vec{x} + \vec{0}_{\mathbb{R}^n} = \vec{x}$$

Definition 7.1.17. The negative of $\vec{x}$ is the vector

$$-\vec{x} = (-1)\vec{x}$$

Example 7.1.18.

$$-\begin{bmatrix} 2 \\ -3 \end{bmatrix} = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$$

Theorem 7.1.19. *For every vector $\vec{x} \in \mathbb{R}^n$, there exists $-\vec{x} \in \mathbb{R}^n$ such that*

$$\vec{x} + (-\vec{x}) = \vec{0}_{\mathbb{R}^n}$$

7.1.7 Algebraic Properties of Vectors

Theorem 7.1.20. *For vectors $\vec{x}, \vec{y}, \vec{z} \in \mathbb{R}^n$ and scalars $a, b \in \mathbb{R}$, the following properties hold:*

1. *Commutative Property:* $\vec{x} + \vec{y} = \vec{y} + \vec{x}$
2. *Associative Property:* $(\vec{x} + \vec{y}) + \vec{z} = \vec{x} + (\vec{y} + \vec{z})$
3. *Additive Identity:* $\vec{x} + \vec{0}_{\mathbb{R}^n} = \vec{x}$
4. *Additive Inverse:* $\vec{x} + (-\vec{x}) = \vec{0}_{\mathbb{R}^n}$
5. *Scalar Associativity:* $a(b\vec{x}) = (ab)\vec{x}$
6. *Scalar Identity:* $1\vec{x} = \vec{x}$
7. *Distributive Property:* $a(\vec{x} + \vec{y}) = a\vec{x} + a\vec{y}$ *and* $(a + b)\vec{x} = a\vec{x} + b\vec{x}$

7.1.8 Standard Basis Vectors

Recall that in $\mathbb{R}^2$ and $\mathbb{R}^3$, we have defined the standard bases vectors.

Example 7.1.21. In $\mathbb{R}^2$, the standard basis vectors are

$$\vec{e}_1 = \begin{bmatrix}1\\0\end{bmatrix}, \quad \vec{e}_2 = \begin{bmatrix}0\\1\end{bmatrix}$$

and in $\mathbb{R}^3$, the standard basis vectors are

$$\vec{e}_1 = \begin{bmatrix}1\\0\\0\end{bmatrix}, \quad \vec{e}_2 = \begin{bmatrix}0\\1\\0\end{bmatrix}, \quad \vec{e}_3 = \begin{bmatrix}0\\0\\1\end{bmatrix}$$

Using the standard basis vectors, we are also able to express a vector as a linear combination of the standard basis vectors.

Example 7.1.22.

$$\begin{bmatrix}3\\5\\2\end{bmatrix} = 3\vec{e}_1 + 5\vec{e}_2 + 2\vec{e}_3$$

Definition 7.1.23. The *standard basis vectors* of $\mathbb{R}^n$ are $\vec{e}_1, \vec{e}_2, ..., \vec{e}_n$ where each vector contains a single 1 and all remaining entries are 0.

7.2 Linear Combinations and Span

In Section 7.1, we introduced vectors in $\mathbb{R}^n$ and studied the basic operations of vector addition and scalar multiplication.

One of the central ideas of linear algebra is that complicated vectors can often be constructed from simpler vectors using these operations.

For example, consider the vectors

$$\vec{u} = \begin{bmatrix}1\\0\end{bmatrix}, \quad \vec{v} = \begin{bmatrix}0\\1\end{bmatrix}$$

By combining these vectors using scalar multiplication and addition, we can produce any vector in $\mathbb{R}^2$

$$\begin{bmatrix}x\\y\end{bmatrix} = x\vec{u} + y\vec{v}$$

This observation leads to two fundamental concepts.

The first is linear combinations, which describe how vectors can be built from other vectors. The second is span, which describes the entire collection of vectors that can be generated from a given set.

These ideas form the foundation for the concepts of linear independence, bases, and dimension that will be developed in the later chapter.

7.2.1 Linear Combinations

We saw in the previous section that we can write any vector in $\mathbb{R}^n$ as a linear combination of the standard basis vectors. However, it could also be possible that the vector could have a different linear combination.

Definition 7.2.1. Let $\vec{v}_1, \vec{v}_2, ..., \vec{v}_k \in \mathbb{R}^n$. vector of the form

$$c_1\vec{v}_2 + c_2\vec{v}_2 + \cdots + c_k\vec{v}_k$$

where $c_1, c_2, ..., c_k \in \mathbb{R}$ is called a *linear combination* of $\vec{v}_1, \vec{v}_2, ..., \vec{v}_k$.

A linear combination is obtained by multiplying vectors by scalars, and adding the results together.

Example 7.2.2. Let $\vec{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$. Find $2\vec{u} - \vec{v}$.

Solution

We compute

$$2\vec{u} - \vec{v} = 2\begin{bmatrix} 1 \\ 2 \end{bmatrix} - \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix} - \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ 5 \end{bmatrix}$$

Example 7.2.3. Let $\vec{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$ and $\vec{v}_2 = \begin{bmatrix} 3 \\ 1 \\ -1 \end{bmatrix}$. Find $3\vec{v}_1 - 2\vec{v}_2$.

Solution

Compute

$$3\vec{v}_1 - 2\vec{v}_2 = 3\begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} - 2\begin{bmatrix} 3 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 6 \end{bmatrix} - \begin{bmatrix} 6 \\ 2 \\ -2 \end{bmatrix} = \begin{bmatrix} -3 \\ -2 \\ 8 \end{bmatrix}$$

A common problem is determining whether a given vector can be constructed from other vectors.

Example 7.2.4. Determine whether $\vec{b} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$ is a linear combination of $\vec{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\vec{v}_2 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$.

Solution

Assume $\vec{b} = c_1\vec{v}_1 + c_2\vec{v}_2$. Then

$$\begin{bmatrix} 5 \\ 1 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

Equate the components

$$\begin{aligned} c_1 + 2c_1 &= 5, \\ 2c_1 - c_2 &= 1 \end{aligned}$$

We solve the system. From the second equation, we have

$$c_2 = 2c_1 - 1$$

Then substitute into the first equation to get

$$\begin{aligned} c + 1 + 2(2c_1 - 1) &= 5 \\ 5c_1 &= 7 \\ c_1 &= \frac{7}{5} \end{aligned}$$

Therefore,

$$c_2 = \frac{9}{5}$$

Since a solution exists, $\vec{b}$ is a linear combination of $\vec{v}_1$ and $\vec{v}_2$. In particular,

$$\vec{b} = \frac{7}{5}\vec{v}_1 + \frac{9}{5}\vec{v}_2$$

7.2.2 Span

Linear combinations lead naturally into the concept of span.

Definition 7.2.5. Let $\vec{v}_1, \vec{v}_2, ..., \vec{v}_k \in \mathbb{R}^n$. The *span* of these vectors is the set of all possible linear combinations and is denoted by

$$\text{span}\{\vec{v}_1, \vec{v}_2, ..., \vec{v}_k\} = \{c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_k\vec{v}_k : c_i \in \mathbb{R}\}$$

The span is the collection of all vectors that can be generated from the given vectors.

Example 7.2.6. Consider $\vec{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. Then

$$\text{span}\{\vec{v}\} = \left\{ c \begin{bmatrix} 2 \\ 1 \end{bmatrix} : c \in \mathbb{R} \right\}$$

All multiples of a single nonzero vector form a line through the origin. Thus, span$\{\vec{v}\}$ is a line.

Example 7.2.7. Consider $\vec{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\vec{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. Then

$$c_1\vec{v}_1 + c_2\vec{v}_2 = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

Since every vector in $\mathbb{R}^2$ has this form,

$$\text{span}\{\vec{v}_1, \vec{v}_2\} = \mathbb{R}^2$$

Two vectors pointing in different directions can generate the entire plane.

Example 7.2.8. Consider the standard basis vectors $\vec{e}_1$, $\vec{e}_2$, and $\vec{e}_3$. Then

$$x\vec{e}_1 + y\vec{e}_2 + z\vec{e}_3 = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Therefore,

$$\text{span}\{\vec{e}_1, \vec{e}_2, \vec{e}_3\} = \mathbb{R}^3$$

The span of vectors may produce different geometric objects.

- One nonzero vector: produces a line through the origin.
- Two nonparallel vectors in $\mathbb{R}^2$: produces the entire plane.
- Two nonparallel vectors in $\mathbb{R}^3$: produces a plane through the origin.
- Three suitable vectors in $\mathbb{R}^3$: produces all of $\mathbb{R}^3$.

7.2.3 Spanning Sets

Definition 7.2.9. A collection of vectors $\vec{v}_1, \vec{v}_2, ..., \vec{v}_k \in \mathbb{R}^n$ is called a *spanning set* for a set S if

$$\text{span}\{\vec{v}_1, \vec{v}_2, ..., \vec{v}_k\} = S$$

Example 7.2.10. The vectors $\begin{bmatrix}1\\0\end{bmatrix}$ and $\begin{bmatrix}0\\1\end{bmatrix}$ form a spanning set for $\mathbb{R}^2$.

Determining whether $\vec{b}$ belongs to $\text{span}\{\vec{v}_1, \vec{v}_2, ..., \vec{v}_k\}$ is equivalent to determining whether the system

$$c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_k\vec{v}_k = \vec{b}$$

has a solution. Equivalently, if

$$A = \begin{bmatrix} \uparrow & \uparrow & & \uparrow \\ \vec{v}_1 & \vec{v}_2 & \cdots & \vec{v}_k \\ \downarrow & \downarrow & & \downarrow \end{bmatrix}$$

then $\vec{b}$ belongs to the span if and only if $A\vec{x} = \vec{b}$ has a solution.

Example 7.2.11. Determine whether $\begin{bmatrix}1\\2\\3\end{bmatrix}$ belongs to $\text{span}\left\{\begin{bmatrix}1\\0\\1\end{bmatrix}, \begin{bmatrix}0\\1\\1\end{bmatrix}\right\}$.

Solution

Set up the linear combination

$$c_1\begin{bmatrix}1\\0\\1\end{bmatrix} + c_2\begin{bmatrix}0\\1\\1\end{bmatrix} = \begin{bmatrix}1\\2\\3\end{bmatrix}$$

This gives a system of equations

$$\begin{aligned} c_1 &= 1, \\ c_2 &= 2 \end{aligned}$$

and $c_1 + c_2 = 3$. The system is consistent. Therefore, the vector belongs to the span.

7.3 Linear Independence

In Section 7.2, we studied linear combinations and spanning sets. We learned that a collection of vectors can be used to generate many other vectors through linear combinations.

However, an important question remains: Are all of the vectors in a spanning set actually necessary?

For example, consider the vectors

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \vec{v}_2 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

Since $\vec{v}_2 = 2\vec{v}_1$, the second vector provides no new information. Any vector that can be generated using $\vec{v}_2$ can already be generated using $\vec{v}_1$.

This motivates one of the most important concepts in linear algebra: linear independence.

Linear independence allows us to determine whether a collection of vectors contains redundant information. It plays a central role in the study of vector spaces and leads naturally to the concept of bases and dimension.

7.3.1 Definition of Linear Independence

Definition 7.3.1. The vectors $\vec{v}_1, \vec{v}_2, ..., \vec{v}_k$ are called *linearly independent* if the equation

$$c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_k\vec{v}_k = \vec{0}$$

has only the trivial solution $c_1 = c_2 = \cdots = c_k = 0$.

The vectors are said to be *linearly dependent* if there exists a nontrivial solution $(c_1, c_2, ..., c_k) \neq (0, 0, ..., 0)$ such that

$$c_1\vec{v}_1 + c_2\vec{v}_2 + \cdots + c_k\vec{v}_k = \vec{0}$$

The idea of independence means that the only way to obtain the zero vector is by using all zero coefficients, while the idea of dependence means that the zero vector can be obtained using some nonzero coefficient.

Example 7.3.2. Determine whether $\vec{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\vec{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ are linearly independent.

Solution

Consider

$$c_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Thus, $c_1 = 0$ and $c_2 = 0$. The trivial solution is the only solution. Therefore, the vectors are linearly independent.

Example 7.3.3. Determine whether $\vec{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\vec{v}_2 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$ are linearly independent.

Solution

Consider

$$c_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} c_1 + 2c_2 \\ 2c_1 + 4c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This gives $c_1 + 2c_2 = 0$. Choosing $c_1 = 1$ and $c_2 = -2$, then

$$(-2)\vec{v}_1 + \vec{v}_2 = \vec{0}$$

Since a nontrivial solution exists, the vectors are linearly dependent.

Linear independence has a simple geometric meaning.

- In $\mathbb{R}^2$, two vectors are linearly independent if they do not lie on the same line through the origin.

Example 7.3.4. The vectors $\vec{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\vec{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ point in different directions. Therefore, they are independent.

Example 7.3.5. The vectors $\vec{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\vec{v}_2 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$ lie on the same line. Therefore, they are dependent.

One of the most useful interpretations is the following.

Theorem 7.3.6. *A collection of vectors is linearly dependent if and only if one of the vectors can be written as a linear combination of the others.*

Example 7.3.7. Consider $\vec{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\vec{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and $\vec{v}_3 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Observe that

$$\vec{v}_3 = \vec{v}_1 + \vec{v}_2$$

Thus, the set is linearly dependent. The third vector is redundant.

Testing independence can be converted into a system of equations.

Theorem 7.3.8. *The vectors $\vec{v}_1, \vec{v}_2, ..., \vec{v}_k$ are linearly independent if and only if the homogeneous system $A\vec{x} = \vec{0}$ has only the trivial solution, where*

$$A = \begin{bmatrix} \uparrow & \uparrow & & \uparrow \\ \vec{v}_1 & \vec{v}_2 & \cdots & \vec{v}_k \\ \downarrow & \downarrow & & \downarrow \end{bmatrix}$$

To test for independence,

- Place the vectors as columns of a matrix.
- Row reduce.
- Solve the homogeneous system.
- Determine whether nontrivial solutions exist.

Example 7.3.9. Determine whether

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \quad \vec{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad \vec{v}_3 = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}$$

are linearly independent.

Solution

First form the coefficient matrix

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & 1 & 2 \end{bmatrix}$$

and row reduce to get

$$\begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & 1 & 2 \end{bmatrix} \to \cdots \to \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

There are only two pivots, so there is a free variable. Hence, the homogeneous system has nontrivial solutions. Thus, the vectors are linearly dependent.

7.3.2 Independence and Number of Vectors

Certain situations immediately imply dependence.

Theorem 7.3.10. *Any collection containing the zero vector is linearly dependent.*

Proof. If $\vec{v}_1 = \vec{0}$, then

$$1\vec{v}_1 + \vec{0} + \cdots + \vec{0} = \vec{0}$$

This is a nontrivial solution. □

Example 7.3.11. $\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right\}$ is linearly dependent.

Theorem 7.3.12. *Any set of more than n vectors in $\mathbb{R}^n$ is linearly dependent.*

Example 7.3.13. Any four vectors in $\mathbb{R}^3$ must be linearly dependent.

To conclude this section, we connect determinants and linearly independent vectors in $\mathbb{R}^n$.

Theorem 7.3.14. *Let A be an $n \times n$ matrix. Then the columns of A are linearly independent if and only if $\det(A) \neq 0$.*

7.4 Subspaces of $\mathbb{R}^n$

In the previous sections, we studied vectors, linear combinations, spanning sets, and linear independence. We learned how vectors can be combined to generate new vectors and how to determine whether a collection of vectors contains redundant information.

Many important collections of vectors encountered in linear algebra share a common feature: they behave like smaller versions of $\mathbb{R}^n$.

For example,

- all vectors lying on a line through the origin.
- all vectors lying in a plane through the origin.
- the solution set of a homogeneous system of equations

all possess the same algebraic properties as $\mathbb{R}^n$. These special subsets are called subspaces.

7.4.1 What is a Subspace?

A subspace is a subset of $\mathbb{R}^n$ that is itself a vector space. Rather than verifying all vector space properties individually, we can use a simpler test.

Definition 7.4.1. A subset $W \subseteq \mathbb{R}^b$ is called a *subspace* of $\mathbb{R}^n$ if

1. The zero vector belongs to W:
2. W is closed under addition: If $\vec{u}, \vec{v} \in W$, then $\vec{u} + \vec{v} \in W$.
3. W is closed under scalar multiplication: If $\vec{u} \in W$ and $c \in \mathbb{R}$, then $c\vec{u} \in W$.

These conditions ensure that vectors in W can be added and scaled without leaving the set. Consequently, all linear combinations of W remain in W. Thus, W behaves like a vector space.

Example 7.4.2. Consider $W = \{\vec{0}\}$. This set only contains the zero vector. Then W is a subspace of $\mathbb{R}^n$.

Solution

The zero vector is included in the set, so $\vec{0} \in W$. Addition also follows because $\vec{0} \in W$ and $\vec{0} + \vec{0} = \vec{0} \in W$. Finally, scalar multiplication also follows because $\vec{0} \in W$ and $c\vec{0} = \vec{0}$. Therefore, $W = \{\vec{0}\}$ is a subspace. This is called the *trivial subspace*.

Example 7.4.3. The entire space $\mathbb{R}^n$ is a subspace of itself.

7.4.2 Lines and Planes Through the Origin

Example 7.4.4. Determine whether the set

$$W = \left\{ \begin{bmatrix} x \\ 2x \end{bmatrix} : x \in \mathbb{R} \right\}$$

is a subspace of $\mathbb{R}^2$.

Solution

We first check the zero vector. By choosing $x = 0$, we have

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} \in \mathbb{R}^2$$

Next, let

$$\vec{u} = \begin{bmatrix} x_1 \\ 2x_1 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} x_2 \\ 2x_2 \end{bmatrix}$$

Then

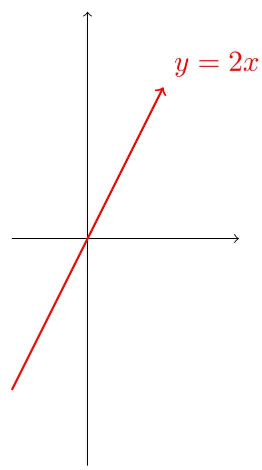
$$\vec{u}+\vec{v}=\begin{bmatrix} x_1+x_2 \\ 2(x_1+x_2) \end{bmatrix}$$

is a vector that belongs to W. Checking scalar multiplication,

$$c\vec{u}=\begin{bmatrix} cx_1 \\ 2(cx_1) \end{bmatrix}$$

Again, this belongs to W. Therefore, W is a subspace of $\mathbb{R}^2$.

In the above example, the W is the line $y=2x$ through the origin.



Example 7.4.5. Consider

$$W=\left\{\begin{bmatrix} x \\ y \\ z \end{bmatrix} : x+y+z=0\right\}$$

Determine whether W is a subsapce of $\mathbb{R}^3$.

Solution

First observe that

$$0+0+0=0$$

Thus, $\vec{0}\in W$.

Next, suppose $x_1+y_1+z_1=0$ and $x_2+y_2+z_2=0$. Then adding gives

$$(x_1+x_2)+(y_1+y_2)+(z_1+z_2)=0$$

Thus, the sum belongs to W.

Finally, if $x+y+z=0$, then $c(x+y+z)=0$. Thus, $cx+cy+cz=0$. Therefore, $c\vec{v}\in W$. Therefore, W is a subspace of $\mathbb{R}^3$.

In the above example, W is a plane passing through the origin.

Not every subset is a subspace.

Example 7.4.6. Determine whether

$$W=\left\{\begin{bmatrix}x\\y\end{bmatrix}:x+y=1\right\}$$

is a subspace of $\mathbb{R}^2$.

Solution

The zero vector is $\begin{bmatrix}0\\0\end{bmatrix}$. However, $0+0\neq 1$. Therefore, $\vec{0}\notin W$, so W is not a subspace.

In the above example, W is a line that does not pass through the origin.

7.4.3 Span Always Produces a Subspace

One of the most important results in linear algebra is the following.

Theorem 7.4.7. *The span of any collection of vectors is a subspace. That is,* $\operatorname{span}\{\vec{v}_1,\vec{v}_2,...,\vec{v}_k\}$ *is always a subspace of* $\mathbb{R}^n$*.*

This is because the span always contains the zero vector, sums of linear combinations, and scalar multiples of linear combinations. Therefore, all subspace conditions hold automatically.

Example 7.4.8. Let $\vec{v}_1=\begin{bmatrix}1\\0\\0\end{bmatrix}$ and $\vec{v}_2=\begin{bmatrix}0\\1\\0\end{bmatrix}$. Then

$$\operatorname{span}\{\vec{v}_1,\vec{v}_2\}=\left\{\begin{bmatrix}x\\y\\0\end{bmatrix}:x,y\in\mathbb{R}\right\}$$

produces a plane through the origin, and is a subspace of $\mathbb{R}^3$.

7.4.4 Subspaces Defined by Homogeneous Systems

Subspaces frequently arise as solution sets of homogeneous systems.

Example 7.4.9. Consider

$$x + y - z = 0$$

The set of all solutions is

$$W = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} : x + y - z = 0 \right\}$$

Since this equation is homogeneous, W forms a subspace of $\mathbb{R}^3$.

7.5 Bases and Dimension

In previous sections, we studied span, linear independence, and subspaces. We learned that a set of vectors can generate a subspace through linear combinations, and we also learned how to detect when vectors are redundant.

We now combine these two ideas.

A basis is a collection of vectors that is both:

- large enough to span a subspace, and
- small enough to avoid redundancy.

Bases are important because they provide an efficient way to describe vector spaces and subspaces. Once we know a basis, we know exactly how many independent directions are needed to describe the space. This number is called the dimension.

7.5.1 Bases

Definition 7.5.1. Let W be a subspace of $\mathbb{R}^n$. A set of vectors $\{\vec{v}_1, \vec{v}_2, ..., \vec{n}_k\}$ is called a *basis* for W if

1. The vectors span W, and
2. The vectors are linearly independent.

That is,

$$\text{span}\{\vec{v}_1, \vec{v}_2, ..., \vec{v}_k\} = W$$

and the vectors contain no redundancy.

A basis is an efficient generating set. It contains enough vectors to describe the whole space, but no unnecessary vectors.

Example 7.5.2. The standard basis vectors for $\mathbb{R}^2$ given by

$$\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \vec{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

form a basis for $\mathbb{R}^2$.

First, they span $\mathbb{R}^2$ as

$$\begin{bmatrix} x \\ y \end{bmatrix} = x\vec{e}_1 + y\vec{e}_2$$

Second, they are linearly independent. Therefore, $\{\vec{e}_1, \vec{e}_2\}$ is a basis for $\mathbb{R}^2$.

Example 7.5.3. The standard basis vectors for $\mathbb{R}^3$ given by

$$\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \vec{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \vec{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

These vectors form a basis for $\mathbb{R}^3$.

Example 7.5.4. Show that

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \vec{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

form a basis for $\mathbb{R}^2$.

Solution

We first check linear independence. Suppose

$$c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This gives the system

$$\begin{aligned} c_1 + c_2 &= 0, \\ c_1 - c_2 &= 0 \end{aligned}$$

Adding the two equations gives $2c_1 = 0$, so $c_1 = 0$. Then $c_2 = 0$. Thus, the vectors are linearly independent.

Since there are two linearly independent vectors in $\mathbb{R}^2$, they span $\mathbb{R}^2$. Therefore,

$$\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}$$

is a basis for $\mathbb{R}^2$.

Example 7.5.5. Let

$$W = \left\{ \begin{bmatrix} x \\ 2x \end{bmatrix} : x \in \mathbb{R} \right\}$$

Find a basis for W.

Solution

Rewrite the general vector as

$$\begin{bmatrix} x \\ 2x \end{bmatrix} = x \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Therefore,

$$W = \operatorname{span} \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$$

The vector $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ is nonzero, so it is linearly independent by itself. Thus, a basis for W is

$$\left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$$

Example 7.5.6. Find a basis for

$$W = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} : x + y + z = 0 \right\}$$

Solution

From $x + y + z = 0$, we get

$$x = -y - z$$

Let $y = s$ and $z = t$ so that

$$x = -s - t$$

Rewriting the general vector

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -s-t \\ s \\ t \end{bmatrix} = s\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + t\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

Thus,

$$W = \operatorname{span}\left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

These two vectors are not scalar multiples of each other, so they are linearly independent. Therefore, a basis for W is

$$\left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

7.5.2 Dimension

Definition 7.5.7. The *dimension* of a subspace W, denoted $\dim(W)$ is the number of vectors in any basis for W.

Example 7.5.8. Since the standard basis for $\mathbb{R}^2$ contains two vectors,

$$\dim(\mathbb{R}^2) = 2$$

Example 7.5.9. Since the standard basis for $\mathbb{R}^3$ contains three vectors,

$$\dim(\mathbb{R}^3) = 3$$

The dimension counts the number of independent directions in a space. For example, a point has dimension 0, a line has dimension 1, a plane has dimension 2, and $\mathbb{R}^3$ has dimension 3. In higher dimensions, the same idea applies algebraically.

7.5.3 Bases and Coordinates

Once a basis is chosen, every vector in the space can be written uniquely as a linear combination of the basis vectors.

Example 7.5.10. Let

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \vec{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Since these form a basis for $\mathbb{R}^2$, every vector in $\mathbb{R}^2$ can be written uniquely as

$$c_1\vec{v}_1 + c_2\vec{v}_2$$

For example, write $\begin{bmatrix} 4 \\ 2 \end{bmatrix}$ as a linear combination of $\vec{v}_1$ and $\vec{v}_2$.

Solution

We solve

$$c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

This gives

$$\begin{aligned} c_1 + c_2 &= 4, \\ c_1 - c_2 &= 2 \end{aligned}$$

Adding gives $2c_1 = 6$, and thus, $c_1 = 3$. Also, we have $c_2 = 1$. Therefore,

$$\begin{bmatrix} 4 \\ 2 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 1 \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

7.5.4 Basis Test in $\mathbb{R}^n$

Theorem 7.5.11. *A set of n vectors in $\mathbb{R}^n$ is a basis for $\mathbb{R}^n$ if and only if the vectors are linearly independent.*

Equivalently, a set of n vectors in $\mathbb{R}^n$ is a basis for $\mathbb{R}^n$ if and only if the vectors span $\mathbb{R}^n$.

Example 7.5.12. Determine whether

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \vec{v}_2 = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

form a basis for $\mathbb{R}^2$.

Solution

Form the coefficient matrix

$$A = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

Computing the determinant

$$\det(A) = (1)(4) - (3)(2) = 4 - 6 = -2 \neq 0$$

Therefore, the columns of A are linearly independent. Therefore, $\{\vec{v}_1, \vec{v}_2\}$ is a basis for $\mathbb{R}^2$.

7.6 Row Space, Column Space, Null Space

Throughout this course, we have studied matrices from several perspectives.

- In Chapter 1, matrices arose naturally when solving systems of linear equations.
- In Chapter 2, we learned how matrices can be manipulated through row operations.
- In Chapter 3, we saw that matrices represent linear transformations.

In this chapter, we introduced vector spaces, subspaces, span, linear independence, bases, and dimension.

We now bring these ideas together by studying three important subspaces associated with a matrix:

- the row space
- the column space
- the null space.

These spaces reveal important structural information about a matrix and provide a geometric understanding of systems of linear equations.

7.6.1 The Row Space

Definition 7.6.1. Let

$$A = \begin{bmatrix} \leftarrow & \vec{r}_1 & \rightarrow \\ \leftarrow & \vec{r}_2 & \rightarrow \\ & \vdots & \\ \leftarrow & \vec{r}_m & \rightarrow \end{bmatrix}$$

be an $m \times n$ matrix. The *row space* of A, denoted row(A) is the span of the rows of A. That is,

$$\text{row}(A) = \text{span}\{\vec{r}_1, \vec{r}_2, ..., \vec{r}_m\}$$

The row space consists of all linear combinations of the rows of A. Since spans are subspaces, the row space is a subspace of $\mathbb{R}^n$.

Example 7.6.2. Consider

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}$$

The rows are

$$\vec{r}_1 = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}, \quad \vec{r}_2 = \begin{bmatrix} 2 & 4 & 6 \end{bmatrix}$$

Notice that $\vec{r}_2 = 2\vec{r}_!$. Therefore,

$$\text{row}(A) = \text{span}\left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \right\}$$

A basis for the row space is $\left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \right\}$ and the dimension is $\dim(\text{row}(A)) = 1$.

7.6.2 The Column Space

Definition 7.6.3. Let

$$A = \begin{bmatrix} \uparrow & \uparrow & & \uparrow \\ \vec{c}_1 & \vec{c}_2 & \cdots & \vec{c}_n \\ \downarrow & \downarrow & & \downarrow \end{bmatrix}$$

The *column space* of A, denoted by $\text{col}(A)$, is the span of the columns of A. That is,

$$\text{col}(A) = \text{span}\{\vec{c}_1, \vec{c}_2, ..., \vec{c}_n\}$$

The column space contains all vectors that can be written as $A\vec{x}$. It is the set of all possible outputs of the matrix transformation.

Example 7.6.4. Consider

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix}$$

The columns are

$$\vec{c}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \vec{c}_2 = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}$$

Since $\vec{c}_2 = 2\vec{c}_1$, the second column is redundant. Thus,

$$\text{col}(A) = \text{span}\left\{\begin{bmatrix}1\\2\\3\end{bmatrix}\right\}$$

A basis is $\left\{\begin{bmatrix}1\\2\\3\end{bmatrix}\right\}$ and

$$\dim(\text{col}(A)) = 1$$

7.6.3 The Null Space

The row and column spaces are generated by rows and columns. The null space is different. It is defined as the solution set of a homogeneous system.

Definition 7.6.5. The *null space* of a matrix A is

$$\text{null}(A) = \{\vec{x} \in \mathbb{R}^n : A\vec{x} = \vec{0}\}$$

The null space consists of all vectors sent to zero vector by the matrix transformation.

Recall the homogeneous system $A\vec{x} = \vec{0}$. The null space is exactly the set of all solutions. Therefore, solving a homogeneous system and finding the null space are the same problem.

Example 7.6.6. Find the null space of

$$A = \begin{bmatrix}1 & 2\end{bmatrix}$$

Solution

We solve $A\vec{x} = \vec{0}$ which gives

$$x_1 + 2x_2 = 0$$

Let $x_2 = t$, so that $x_1 = -2t$. Thus,

$$\vec{x} = \begin{bmatrix}-2t\\t\end{bmatrix} = t\begin{bmatrix}-2\\1\end{bmatrix}$$

Therefore,

$$\text{null}(A) = \text{span}\left\{\begin{bmatrix}-2\\1\end{bmatrix}\right\}$$

The vectors multiplying the free parameters form a basis.

Example 7.6.7. Find a basis for the null space of

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}$$

Solution

First row reduce to get

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

Then we solve

$$x_1 + 2x_2 + 3x_3 = 0$$

Let $x_2 = s$ and $x_3 = t$. Then $x_1 = -2s - 3t$. Writing the general solution gives

$$\vec{x} = s \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}$$

Therefore, a basis for it is

$$\left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} \right\}$$

and the dimension is 2.

7.6.4 Finding a Basis

A convenient theorem simplifies calculations.

Theorem 7.6.8. *The nonzero rows of the row-echelon form of a matrix form a basis for the row space. Similarly, the pivot columns of the original matrix form a basis for the column space.*

Example 7.6.9. Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}$$

Row reducing gives

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

A basis for the row space is

$$\left\{ \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 2 \end{bmatrix} \right\}$$

Example 7.6.10. Consider

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 1 & 2 & 2 \end{bmatrix}$$

Row reducing gives

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

The pivot columns are 1 and 3. Thus, a basis for the column space is

$$\left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} \right\}$$

7.6.5 Rank of a Matrix

The dimension of the row space and column space is extremely important.

Definition 7.6.11. The *rank* of a matrix is the dimension of its row space. Equivalently $\operatorname{rank}(A) = \dim(\operatorname{row}(A))$.

Theorem 7.6.12. *For every matrix A,*

$$\dim(\operatorname{row}(A)) = \dim(\operatorname{col}(A))$$

Thus, $\operatorname{rank}(A) = \dim(\operatorname{col}(A))$.

For a matrix $A : \mathbb{R}^n \to \mathbb{R}^m$, the column space describes all outputs, while the null space describes all inputs sent to zero. Together, these spaces describe how the matrix transformation behaves.

Chapter 8

Complex Numbers $\mathbb{C}$

Throughout the course, we have worked almost exclusively with real numbers. Systems of equations, matrices, determinants, eigenvalues, vectors, and geometric objects have all been studied within the familiar setting of the real number system $\mathbb{R}$.

However, not every polynomial has a real solution. For example, the equation

$$x^2 + 1 = 0$$

has no solution in $\mathbb{R}$, since no real number satisifies

$$x^2 = -1$$

To overcome this limitation, mathematicians introduced a larger numebr system called the *complex numbers*, denoted by $\mathbb{C}$.

Complex numbers extend the real numbers by introducing the imaginary unit

$$i = \sqrt{-1}$$

which satisfies

$$i^2 = -1$$

Although complex numbers were originally developed to solve polynomial equations, they have become one of the most important mathematical tools in science and engineering.

One remarkable fact is that every polynomial with real coefficients has all of its roots in $\mathbb{C}$. Thus result, known as the Fundamental Theorem of Algebra, makes the complex numbers a natural setting for many areas of mathematics.

In this chapter, we introduce the algebra and geometry of complex numbers and explore several applications to linear algebra.

8.1 Complex Numbers

Throughout this course, we have worked primarily with real numbers. Real numbers are sufficient for solving many equations, describing vectors, and analyzing matrices.

However, some algebraic equations have no real solutions. For example

$$x^2 + 1 = 0$$

has no solution in $\mathbb{R}$, since every real number satisfies $x^2 \geq 0$.

To solve equations such as this, mathematicians introduced a larger number system called the complex numbers.

Complex numbers extend the real numbers by introducing a new number i, called the imaginary unit, which satisfies

$$i^2 = -1$$

The introduction of i allows every polynomial equation to have solutions in a sufficiently large number system.

8.1.1 The Imaginary Unit i

Definition 8.1.1. The *imaginary unit* is the unit $i = \sqrt{-1}$ satisfying $i^2 = -1$.

Since $i^2 = -1$, higher powers repeat in cycle:

$$\begin{aligned} i^1 &= i, \\ i^2 &= -1, \\ i^3 &= i^2(i) = (-1)i = -i, \\ i^4 &= (i^2)^2 = (-1)^2 = 1 \\ i^5 &= i \end{aligned}$$

and the pattern repeats. Thus, there is a pattern that repeats every four powers:

$$i, \quad -1, \quad -i, \quad 1$$

Example 8.1.2. Compute i^{27}.

Solution

Divide 27 by 4 to get

$$27 = 4(6) + 3$$

Therefore,

$$i^{27} = i^3 = -i$$

8.1.2 Definition of a Complex Number

Definition 8.1.3. A *complex number* is a number of the form

$$z = a + bi$$

where $a, b \in \mathbb{R}$. The number a is called the *real part*, and b is called the *imaginary part*.

If $z = a + bi$, then $\mathrm{Re}(z) = a$ and $\mathrm{Im}(z) = b$, so

$$z = \mathrm{Re}(z) + i\mathrm{Im}(z)$$

Example 8.1.4. The following examples are complex numbers

$$3 + 2i, \quad -5 + 7i, \quad 4 = 4 + 0i, \quad -2i = 0 - 2i$$

8.1.3 Equality of Complex Numbers

Two complex numbers are equal when their corresponding real and imaginary parts are equal.

Theorem 8.1.5. *If $a + bi = c + di$, then $a = c$ and $b = d$.*

Example 8.1.6. Find x and y if

$$(2x - 1) + (y + 3)i = 5 + 7i$$

Solution

Equating the real parts, we have

$$\begin{aligned} 2x - 1 &= 5 \\ x &= 3 \end{aligned}$$

Equating the imaginary parts, we have

$$\begin{aligned} y + 3 &= 7 \\ y &= 4 \end{aligned}$$

Therefore, $x = 3$ and $y = 4$.

8.1.4 Addition and Subtraction

Complex numbers are added componentwise.

Definition 8.1.7. If $z = a + bi$ and $w = c + di$, then

$$z + w = (a + c) + (b + d)i \tag{8.1}$$

Similarly,

$$z - w = (a - c) + (b - d)i$$

Example 8.1.8. Compute $(3 + 2i) + (1 - 5i)$.

Solution

Adding componentwise, we have

$$(3 + 2i) + (1 - 5i) = (3 + 1) + (2 - 5)i = 4 - 3i$$

Example 8.1.9. Compute $(5 + 3i) - (2 - 4i)$.

Solution

Subtracting componentwise, we have

$$(5 + 3i) - (2 - 4i) = (5 - 2) + (3 + 4)i = 3 + 7i$$

8.1.5 Multiplication

Complex numbers are multiplied using ordinary algebra and the relation $i^2 = -1$.

Example 8.1.10. Compute $(2 + 3i)(1 + 4i)$.

Solution

Expand

$$(2 + 3i)(1 + 4i) = 2 + 8i + 3i + 12i^2 = 2 + 11i - 12 = -10 + 11i$$

8.1.6 Multiplying Complex Conjugates

An important special case occurs when multiplying conjugates.

Definition 8.1.11. The *complex conjugate* of $z = a + bi$ is

$$\bar{z} = a - bi$$

Example 8.1.12.

$$\overline{3+4i} = 3 - 4i,$$
$$\overline{-2+i} = -2 - i$$

8.1.7 Properties of Conjugates

Theorem 8.1.13. *If $z = a + bi$, then $z\bar{z} = a^2 + b^2$.*

Proof. Expanding gives

$$z\bar{z} = (a + bi)(a - bi) = a^2 - abi + abi - b^2i^2 = a^2 + b^2$$

□

Example 8.1.14. Compute $(3 + 4i)(3 - 4i)$.

Solution

We have

$$(3 + 4i)(3 - 4i) = 3^2 + 4^2 = 25$$

8.1.8 Division of Complex Numbers

To divide complex numbers, multiply the numerator and denominator by the conjugate of the denominator.

Example 8.1.15. Compute $\frac{2+3i}{1-2i}$.

Solution

Multiply top and bottom by $1 + 2i$

$$\frac{2+3i}{1-2i} \cdot \frac{1+2i}{1+2i} = \frac{2+4i+3i+6i^2}{5} = \frac{-4+7i}{5} = -\frac{4}{5} + \frac{7}{5}i$$

8.1.9 The Complex Plane

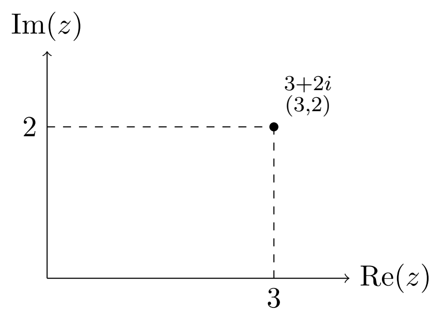
Complex numbers can be represented geometrically.

Definition 8.1.16. The *complex plane* associates $z = a + bi$ with the point (a, b). The horizontal axis is called the *real axis*, and the vertical axis is called the *imaginary axis*.

Example 8.1.17. Plot $z = 3 + 2i$.

Solution

This corresponds to the point $(3, 2)$ on the complex plane.



8.1.10 Modulus of a Complex Number

The distance from the origin to the point representing a complex number is called its modulus.

Definition 8.1.18. If $z = a + bi$, then the *modulus* of z is

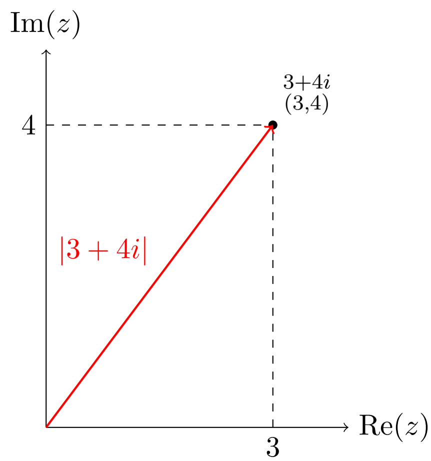
$$|z| = \sqrt{a^2 + b^2}$$

The modulus is exactly the length of the vector from the origin to (a, b).

Example 8.1.19. Find $|3 + 4i|$.

Solution

$|3 + 4i| = \sqrt{3^2 + 4^2} = 5$



8.1.11 Relationship Between Modulus and Conjugates

Theorem 8.1.20. *For any complex number z,*

$$z\bar{z} = |z|^2$$

Example 8.1.21. Let $z = 2 - 3i$. Then

$$z\bar{z} = (2 - 3i)(2 + 3i) = 13$$

Also, $|z| = \sqrt{13}$ so $|z|^2 = 13$.

8.1.12 Properties of the Modulus

Theorem 8.1.22. *For complex numbers z, w,*

1. $|zw| = |z||w|$
2. $\left|\frac{z}{w}\right| = \frac{|z|}{|w|}$, *provided that $w \neq 0$.*

8.2 Polar Form and Euler's Formula

In the previous section, we introduced complex numbers in the form $z = a + bi$ which is called the *rectangular form* (or Cartesian form) of a complex number.

While rectangular form is convenient for addition and subtraction, it is often cumbersome when multiplying dividing, or raising complex numbers to powers.

Because every complex number corresponds to a point in the complex plane, we can also describe it using geometric quantities:

- its distance from the origin, and
- the angle it makes with the positive real axis.

This leads to the polar form of a complex number.

Polar form provides a powerful geometric interpretation of complex multiplication and allows us to derive one of the most beautiful formulas in mathematics:

$$\cos(\theta) + i\sin(\theta)$$

This result, known as Euler's formula, connects algebra, geometry, trigonometry, and calculus in a single equation.

8.2.1 Polar Coordinates in the Complex Plane

Recall that every complex number $z = a + bi$ corresponds to the point (a, b) in the complex plane.

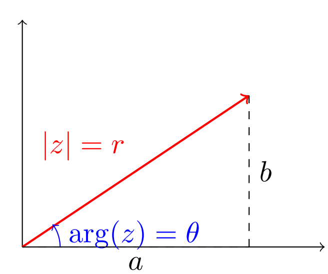
Instead of using coordinates (a, b), we can describe the same point using its distance from the origin and its angle from the positive real axis.

Definition 8.2.1. Let $z = a + bi$. The *modulus* of z is

$$|z| = \sqrt{a^2 + b^2}$$

The *argument* of z, denoted by $\arg(z)$, is the angle θ measured from the positive real axis to the line segment joining the origin to z.

A complex number is represented by its modulus r and argument θ.



Thus, every nonzero complex number corresponds to polar coordinates (r, θ).

8.2.2 Converting to Polar Form

Using trigonometry, in particular SOH and CAH

$$\cos(\theta) = \frac{a}{r}, \quad \sin(\theta) = \frac{b}{r}$$

Rearranging

$$a = r\cos(\theta), \quad b = r\sin(\theta)$$

Therefore, substituting into $z = a + bi$, we have

$$z = r\cos(\theta) + ir\sin(\theta) = r(\cos(\theta) + i\sin(\theta))$$

Definition 8.2.2. The *polar form* of a complex number is

$$z = r(\cos(\theta) + i\sin(\theta))$$

Example 8.2.3. Express $z = 1 + i$ in polar form.

Solution

We first find the modulus

$$r = |1 + i| = \sqrt{1^2 + 1^2} = \sqrt{2}$$

and we find the argument. Since

$$\tan(\theta) = \frac{1}{1} = 1$$

then

$$\theta = \tan^{-1}(1) = \frac{\pi}{4}$$

Therefore, with $r = \sqrt{2}$ and $\theta = \frac{\pi}{4}$, substitute

$$z = \sqrt{2}\left(\cos\left(\frac{\pi}{4}\right) + i\sin\left(\frac{\pi}{4}\right)\right)$$

Example 8.2.4. Express $z = -2 + 2i$ in polar form.

Solution

Computing the modulus,

$$r = |-2 + 2i| = \sqrt{(-2)^2 + 2^2} = \sqrt{8} = 2\sqrt{2}$$

Then determine the argument. Note that the point $(-2, 2)$ lies in Quadrant II, so the reference angle is $\frac{\pi}{4}$. Thus,

$$\theta = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$$

Therefore,

$$z = 2\sqrt{2}\left(\cos\left(\frac{3\pi}{4}\right) + i\sin\left(\frac{3\pi}{4}\right)\right)$$

8.2.3 Converting Back to Rectangular Form

We use

$$a = r\cos(\theta), \quad b = r\sin(\theta)$$

to convert from polar form to rectangular form.

Example 8.2.5. Convert

$$2\left(\cos\left(\frac{\pi}{3}\right) + i\sin\left(\frac{\pi}{3}\right)\right)$$

to rectangular form.

Solution

$$2\left(\cos\left(\frac{\pi}{3}\right) + i\sin\left(\frac{\pi}{3}\right)\right) = 2\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) = 1 + i\sqrt{3}$$

8.2.4 Euler's Formula

One of the most remarkable formulas in mathematics is Euler's formula.

Theorem 8.2.6. *For every real number θ*

$$e^{i\theta} = \cos(\theta) + i\sin(\theta)$$

Substituting into polar form gives

$$z = re^{i\theta}$$

This is the most compact representation of a complex number.

Example 8.2.7. Express

$$z = 3\left(\cos\left(\frac{\pi}{6}\right) + i\sin\left(\frac{\pi}{6}\right)\right)$$

using Euler's formula.

Solution

$$z = 3e^{i\frac{\pi}{6}}$$

8.2.5 Euler's Identity

Setting $\theta = \pi$ in Euler's formula gives

$$e^{i\pi} = \cos(\pi) + i\sin(\pi) = -1$$

Therefore,

$$e^{i\pi} + 1 = 0$$

This equation is known as *Euler's identity.* It connects five fundamental mathematical constants

$$0, \quad 1, \quad e, \quad i, \quad \pi$$

8.2.6 Multiplication Form in Polar Form

Polar form greatly simplifies multiplication.

Theorem 8.2.8. *If*

$$z_1 = r_1(\cos(\theta_1) + i\sin(\theta_1)),$$
$$z_2 = r_2(\cos(\theta_2) + i\sin(\theta_2)),$$

then

$$z_1 z_2 = r_1 r_2(\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2))$$

When multiplying complex numbers, we multiply the moduli, and add the arguments.

Example 8.2.9. Multiply

$$z_1 = 2\left(\cos\left(\frac{\pi}{4}\right) + i\sin\left(\frac{\pi}{4}\right)\right),$$
$$z_2 = 3\left(\cos\left(\frac{\pi}{6}\right) + i\sin\left(\frac{\pi}{6}\right)\right)$$

Solution

Multiply the moduli

$$(2)(3) = 6$$

and add the angles

$$\frac{\pi}{4} + \frac{\pi}{6} = \frac{5\pi}{12}$$

Therefore,

$$z_1 z_2 = 6\left(\cos\left(\frac{5\pi}{12}\right) + i\sin\left(\frac{5\pi}{12}\right)\right)$$

8.2.7 Division in Polar Form

Theorem 8.2.10. *If*

$$z_1 = r_1 e^{i\theta_1}, \quad z_2 = r_2 e^{i\theta_2}$$

then

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}$$

That is, when dividing complex numbers, divide the moduli, and subtract the arguments.

Example 8.2.11. Divide

$$z_1 = 2\left(\cos\left(\frac{\pi}{4}\right) + i\sin\left(\frac{\pi}{4}\right)\right),$$
$$z_2 = 3\left(\cos\left(\frac{\pi}{6}\right) + i\sin\left(\frac{\pi}{6}\right)\right)$$

Solution

Divide the moduli

$$\frac{r_1}{r_2} = \frac{2}{3}$$

Then subtract the arguments

$$\frac{\pi}{4} - \frac{\pi}{6} = \frac{\pi}{12}$$

Therefore,

$$\frac{z_1}{z_2} = \frac{2}{3}\left(\cos\left(\frac{\pi}{12}\right) + i\sin\left(\frac{\pi}{12}\right)\right)$$

8.2.8 De Moivre's Formula

Repeated multiplication leads to a powerful theorem.

Theorem 8.2.12 (**De Moivre's Formula**)**.** *For any real number α,*

$$(\cos(\theta) + i\sin(\theta))^{\alpha} = \cos(\alpha\theta) + i\sin(\alpha\theta)$$

This is particularly useful for when α is an integer. In which case, if n is an integer, then

$$(\cos(\theta) + i\sin(\theta))^{n} = \cos(n\theta) + i\sin(n\theta)$$

Using Euler's formula,

$$(e^{i\theta})^n = e^{i(n\theta)}$$

Example 8.2.13. Compute

$$\left(\cos\left(\frac{\pi}{6}\right) + i\sin\left(\frac{\pi}{6}\right)\right)^5$$

Solution

Using De Moivre's formula,

$$\left(\cos\left(\frac{\pi}{6}\right) + i\sin\left(\frac{\pi}{6}\right)\right)^5 = \cos\left(\frac{5\pi}{6}\right) + i\sin\left(\frac{5\pi}{6}\right) = -\frac{\sqrt{3}}{2} + \frac{1}{2}i$$

Example 8.2.14. Compute $(1+i)^8$.

Solution

Convert to polar form

$$1 + i = \sqrt{2}\left(\cos\left(\frac{\pi}{4}\right) + i\sin\left(\frac{\pi}{4}\right)\right)$$

Then applying De Moivre's Formula

$$(1+i)^8 = (\sqrt{2})^8\left(\cos(2\pi) + i\sin(2\pi)\right) = 16$$

8.3 Roots of a Complex Equation

In the previous section, we introduced the polar form of a complex number and developed Euler's formula

$$\cos(\theta) + i\sin(\theta)$$

We also learned that polar form simplifies multiplication and division, and powers of complex numbers through De Moivre's formula.

In this section, we reverse that process. Instead of computing powers such as z^n, we ask the opposite question: Given a complex number w, what complex numbers satisfy $z^n = w$.

Problems of this form arise naturally when solving polynomial equations. For example,

$$z^2 + 1 = 0, \quad z^3 = 8, \quad z^4 = 16, \quad z^n = a$$

Using polar form, these equations can be solved systematically.

8.3.1 Motivation

Consider the equation

$$z^2 = 1$$

There are two square roots.

Similarly,

$$z^3 = 1$$

has three distinct solutions.

More generally,

$$z^n = w$$

has n distinct complex solutions whenever $w \neq 0$. Our goal is to determine all of them.

8.3.2 Roots of Unity

We begin with one of the most important examples.

Definition 8.3.1. The solutions of

$$z^n = 1$$

are called the *nth roots of unity*.

Write

$$1 = e^{i(2k\pi)}$$

for any integer k. Suppose $z = e^{i\theta}$. Then

$$z^n = e^{in\theta} = 1 = e^{i(2k\pi)}$$

Therefore,

$$n\theta = 2k\pi$$

Therefore,

$$\theta = \frac{2k\pi}{n}$$

Theorem 8.3.2. *The nth roots of unity are*

$$z_k = e^{\frac{2k\pi i}{n}}, \quad k = 0, 1, 2, ..., n-1$$

Example 8.3.3. Find all solutions of $z^4 = 1$.

Solution

Using

$$z_k = e^{i\frac{2k\pi}{4}}$$

we obtain

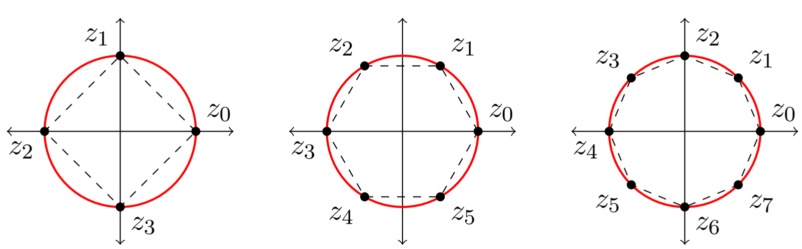
$$z_0 = e^0 = 1, \quad z_1 = e^{i\frac{\pi}{2}} = i, \quad z_2 = e^{i\pi} = -1, \quad z_3 = e^{i\frac{3\pi}{2}} = -i$$

8.3.3 Geometric Interpretation

The roots of unity lie on the unit circle. Moreover, they are equally spaced.

Theorem 8.3.4. *The nth roots of unity form the vertices of a regular n-gon inscribed in the unit circle.*

Example 8.3.5. The fourth roots of unity form a square. The sixth roots of unity form a regular hexagon. The eighth roots of unity form a regular octagon.



8.3.4 General nth Roots

Now consider the complex equation

$$z^n = w$$

where $w \neq 0$. Write w in its polar form

$$w = re^{i\theta}$$

Assume $z = \rho e^{i\phi}$. Then $z^n = \rho^n e^{in\phi}$. Then comparing moduli and arguments give

$$\rho^n = r, \quad n\phi = \theta = 2k\pi$$

Theorem 8.3.6. *Let $w = re^{i\theta}$. The solutions of $z^n = w$ are*

$$z_k = r^{\frac{1}{n}} e^{i\frac{\theta+2k\pi}{n}}, \quad k = 0, 1, 2, ..., n-1$$

It is important to observe that there are exactly n distinct roots.

Example 8.3.7. Find all solutions to $z^3 = 8$.

Solution

Write 8 in polar form

$$8 = 8e^{i0}$$

Apply the formula,

$$z_k = 8^{\frac{1}{3}} e^{i\frac{0+2k\pi}{3}} = 2e^{i\frac{2k\pi}{3}}, \quad k = 0, 1, 2$$

Computing the roots,

$$\begin{aligned} z_0 &= 2, \\ z_1 &= -1 + \sqrt{3}i, \\ z_2 &= -1 - \sqrt{3}i \end{aligned}$$

Example 8.3.8. Find all solutions of $z^4 = 16$.

Solution

Write

$$16 = 16e^{i0}$$

Then

$$z_k = 2e^{i\frac{k\pi}{2}}, \quad k = 0, 1, 2, 3$$

The four roots are

$$z_0 = 2, \quad z_1 = 2i, \quad z_2 = -2, \quad z_3 = -2i$$

8.4 Applications of Complex Numbers in Linear Algebra

Throughout this course, we have studied matrices, systems of linear equations, determinants, linear transformations, and eigenvalues. Up to this point, our work has largely been restricted to matrices with real entries and vectors whose components are real numbers.

However, many matrices naturally possess eigenvalues that are not real numbers.

For example, consider the matrix

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

Its characteristic polynomial is

$$\lambda^2 + 1$$

which has roots $\pm i$.

Thus, even though the entries of the matrix are real, its eigenvalues are complex.

The complex numbers provide a larger number system in which every polynomial equation has solutions. Consequently, every square matrix has all of its eigenvalues in $\mathbb{C}$.

In this section, we revisit several concepts from linear algebra and extend them to matrices with complex entries.

8.4.1 Systems of Linear Equations with Complex Coefficients

The techniques developed in Chapters 1 and 2 continue to work when coefficients are complex.

Example 8.4.1. Solve

$$\begin{aligned} x + iy &= 1, \\ (1+i)x + y &= 2 \end{aligned}$$

Solution

Write the augmented matrix

$$\left[\begin{array}{cc|c} 1 & i & 1 \\ 1+i & 1 & 2 \end{array}\right]$$

Perform $R_2 \to R_2 - (1+i)R_1$. This gives

$$\left[\begin{array}{cc|c} 1 & i & 1 \\ 0 & 2 & 1-i \end{array}\right]$$

Thus,

$$y = \frac{1-i}{2}$$

Substituting into the first equation,

$$x = 1 - i\left(\frac{1-i}{2}\right)$$

Simplifying

$$x = \frac{1-i}{2}$$

Gaussian elimination works exactly as before. The only difference is that arithmetic involving i may be required.

Example 8.4.2. Row reduce

$$A = \begin{bmatrix} 1 & i \\ 2 & 1+i \end{bmatrix}$$

Solution

Perform $R_2 \to R_2 - 2R_1$ to get

$$\begin{bmatrix} 1 & i \\ 0 & 1-i \end{bmatrix}$$

Multiply the second row by

$$\frac{1}{1-i} = \frac{1+i}{2}$$

Therefore,

$$\begin{bmatrix} 1 & i \\ 0 & 1-i \end{bmatrix}$$

Finally, perform $R_1 \to R_1 - iR_2$ to get

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

All row operations remain valid over $\mathbb{C}$.

8.4.2 Determinants of Complex Matrices

Determinants are computed exactly as before.

Example 8.4.3. Compute

$$\det \begin{bmatrix} 1 & i \\ 2 & 1-i \end{bmatrix}$$

Solution

$$\det \begin{bmatrix} 1 & i \\ 2 & 1-i \end{bmatrix} = (1)(1-i) - 2i = 1 - 3i$$

8.4.3 Complex Eigenvalues and Eigenvectors

One of the most important applications of complex numbers in linear algebra occurs when finding eigenvalues.

Example 8.4.4. Consider

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

This matrix represents a rotation by 90°. Find the eigenvalues of A.

Solution

Compute the characteristic polynomial

$$\det(A - \lambda I) = \begin{bmatrix} -\lambda & -1 \\ 1 & -\lambda \end{bmatrix} = \lambda^2 + 1$$

Then solve $\lambda^2 + 1 = 0$, which gives $\lambda = i$ and $\lambda = -i$

Recall that the matrix has no real eigenvalues. Complex numbers are required to fully understand its behaviour.

The process to finding complex eigenvectors is identical to the real case.

Example 8.4.5. For each eigenvalue found above, find a corresponding eigenvector.

Solution

For $\lambda = i$, we have

$$\begin{bmatrix} -i & -1 \\ 1 & -i \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

From the first equation,

$$-ix - y = 0$$

Thus,

$$y = -ix$$

Choosing $x = 1$, $y = -i$. Therefore, an eigenvector is

$$\vec{v}_1 = \begin{bmatrix} 1 \\ -i \end{bmatrix}$$

8.4.4 Complex Conjugate Eigenvalues

Complex eigenvalues of real matrices always occur in conjugate pairs.

Theorem 8.4.6. *If a matrix has eigenvalue $\lambda = a + bi$, then it also has eigenvalue $\bar{\lambda} = a - bi$.*

Example 8.4.7. If $\lambda = 2+3i$ is an eigenvalue of a real matrix, then $\lambda = 2-3i$ must also be an eigenvalue.

Example 8.4.8. Consider

$$A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

The characteristic polynomial is

$$\det(A - \lambda I) = (1-\lambda)^2 + 1 = \lambda^2 - 2\lambda + 2$$

Solving by the quadratic formula,

$$\lambda = \frac{2 \pm \sqrt{-4}}{2} = 1 \pm i$$

Therefore, the eigenvalues are $\lambda_1 = 1 + i$ and $\lambda_2 = 1 - i$.

8.4.5 Diagonalization Over $\mathbb{C}$

Some matrices that cannot be diagonalized over $\mathbb{R}$ become diagonalizable over $\mathbb{C}$.

Example 8.4.9. The rotation matrix

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

cannot be diagonalized over $\mathbb{R}$. However, over $\mathbb{C}$,

$$A = P \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix} P^{-1}$$

Observe that the complex numbers often simplify matrix computations.

8.4.6 Complex Vector Spaces

Up to now, vectors have belonged to $\mathbb{R}^n$. More generally, vectors may also belong to $\mathbb{C}^n$.

Example 8.4.10. $\begin{bmatrix} 1+i \\ 2 \\ -3i \end{bmatrix}$ is a vector in $\mathbb{C}^3$.

All concepts from Chapter 7 extend naturally, including linear combinations, span, independence, bases, and dimension.

Appendix A

Trigonometry

In this appendix, we review the basic trigonometric ideas that are useful throughout this course. In particular, trigonometry appears when we study angles between vectors, projections, work done by a force, torque, and the polar form of complex numbers.

This appendix is not meant to replace a full course in trigonometry. Instead, it is intended to provide a quick, but detailed reference for the main tools students will need in this course.

A.1 Angles and Units

An angle measures the amount of rotation between two rays that share a common endpoint. There are two common ways to measure angles: degrees and radians.

A full rotation is equal to 360°. In radians, a full rotation is equal to 2π. Therefore,

$$360^\circ = 2\pi \text{ radians}$$

Dividing both sides by 2, we get

$$180^\circ = \pi \text{ radians}$$

Therefore, to convert from degrees to radians, we multiply by $\frac{\pi}{180}$. To convert from radians to degrees, we multiply by $\frac{180}{\pi}$.

Example A.1.1. Convert 60° to radians.

Solution

We multiply by $\frac{\pi}{180}$.

$$60^\circ = 60 \cdot \frac{\pi}{180} = \frac{\pi}{3}$$

Example A.1.2. Convert $\frac{5\pi}{6}$ radians to degrees.

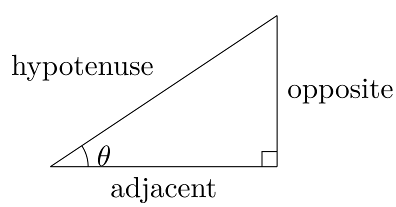
Solution

We multiply by $\frac{180}{\pi}$

$$\frac{5\pi}{6} = \frac{5\pi}{6} \cdot \frac{180}{\pi} = 3 \cdot 30 = 150^\circ$$

A.2 Right Angle Trigonometry

Consider a right triangle with an angle θ. Relative to the angle θ, the three sides of the triangle are called the opposite side, adjacent side, and hypotenuse.



The three main trigonometric ratios are sine, cosine, and tangent:

$$\sin(\theta) = \frac{\text{opposite}}{\text{hypotenuse}}, \quad \cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}}, \quad \tan(\theta) = \frac{\text{opposite}}{\text{adjacent}}$$

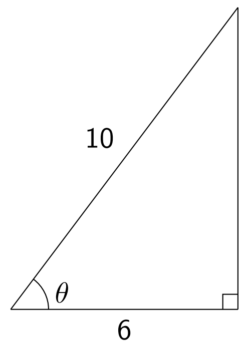
A common way to remember these ratios is SOH CAH TOA.

Example A.2.1. Suppose a right triangle has hypotenuse 10 and an angle θ such that the side adjacent to θ has length 6. Find $\cos(\theta)$.

Solution

Using the definition of cosine,

$$\cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{6}{10} = \frac{3}{5}$$



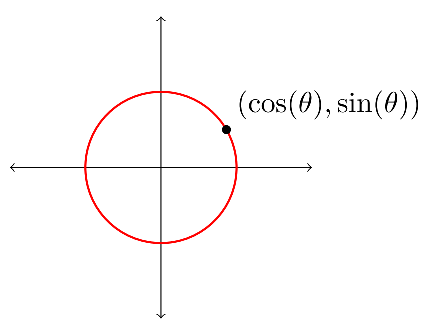
A.3 The Unit Circle

The unit circle is the circle of radius 1 centered at the origin in the plane. Its equation is

$$x^2 + y^2 = 1$$

If a point on the unit circle corresponds to an angle θ, then the coordinates of the point are

$$(\cos(\theta), \sin(\theta))$$



Therefore, cosine gives the x-coordinate and sine gives the y-coordinate. Some important exact values are

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\cos(\theta)$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\sin(\theta)$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1

A.4 Useful Trigonometric Identities

The most important identity for this course is the Pythagorean identity

$$\cos^2(\theta) + \sin^2(\theta) = 1$$

This identity follows directly from the equation of the unit circle.

We will also sometimes use the angle sum identities

$$\begin{aligned}\cos(\alpha \pm \beta) &= \cos(\alpha)\cos(\beta) \mp \sin(\alpha)\sin(\beta),\\ \sin(\alpha \pm \beta) &= \sin(\alpha)\cos(\beta) \pm \cos(\alpha)\sin(\beta)\end{aligned}$$

These are especially useful when working with complex numbers in polar form.

A.5 Inverse Trigonometric Functions

The inverse trigonometric functions allow us to recover angles from trigonometric ratios.

For example, if

$$\cos(\theta) = \frac{1}{2}$$

then

$$\theta = \cos^{-1}\left(\frac{1}{2}\right)$$

In this case, if we are looking for an angle between 0 and π, then

$$\theta = \frac{\pi}{3}$$

In linear algebra, inverse cosine is especially important because the angle between two nonzero vectors $\vec{u}$ and $\vec{v}$ is given by

$$\theta = \cos^{-1}\left(\frac{\vec{u}\cdot\vec{v}}{\|\vec{u}\|\|\vec{v}\|}\right)$$

A.6 Related Acute Angles and Principle Angles

So far, many of our examples have involved acute angles meaning angles between 0 and $\frac{\pi}{2}$. However, in many applications, especially when working with vectors and complex numbers, we need to compute trigonometric ratios for angles larger than $\frac{\pi}{2}$.

A.6.1 Related Acute Angles

A *related acute angle* is the acute angle formed between the terminal arm of an angle and the nearest part of the x-axis.

For example, consider the angle $\frac{5\pi}{6}$. This angle lies in Quadrant II. Its terminal arm is $\frac{\pi}{6}$ away from the negative x-axis. Therefore, the related acute angle is

$$\pi - \frac{5\pi}{6} = \frac{\pi}{6}$$

So $\frac{5\pi}{6}$ has related acute angle $\frac{\pi}{6}$.

Similarly,

- $\frac{7\pi}{6}$ has related acute angle $\frac{\pi}{6}$ since

$$\frac{7\pi}{6} - \pi = \frac{\pi}{6}$$

- $\frac{11\pi}{6}$ has related acute angle $\frac{\pi}{6}$ since

$$2\pi - \frac{11\pi}{6} = \frac{\pi}{6}$$

The related acute angle helps us determine the size of the trigonometric ratio, while the quadrant determines the sign.

A.6.2 Signs of Trigonometric Ratios in Each Quadrant

The signs of $\sin(\theta)$, $\cos(\theta)$, and $\tan(\theta)$ depend on the quadrant where the terminal arm of θ lies.

Quadrant	$\sin(\theta)$	$\cos(\theta)$	$\tan(\theta)$
I	$+$	$+$	$+$
II	$+$	$-$	$-$
III	$-$	$-$	$+$
IV	$-$	$+$	$-$

A common way to remember is: **A**ll **S**tudents **T**ake **C**alculus.

This means

- In Quadrant I, all three are positive.
- In Quadrant II, sine is positive.
- In Quadrant III, tangent is positive.
- In Quadrant IV, cosine is positive.

A.6.3 Computing Trigonometric Ratios Beyond $\frac{\pi}{2}$

To compute a trigonometric ratio for an angle beyond $\frac{\pi}{2}$, we can use the following steps:

1. Determine the quadrant of the angle.
2. Find the related acute angle.
3. Compute the trigonometric ratio using the related acute angle.
4. Choose the correct sign based on the quadrant.

Example A.6.1. Compute $\sin\left(\frac{5\pi}{6}\right)$, $\cos\left(\frac{5\pi}{6}\right)$, and $\tan\left(\frac{5\pi}{6}\right)$.

Solution

The angle $\frac{\pi}{6}$ lies in Quadrant II. Its related acute angle is

$$\pi - \frac{5\pi}{6} = \frac{\pi}{6}$$

In Quadrant II, only sine is positive. Therefore,

$$\begin{aligned} \sin\left(\frac{5\pi}{6}\right) &= \sin\left(\frac{\pi}{6}\right) = \frac{1}{2}, \\ \cos\left(\frac{5\pi}{6}\right) &= -\cos\left(\frac{\pi}{6}\right) = -\frac{\sqrt{3}}{2}, \\ \tan\left(\frac{5\pi}{6}\right) &= -\tan\left(\frac{\pi}{6}\right) = -\frac{1}{\sqrt{3}} \end{aligned}$$

Example A.6.2. Compute $\sin\left(\frac{5\pi}{4}\right)$, $\cos\left(\frac{5\pi}{4}\right)$, and $\tan\left(\frac{5\pi}{4}\right)$.

Solution

The angle $\frac{5\pi}{4}$ is in Quadrant III. Its related acute angle is

$$\frac{5\pi}{4} - \pi = \frac{\pi}{4}$$

In Quadrant III, only tangent is positive. Therefore,

$$\begin{aligned} \sin\left(\frac{5\pi}{4}\right) &= -\sin\left(\frac{\pi}{4}\right) = -\frac{1}{\sqrt{2}}, \\ \cos\left(\frac{5\pi}{4}\right) &= -\cos\left(\frac{\pi}{4}\right) = -\frac{1}{\sqrt{2}}, \\ \tan\left(\frac{5\pi}{4}\right) &= \tan\left(\frac{\pi}{4}\right) = 1 \end{aligned}$$

Example A.6.3. Compute $\sin\left(\frac{5\pi}{3}\right)$, $\cos\left(\frac{5\pi}{3}\right)$, and $\tan\left(\frac{5\pi}{3}\right)$.

Solution

The angle $\frac{5\pi}{3}$ is in Quadrant IV. Its related acute angle is

$$2\pi - \frac{5\pi}{3} = \frac{\pi}{3}$$

In Quadrant IV, only cosine is positive. Therefore,

$$\begin{aligned}\sin\left(\frac{5\pi}{3}\right) &= -\sin\left(\frac{\pi}{3}\right) = -\frac{1}{2},\\ \cos\left(\frac{5\pi}{3}\right) &= \cos\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2},\\ \tan\left(\frac{5\pi}{3}\right) &= -\tan\left(\frac{\pi}{3}\right) = -\sqrt{3}\end{aligned}$$

A.6.4 Principal Angles

When an angle is larger than 2π or negative, we often replace it with an equivalent angle between 0 and 2π. This angle is called the *principal angle*.

Two angles are coterminal if they have the same terminal arm. Since a full rotation is 2π, coterminal angles differ by multiples of 2π. For example,

$$\frac{13\pi}{6} = 2\pi + \frac{\pi}{6}$$

Therefore, $\frac{13\pi}{6}$ and $\frac{\pi}{6}$ are coterminal. The principal angle is $\frac{\pi}{6}$.

Similarly,

$$-\frac{\pi}{4} + 2\pi = \frac{7\pi}{4}$$

Therefore, $-\frac{\pi}{4}$ and $\frac{7\pi}{4}$ are coterminal. The principle angle is $\frac{7\pi}{4}$.

Example A.6.4. Find the principal angle for $\frac{25\pi}{6}$.

Solution

We subtract multiples of 2π until the result is between 0 and 2π.

$$\frac{25\pi}{6} - 2\pi = \frac{13\pi}{6}$$

This is still larger than 2π, so we subtract 2π again

$$\frac{13\pi}{6} - 2\pi = \frac{\pi}{6}$$

Therefore, the principal angle is $\frac{\pi}{6}$.

Example A.6.5. Find the principal angle for $-\frac{2\pi}{3}$.

Solution

Since this angle is negative, we add 2π.

$$-\frac{2\pi}{3} + 2\pi = \frac{4\pi}{3}$$

Therefore, the principal angle is $\frac{4\pi}{3}$.

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